



**GREEN
CLIMATE
FUND**

Meeting of the Board
17 – 20 October 2018
Manama, Bahrain
Provisional agenda item 17

GCF/B.21/10/Add.13

26 September 2018

Consideration of funding proposals - Addendum XIII

Funding proposal package for FP094

Summary

This addendum contains the following three parts:

- a) A funding proposal summary titled “Ensuring climate resilient water supplies in the Comoros Islands”;
- b) No-objection letter issued by the national designated authority(ies) or focal point(s); and
- c) Environmental and social report(s) disclosure;

The funding proposal package for FP094 is being submitted for the first time for the Board's consideration at its twenty-first meeting.



GREEN
CLIMATE
FUND



Funding Proposal

Version 1.1

The Green Climate Fund (GCF) is seeking high-quality funding proposals.

Accredited entities are expected to develop their funding proposals, in close consultation with the relevant national designated authority, with due consideration of the GCF's Investment Framework and Results Management Framework. The funding proposals should demonstrate how the proposed projects or programmes will perform against the investment criteria and achieve part or all of the strategic impact results.

Project/Programme Title: Ensuring climate resilient water supplies in the Comoros Islands

Country/Region: Comoros

Accredited Entity: United Nations Development Programme

Date of Submission: 16/08/2018

Contents

Section A	PROJECT / PROGRAMME SUMMARY
Section B	FINANCING / COST INFORMATION
Section C	DETAILED PROJECT / PROGRAMME DESCRIPTION
Section D	RATIONALE FOR GCF INVOLVEMENT
Section E	EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA
Section F	APPRAISAL SUMMARY
Section G	RISK ASSESSMENT AND MANAGEMENT
Section H	RESULTS MONITORING AND REPORTING
Section I	ANNEXES

Note to accredited entities on the use of the funding proposal template

- Sections **A, B, D, E** and **H** of the funding proposal require detailed inputs from the accredited entity. For all other sections, including the Appraisal Summary in section F, accredited entities have discretion in how they wish to present the information. Accredited entities can either directly incorporate information into this proposal, or provide summary information in the proposal with cross-reference to other project documents such as project appraisal document.
- The total number of pages for the funding proposal (excluding annexes) is expected not to exceed 50.

Please submit the completed form to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"[FP]-[Agency Short Name]-[Date]-[Serial Number]"

A.1. Brief Project / Programme Information		
A.1.1. Project / programme title		Ensuring climate resilient water supplies in the Comoros Islands
A.1.2. Project or programme		Project
A.1.3. Country (ies) / region		Comoros
A.1.4. National designated authority (ies)		Ministry of Energy, Agriculture, Fisheries, Environment, Country Planning and Urbanism
A.1.5. Accredited entity		United Nations Development Programme
A.1.5.a. Access modality		☐ Direct ☒ International
A.1.6. Executing entity / beneficiary		Executing Entity: Government of Comoros acting through the Ministry of Energy, Agriculture, Fisheries, Environment, Country Planning and Urbanism (MEAPEATU) Beneficiary: 450,000 direct beneficiaries in 103 communities 800,000 indirect beneficiaries
A.1.7. Project size category (Total investment, million USD)		☐ Micro (≤ 10) ☐ Small ($10 < x \leq 50$) ☒ Medium ($50 < x \leq 250$) ☐ Large (> 250)
A.1.8. Mitigation / adaptation focus		☐ Mitigation ☒ Adaptation ☐ Cross-cutting
A.1.9. Date of submission		24/05/2018; 10/08/2018; 16/08/2018
A.1.10. Project contact details	Contact person, position	Henry Rene Diouf; Regional Technical Specialist
	Organization	UNDP
	Email address	henry.rene.diouf@undp.org
	Telephone number	+251 912503321
	Mailing address	UNDP – Global Environment Finance Unit Bureau for Policy and Programme Support Addis Ababa Regional Hub, Kirkos Sub City, Kebele 01, House No 119, Po Box 60130, ETHIOPIA

A.1.11. Results areas (mark all that apply)	
Reduced emissions from:	
☐	Energy access and power generation (E.g. on-grid, micro-grid or off-grid solar, wind, geothermal, etc.)
☐	Low emission transport (E.g. high-speed rail, rapid bus system, etc.)
☐	Buildings, cities and industries and appliances (E.g. new and retrofitted energy-efficient buildings, energy-efficient equipment for companies and supply chain management, etc.)
☐	Forestry and land use (E.g. forest conservation and management, agroforestry, agricultural irrigation, water treatment and management, etc.)
Increased resilience of:	
☐	Most vulnerable people and communities (E.g. mitigation of operational risk associated with climate change – diversification of supply sources and supply chain management, relocation of manufacturing facilities and warehouses, etc.)

- ☒ Health and well-being, and food and water security
(E.g. climate-resilient crops, efficient irrigation systems, etc.)
- ☐ Infrastructure and built environment
(E.g. sea walls, resilient road networks, etc.)
- ☐ Ecosystem and ecosystem services
(E.g. ecosystem conservation and management, ecotourism, etc.)

A.2. Project / Programme Executive Summary (max 300 words)

The proposed project supports Comoros, one of only two LDC, African Small Island Developing State (SIDS), to adapt to increasing **extreme climate risks** (including droughts, flooding and water quality impacts from landslides/erosion) **that impact the country's drinking and irrigation water supply.**

Comoros has a very small national land area of only 2,612 km² consisting of steep volcanic terrain, with no land further than 7 km from the coast. It therefore has very small watersheds and aquifers which have little natural water storage capacity, and consequently are highly vulnerable to climate change magnified rainfall variability - as is the rural population reliant on only rainwater harvesting - resulting in predicted increases in water scarcity due to drought, flood and salinization impacts on the nations' water supplies.

In conjunction with national and state governments, water service providers, water user associations and communities, and their development partners (China, Arab Fund for Economic and Social Development and UNDP) who provide the co-financing for this project, GCF resources will be used to address critical technical, institutional and financial barriers impeding the improvement of climate resilience of the country's water resources and water supplies.

The project will achieve a national paradigm shift in strengthening the climate resilience of water supply by mainstreaming systematic climate risk reduction approaches into the governance and delivery of water resources, watersheds, water supply infrastructure and water user management, including in planning, investment, design, operation & maintenance.

Specifically, the project will invest in:

- Reinforcing the management of climate resilient water supply by strengthening the water sector enabling environments, for medium to long-term climate adaptation planning. This will be achieved by integration of climate information into the recently revised national water legislation reforms, training on risk-based water management practices, and upgrading tariff reforms to include the additional costs of climate risk reduction;
- Protecting water quality and moderating extreme high and low water resource flows using integrated watershed management improvements in 32 watersheds (informed by water resources monitoring); and using water resources monitoring to provide early warnings and forecasts of climate risks to improve water supply resilience; and
- Increasing the climate resilience of water supply infrastructure through diversifying the water supply sources for 450,000 people (rainwater, surface water and groundwater); and designing and constructing climate-change risk informed infrastructure to protect from flood risks and sized to withstand drought periods.

The project is consistent with the priorities identified in the NAPA and has a no objection letter from the NDA.

A.3. Project/Programme Milestone

Expected approval from accredited entity's Board (if applicable)	23/05/2018
Expected financial close (if applicable)	TBD [date of agreement on the FAA between UNDP and GCF]
Estimated implementation start and end date	Start: <u>01/12/2018</u> End: <u>30/11/2026</u>

Project/programme lifespan	8 years' project delivery Outcome lifespan– 25 years
----------------------------	---

B.1. Description of Financial Elements of the Project / Programme

Grant financing is requested from the GCF to build the resilience of the Comoros water supply to the impacts of climate change, more specifically, to manage its limited water resources and watersheds, protect its water supply infrastructure and strengthen the adaptation capacity of its institutions and communities to plan and operate in increasing climatic extremes. While there are other types of financial instruments available, the proposed traditional grant is being requested due to: the nature of interventions (in a public good); the extreme poverty status of Comoros which contributes to its lack of resources; and the urgent need for climate adaptation finance in Comoros as a LDC, SIDS and African state. First, the grant will be used to meet the additional costs of building the resilience of water resources, as a public good, to the current and project impacts of climate change, which in the absence of climate change, would not be required. Second, as one of the poorest countries in the world, Comoros faces severe limitations in financial resources due to the extremely high levels of poverty and unemployment (with 80% of the rural population poverty-stricken, 45% of the population in absolute poverty and 14.3% of the population unemployed). Third, as one of the 2 countries in the world to be classified as a LDC, African State and SIDS, Comoros is also highly exposed and vulnerable to current and project impacts of climate change which has created urgent needs for climate change action and finance. Additionally, the investment shall not lead to creation of distorting subsidies given the general lack of private sector financing of water supply and management in Comoros.

The current water supply tariff system is undergoing national reform as part of an on-going sector wide governance strengthening initiative. This tariff reform process does not however include or allow for increased Operation and Maintenance costs or capital renewal that would be needed for climate change induced risks, impacts and damages to the water supply schemes. One of the paradigm shifting aspirations of this project is to change that for the future. The project will support the GoC to establish an appropriate and accepted climate resilience tariffing system beyond Year 7 of the project. However, because of the current gap between the costs for providing water and the revenue generation from the water supply system combined with the poor economic conditions of the majority of water users in Comoros, it is more appropriate to plan for a progressive tariff adjustment system that will aim to cover first the O&M costs and potentially progress towards the full cost recovery. For this reason, the water tariff would at the beginning (in 7 years) aim to cover only the O&M budget and progressively integrate the costs of infrastructure replacement and after the costs of capital. This means that the Government intervention to complement tariff revenue (see Operation & Maintenance Plan in Annex XIIIb) will remain until full cost recovery.

The GoC's contribution is made up of a cash co-financing of USD 3,819,270; in-kind co-financing of US\$9,381,165 in lands donated by the government to host the climate resilient water supply infrastructures; and a further \$12,034,399, to support the Operation and Maintenance (O&M) needs for the established system for 25 years (out of which \$1,397,033 is for the 8 year period of project implementation and is counted as in-kind co-finance in table B.2). The GoC cash contribution will support the establishment of the water tariff, the installation of meters to improve the water pricing, capacity building for the integration of climate change in the management of the water supply systems, support to integrated water resources management (IWRM) committees, the integration of climate resilient management of water resources in the university curricula and the project management costs. The GoC's financing and maintenance commitments are provided in Annex IV.

In addition, the China Geo-Engineering Corporation will provide co-financing of US\$1,940,856 for infrastructure resilience upgrades on Grand Comore, including construction of the pipeline and acquisition of the water treatment, pumping and supply system for 19 villages. UNDP will provide co-financing of US\$2 million for project management as well as for project activities - please see detailed budget/annex). The Arab Fund for Economic and Social Development (FADES) will provide co-financing of US\$ 293,363 for infrastructure resilience upgrades on Grand Comore, including support for drilling testing and commissioning of new boreholes and the testing of the salinity of the existing boreholes to determine optimum pumping rate. All co-financing amounts are guaranteed and are sufficient to ensure the satisfactory completion of all proposed activities for which they are designated.

Table 1: Financial Elements per project outputs							
Output	Activity	GCF funding amount	Co-financing amount				Project Total
			GoC	China GEC	FADES	UNDP	
		(million USD)					
Output 1. Climate informed water supply planning and management	Activity 1.1 Prepare recommendations and legal guidance on the integration of climate change adaptation into the national (federal) and regional (state) water sector agencies governance frameworks, regulations and operations	0.328737					0.328737
	Activity 1.2 Develop water sector climate change risk reduction awareness raising programme for national and state agencies and establish CCA knowledge and information exchange mechanisms	0.1000000					0.1000000
	Activity 1.3 Develop and apply criteria for assessing socially sensitive water pricing mechanisms ensuring prices take into account the actual costs of production, storage and processing required in view of the projected climate stresses		0.175000				0.175000
	Activity 1.4 Develop planning guidance on source protection and water quality standards in view of climate change, operating procedures during periods of drought/floods; and safety plans	0.189400					0.189400
	Activity 1.5 Design and conduct trainings on best practices and gender-sensitive techniques of climate change adaptation in the context of water management, health and nutrition among national, regional and local water stakeholders	0.505082					0.505082
	Activity 1.6 Strengthen decentralized water resources management	0.371887	0.130000				0.501887

	capacities to undertake climate risk reduction assessments and develop and deliver awareness campaigns and training programmes to Water Management Committees and users							
	Total Output 1	1.495163	0.305000				1.800163	
Output 2: Climate Informed Water Resources and Watershed Management including forecasting and early warnings of climate risks	<i>Activity 2.1</i> Establish climate resilience focused IWRM Committees and Watershed Risk Reduction Action Plans in the project intervention areas	0.313606				0.214500	0.528106	
	<i>Activity 2.2</i> Implement the water protection and risk mitigating measures on the ground/operationalize the risk reduction plans	0.190000					0.190000	
	<i>Activity 2.3</i> Support IWRM Management Committees to establish water source protection zones and raise public awareness on climate risk reduction benefits of watershed management	0.477205	0.140000				0.617205	
	<i>Activity 2.4</i> Establish water resource monitoring network and upgrade the existing monitoring infrastructure to enable the collection of the required climate/weather data	2.093501	0.103020				2.196521	
	<i>Activity 2.5</i> Build the capacities of the meteorological services to analyse and produce drought and flood forecasts for targeted users, including for flood early warning system	0.388000					0.388000	
	<i>Activity 2.6</i> Build the capacity of the key government, local authorities and committees to interpret the climate information and raise awareness of the local population to act upon the forecasts and EWS		0.340000			1.402700	1.742700	
	Total Output 2	3.462312	0.583020			1.617200	5.662532	

Output 3: Climate Resilient Water Supply Infrastructure	Activity 3.1 Undertake climate risk assessments of existing groundwater abstraction wells to develop risk reduction pumping strategies, and construction of additional boreholes in zones at risk of drought water scarcity in Grande Comore	3.950148			0.293363		4.243511
	Activity 3.2 Build infrastructure to increase resilience of water supply facilities to extended duration low flow periods, greater intensity flood flow damage and flood flow higher turbidity and bacteria loadings (Grande Comore, Anjouan island and Moheli island)	31.149959	10.778198	1.940856			43.869013
	Activity 3.3 Installation of flowmeters to support climate resilient tariff adjustments, and leakage reduction programmes to improve the water pricing and management system taking into account the additional costs associated with climatic hazards		1.944050				1.944050
	Total Output 3	35.100107	12.722248	1.940856	0.293363		50.056574
Total Project Management Costs		1.862226	0.987200			0.382800	3.232226
Project Total		41.919808	14.597468	1.940856	0.293363	2.000000	60.751495

- a breakdown of cost/budget by expenditure type (project staff and consultants, travel, goods, works, services, etc.) and disbursement schedule in project/programme confirmation (term sheet) as included in section I, Annexes.

The budget figures presented in this proposal exclude the fee (please refer to the term sheet).

B.2. Project Financing Information

	Financial Instrument	Amount	Currency	Tenor	Pricing
(a) Total project financing	(a) = (b) + (c)	60.751495	million USD (\$)		

(b) GCF financing to recipient	(i) Senior Loans		<u>Options</u>	() years	() %		
	(ii) Subordinated Loans		<u>Options</u>	() years	() %		
	(iii) Equity		<u>Options</u>		() % IRR		
	(iv) Guarantees		<u>Options</u>				
	(v) Reimbursable grants *		<u>Options</u>				
	(vi) Grants *	41.919808	<u>million USD</u> (\$)				
* Please provide economic and financial justification in section F.1 for the concessionality that GCF is expected to provide, particularly in the case of grants. Please specify difference in tenor and price between GCF financing and that of accredited entities. Please note that the level of concessionality should correspond to the level of the project/programme's expected performance against the investment criteria indicated in section E.							
Total requested (i+ii+iii+iv+v+vi)		41.919808	<u>million USD</u> (\$)				
(c) Co-financing to recipient	Financial Instrument	Amount	Currency	Name of Institution	Tenor	Pricing	Seniority
	<u>Grant</u>	2.000000	<u>million USD</u> (\$)	UNDP	N/A	N/A	<u>Options</u> <u>Options</u> <u>Options</u> <u>Options</u>
	<u>Grant</u>	3.819270		Gov of Comoros			
	<u>In-kind</u>	9.381165		Gov of Comoros			
	<u>In-kind</u>	1.397033		Gov of Comoros			
	<u>In-kind</u>	1.940856		China Geo-Engineering Corporation			
	<u>Grant</u>	0.293363		FADES (Arab Fund for Economic and Social Development)			
	Lead financing institution: N/A						
* Please provide a confirmation letter or a letter of commitment in section I issued by the co-financing institution.							
(d) Financial terms between GCF and AE (if applicable)	N/A						
B.3. Financial Markets Overview (if applicable)							
N/A							

Please fill out applicable sub-sections and provide additional information if necessary, as these requirements may vary depending on the nature of the project / programme.

C.1. Strategic Context

Comoros is one of only 2 countries in the world to be classified as a Least Developed Country, an African State and a Small Island Developing State (SIDS) by the UN System. The hydro-physical features of Comoros significantly contribute to its high vulnerability to climate change impacts. The entire country, separated into three islands, has a land area of only 2,612 km² and no land further than 7km from the coast. Comoros therefore has very small watersheds and aquifers with very limited natural storage which respond rapidly to changes in rainfall and are consequently highly vulnerable to rainfall variability and intensity, resulting in significant drought, flood erosion and salinization impacts.

Climate change predictions for Comoros include an increase in rainfall variability, lengthening of droughts and increasing frequency and intensity of storm floods and resulting erosion.

The main island of Grand Comore has no surface water, requiring coastal towns to exploit marginally fresh groundwater resources, whilst the rural upland communities, making up 50% of the island's population, have to rely solely on rainwater harvesting. On the two more remote and poorer islands of Anjouan and Moheli there are no proven groundwater resources and therefore are completely reliant on the seasonally variable streams.

Comoros is also one of the poorest countries in the world, with an estimated 80% of the rural population considered poverty-stricken and 46% of the population living in absolute poverty (<\$1.25/person/day). This severely constrains its national adaptation capacity.

One of the most urgent needs in Comoros, as stated by the NAPA, is to build the resilience of their water supply to the impacts of climate change. More specifically Comoros needs to increase the resilience of its limited water resources and watersheds, protect its water supply infrastructure and strengthen the adaptation capacity of its institutions and communities to plan and operate in increasing climatic extremes.

National capacity to adapt to climate change risks in Comoros is extremely limited, as it is for many SIDS, but particularly those that are also LDCs. At least 14.3% of the population is unemployed. The unemployment rate among those aged 15-24 is very high at 50.5%.¹ Between 70-80% of the Comorian population are small-scale farmers that are dependent on rain-fed water resources for subsistence agriculture. National food security is therefore closely linked to water security and to climate change impacts and their successful adaptation. More widely, poverty issues and limited employment opportunities are severely hindering the country from self-sustaining economic growth.²

Comoros is therefore not only intrinsically vulnerable to climate change impacts but has little current capacity to strengthen its adaptive capacity to be resilient to these impacts. This lack of resilience extends as far as the capacity to submit grant applications to the global climate change adaptation funds.

Climate Change Impacts and Vulnerabilities

Climate Change Projections for Comoros

The IPCC AR5³ indicates that for the southern Indian Ocean (including the Comoros Islands) there will be:

- a decline in rainfall - that will cause a reduction in surface water flows and groundwater recharge and an increase in pollution due to reduced dilution potential;
- an increase in mean temperatures - that will reduce water supplies via evapotranspiration and increase water demand;
- continued sea level rise that will contribute to further saltwater intrusion into coastal aquifers; and
- an increase in the frequency and intensity of natural hazards such as cyclones, landslides and droughts.

¹ Programme Pays pour le Travail Décent 2015-2019

² Comoros' Country Work Programme: Accelerated Growth and Sustainable Development - SCA2D (2015-2019)

³ Commission de L'Océan Indien. 2012. Synthèse des travaux du projet Acclimate.

The UNDP Climate Change Country Profile for Comoros⁴ developed by the University of Oxford used refined Global Climate Models to evaluate climate change predictions for the country. This work identified up to 47% reduction in rainfall during the dry season by 2090, with up to a 45% increase in rainfall during the wet season and a further 14% increase of rainfall in extreme events. This does not include the potential increase in cyclone activity – Comoros is in the Indian Ocean cyclone path – which may not be captured by GCM projections. This work also identified a rise in sea levels of up to 0.56m by 2090.

A specific review of Global Climate Model predictions for Mutsamudu in Anjouan⁵ reports a 5% reduction in annual rainfall to 2065, but more importantly up to an 80% reduction in dry season rainfall (June to October) is predicted and up to 100% increases in monthly rainfall for some GCM predictions in some wet season months. Data from the existing rain gauges has indicated that mean annual reduction of precipitation on all the islands is 5% per year.⁶ With these projections, it can be reasonably expected that there will be increases in the duration, intensity and frequency of both droughts and floods in Comoros.

Impact of Climate Change on Rainfall in Comoros

A detailed analysis of local monthly and daily rainfall analysis (see Feasibility Study Section 1.2 and its Appendix 3) has been undertaken to investigate evidence of climate change influences on rainfall. Monthly data has been used to look at potential increases in climate change aggravated drought, and daily rainfall used to consider potential increased in flooding risk.

A ranking analysis using monthly data has revealed a clear downward trend of annual rainfall over the last 60 years, with the **four driest years in the last 60 years to have been 1999, 2003, 2007 and 2013**, with the most extreme driest (1%) periods to all be in the last 1-2 decades. This is clear evidence of an increasing trend of extended low rainfall periods.

The daily rainfall analysis revealed the **longest consecutive sequence of dry days** (<1mm/d) currently **averages 34 days** per year across the country, with an increased frequency of the occurrence of 10 consecutive dry days over the last decade alone.

The above analysis was repeated for Effective Dry Days *i.e.* when rainfall minus evapotranspiration is less than 0 mm/d, which is a more useful indicator for agriculture. In this analysis the longest consecutive effective dry period in the country averaged across the three islands was 92 days – almost 3 months.

Daily rainfall analysis also revealed maximum daily rainfall to have exceeded more than 100mm/d in 8 of the last 9 years, with the most extreme rainfall event on record being in April 2012 (718 mm) where more than 1, 942mm fell over 6 days. When considered alongside GCM predictions of 100% increases in wet season monthly rainfall, this demonstrates an increasing likelihood of extreme flood events.

An analysis of the GCM results on the World Bank Climate Portal⁷ has been undertaken as part of the feasibility study for this project design⁸. This work concludes there will be an **average increase across the country in consecutive effective dry** (*i.e.* (<5mm/d rainfall) days over the duration of the project investment lifecycle (2043) **of 48 days** (at the 95%ile) **solely attributable to climate change**. The actual maximum consecutive dry days (<5mm/d) observed across the country in the last decade varied between 54 and 132 days at the 95%ile, depending on location.

Climate Change Rainfall Impacts On Water Resources and Water Supply

Impact of Reduced Rainfall on Rainwater Harvesting Supply Security (primarily Grand Comore)

⁴ Comoros UNDP Country Profile. 2006. University of Oxford.

⁵ Raphael Tshimanga, 2015. Mutusmudu Watershed Integrated Management Plan. UNEP IWRM SIDS Programme

⁶ African Development Bank, Strategy and Programme of PEAPA in the Comoros, Annex 1: Hydrologic and water resources balance

⁷ <http://sdwebx.worldbank.org/climateportal>

⁸ GCF Submission Feasibility Study Section 1.3.3

More than 50% of the population in Grand Comore currently relies on rainwater harvesting. Rainwater harvesting analysis⁹ of average household rainwater tanks¹⁰ shows they currently go dry for as long as 35 days (using a government rural demand estimate¹¹ of 35 lpd for a household of 8 persons), although the analysis shows this can be reduced to 15 days with household rainwater rationing to 20 lpd, if adequate drought forecasting was available.

The use of community shared rainwater harvesting¹⁰ does provide some added water security, but the analysis⁹ shows even using drought rationing these are unlikely to provide an additional 21 days of supply.

The feasibility study rainwater harvesting analysis therefore identifies current dry season rainwater harvesting practices in the last decade are on the limits of their reliability – with combined average household and community systems only just getting rural communities through the most recent decade of climate change extended droughts, by 5-7 days, whereas poorer families in smaller houses and communities in drier parts of the island now have to buy water in the towns. Under the current climatic conditions therefore, the **existing water schemes** and ad hoc backup supplies are **just sufficient (by 5-7 days) to provide water supply through the currently experienced dry periods**.

However, the predicted increase in the length of **dry periods by up to 48 days due to climate change** will therefore have **catastrophic impacts on rural community water supplies** right across Grand Comore, due to the island wide failure of the rainwater harvesting systems.

Impact of Reduced Rainfall on Surface Water Flows and Water Supplies (primarily Anjouan and Mohéli)

The small and steep nature of watershed catchments in Comoros and lack of lakes and other natural reservoirs means the stream flows are very variable and many watersheds dry out entirely during dry periods, limiting those which can be used for year-round water supply. Limited hydrological information indicates dry season flows in perennial streams are only sustained by groundwater and are typically 10% of average wet season flows.

Reduced annual and seasonal rainfall will result in lower average river flows as well as less groundwater recharge and therefore reduced dry season groundwater fed base flows. This is consistent with reports that although permanent rivers exist on two of the islands, Anjouan and Mohéli, these rivers are becoming increasingly ephemeral; the number of permanent rivers on Anjouan is reported to have declined from 45 in 1950 to 12 today.¹²

Climate change can therefore be expected to reduce the number of perennial rivers that will be able to supply water for the two islands of Anjouan and Mohéli, which are reliant on surface water intakes. This will have both water supply quantity (increase the frequency and duration of low flow and/or no flow periods) and water quality impacts (increased temperature and flow stagnation increasing both microbiological activity and macro-biological presence e.g. parasites).

These surface water supply systems are highly vulnerable to such impacts as they have little on-line water storage and few have any water treatment, only delivering raw water.

Impacts of Reduced Rainfall on Groundwater Resources and Supplies

Ordinarily groundwater aquifers contain multiple years of recharge and therefore require multiple dry years in order to be impacted upon by reducing rainfall. Groundwater resources are therefore generally considered to be much more resilient to dry periods.

In the most populated island of Grande Comore, groundwater is the main water source for 30% of the population – primarily in the coastal towns and villages. All boreholes are close to the coast and many exhibit elevated salinity year-round¹³. Reduced annual aquifer recharge and reduced groundwater throughflow will be at least equivalent to the predicted annual rainfall reduction (-15%¹⁴) but are likely to be reduced by a greater percentage due to increased

⁹ GCF Submission Feasibility Study Section 4.1.1 and 4.1.2, 2018

¹⁰ MP2L, 2015. Schema Directeur AEP Mbadjini Est

¹¹ Stratégie et Programme National d'AEPA aux Iles Comores, 2013

¹² NAPA 2006

¹³ Feasibility Study Section 4.2.1.

¹⁴ Comoros UNDP Country Profile. 2006. University of Oxford.

temperatures and higher evapotranspiration. This will therefore increase salinity in pumping wells, reducing the availability of dry season groundwater supply where boreholes are already only marginally fresh.

Water supplies from boreholes beyond the capital of Moroni in Grand Comore are particularly vulnerable to increases in salinity (see Figure 18 of the Feasibility study) as their salinity is naturally elevated to an extent that their water quality is only just potable. Hence even small future increases beyond current seasonal salinity fluctuations therefore can change groundwater from being potable to undrinkable, necessitating increased operational monitoring and minimising pumping rates.

Impacts of Reduced Rainfall on Water for Agriculture

Agricultural dry seasons already limit rain-fed cropping – with effective dry periods (*i.e.* rainfall – evaporation) of 3 months' duration¹⁵. Rural farmers are therefore already reliant on streams and non-potable water rainfall structures to sustain crops through these periods. The predicted increase in effective dry days by 48 days, will however extend the dry period requiring irrigation to beyond 5 months. This explains why dry season crop failure is such a high priority issue (NAPA, 2006).

As detailed above, reductions in rainfall will reduce both rainfall capture and stream flows, reducing rural food security.

Impacts of Increased Storm Floods on Water Resources and Water Supplies

Intense rainfall is predicted¹⁶ to increase with **climate change**, both in terms of wet season rainfall (+45%) and rainfall falling as storms (+14%). The amount¹⁷ (at the 95%ile) of **very wet days** per annum is predicted to **increase by 30 days** per year, with the daily maximum increasing by 44mm (a 26% increase), the 5 day cumulative rainfall increasing by 120mm (a 34% increase) and the largest monthly rainfall increasing by 292mm.

These increases in storm event rainfall intensity represent an increased risk to both surface water and groundwater resources and their associated abstraction infrastructure due to increased peak and total flood flows causing both direct physical damage and also increased catchment erosion resulting in higher turbidity raw water entering the water supply systems

Because rainfall is concentrated into water courses, and because the catchments in Comoros are so small and so steep, the percentage increase in peak flood flows – *i.e.* those flows most likely to be creating damage, is greater than the percentage increase in rainfall. So **peak flood flows will increase at least by 34 – 45%** but in reality considerably more.

Furthermore because of the 'feedback' process of erosion, once these climate change enhanced peak flows increase erosion and start to destabilize the catchment, the climate change derogated watershed is then more vulnerable to erosion in later smaller flow events than it was before. Thus turbidity increases all year round due to the climate change enhanced flood flow initiation of erosion, increasing monitoring, treatment and maintenance requirements.

These observations are supported by research¹⁸ on watershed erosion rates that show a doubling of the rainfall intensity rate increases the erosion rate by an order of magnitude. Furthermore, the same research shows the erosion rate is an order of magnitude higher for poorly vegetated terrains compared to mature vegetation canopy. Therefore, **storm damage to steep watersheds** which removes mature canopy during the storm events not only can **increase erosion rates rapidly by two orders of magnitude** but then leaves the watershed more erodible by an order of magnitude, to subsequent non-storm events.

The April 2012 floods (>1250mm over 6 days, >400mm on one day) caused much damage in both islands. Stream water supply intakes were both physically damaged by vegetation and debris as well as received large volumes of sediment both at the intakes and further into the system. The groundwater supplies on Moroni however did not escape the flood damage. Several of the main water supply boreholes are located in close proximity to storm run-off routes off the volcano

¹⁵ Feasibility Study Section 1.3.3 – Daily Rainfall Analysis.

¹⁶ Comoros UNDP Country Profile. 2006. University of Oxford.

¹⁷ <http://sdwebx.worldbank.org/climateportal>

¹⁸ USGS 1984. Physical Basis And Potential Estimation Techniques For Soil Erosion Parameters In The Precipitation-Runoff Modeling System.

- as these are recognized as being zones of enhanced groundwater recharge into the underlying aquifer. However, this also puts the boreholes at risk from storm run-off.

Climate Vulnerability Variance and Distribution

The Union of Comoros consists of three islands – Grand Comore, Anjouan and Moheli. The most populated is Grand Comore, then Anjouan, then Moheli, with the capital Moroni in south-west Grand Comore.

Grand Comore has groundwater resources, although the steep topography limits exploitation to locations within 3km of the coast, where saline intrusion is problematic. The population away from the coast (50% of the island) is solely reliant on rainwater harvesting.

Anjouan and Moheli rely on stream flows from small, steep, highly erodible volcanic watersheds. The catchment flows respond quickly to rainfall variations, both drying out in extended dry periods and generating rapid and turbid storm flood flows after heavy rainfall.

The islands therefore have differing water resources, but are vulnerable in differing ways to the increasing climate change enhanced rainfall variability– drought primarily in Grand Comore, whereas flood damage and turbid water quality in Anjouan and Moheli .

The lack of climate change resilience is therefore endemic at the national scale, irrespective of whether the climate risk is water supply scarcity due to extended drought or flood damaged/polluted water infrastructure. There is no climate risk reduction legislation to require government agencies to address the issue, no understanding of water resources vulnerability to climatic extremes, no technical capacity to identify and treat climate risks to watershed or to water supply infrastructure nor to forecast and provide warnings of climatic extremes, and little public awareness of how to develop and adapt to climate change at the community level.

Without a paradigm shift in the national enabling environment to adapt to climate change, any interventions to support the most vulnerable communities, be they rural farmers or peri-urban town hawkers will be unsustainable.

National Climate Change Policies and Strategies

The Comoros national sustainable development strategy, the ***Accelerated Growth and Sustainable Development Strategy, SCA2D, (2015-2019)***, articulates GoC's vision to reduce poverty and emphasizes the promotion and expansion of access to **resilient** drinking water and sanitation, especially for the most vulnerable people. The SCA2D also prioritizes climate change resilience through initiatives that include the establishment of appropriate monitoring networks to detect climate change impacts and setting up climate sensitive water and sanitation projects.

Both the Second National Communication on Climate Change (2012) and the NAPA (2006) identify specific climate change impacts (most notably drought, but also heavy rains and cyclones) on the water sector (see Section E.5). necessitating:

- strengthening water resources management and environmental monitoring;
- improving groundwater management and preservation;
- expanding hydrological and meteorological monitoring infrastructure;
- protecting ecosystems and regulating stream flow; and
- integrating local populations into water resources management.

The National Water Code, recently revised in 2015 and awaiting ratification, includes detailed plans on how the more decentralized, current institutional reforms will improve water management. The National Water Code does not however make explicit reference to climate change risks and the need for water sector's adaptation to these risks.

The vision of the **National Water Strategy** is to increase water supply to 100% of the population by 2030¹⁹ and a goal of Comoros' **INDC (2015)** is to provide irrigation water so that 100% of country farmers can have access to water management systems adapted to the evolving impacts of climate change.²⁰

This project has been designed to support these complimentary goals by improving the country's ability to reduce climatic risks to its water supply sector through: mainstreaming climate risk reduction into water supply infrastructure planning and financing; improving water resources climate impact monitoring and risk forecasting and watershed adaptation risk reduction approaches; and ensuring climate risk reduction measures are integral to water supply infrastructure design, construction and operation & maintenance.

C.2. Project / Programme Objective against Baseline

1. Baseline Scenario: Current context of water insecurity and scarcity

Water Resources

The types of water resources available and exploited in Comoros are island specific. Watersheds on Anjouan and Moheli islands are small, steep and highly vulnerable to erosion. These catchments respond very quickly to intense rainfall events (typically within 1-2 days), generating rapid peak flows, yet also have limited retention times and can drain out within 1-2 months²¹). River water intakes are therefore susceptible to extreme weather periods, including damage from peak flood flows, are compromised by highly turbid flood flows and some have dry season flow shortfalls. – The water supply schemes continue however to function. There is no current monitoring programme of stream flows in the country. Seven of the project target areas are in Anjouan and two of the target areas are on Moheli (see Feasibility Study Appendix 8).

On Grande Comore, in contrast there is an almost complete absence of surface water. Water supplies have to be sourced from groundwater resources and through rainwater harvesting.²²

Groundwater sources had recent significant issues with pollution inundation from the 2012 extreme storm flood event run-off debris. Retrospective protection measures have now been put in place but there is no current effective pro-active climate risk reduction strategy.

The fresh groundwater resources on Grand Comore are not uniform around the island with 31 of the 50 boreholes too saline for drinking use.²³ Hydrogeological studies²⁴ demonstrate the most substantial fresh groundwater resources to be on the south-west side of Grand Comore immediately south of the capital Moroni. There are 6 project target areas on Grand Comore.

Limited groundwater exploitation has also occurred on Anjouan and Moheli, with high level springs supporting stream water abstractions and small numbers of wells reported near the coast, on the central plateau and in river gravels²⁵. There is no groundwater monitoring in the country at this time, other than operational monitoring of production boreholes.

Grand Comore residents also therefore have to exploit rainwater harvesting systems. Estimates for Grand Comore are approximately 50% of the population island wide uses household rainwater harvesting, but this is reported to vary between 40 and 90% for individual villages²⁶.

The daily water balance calculations (see Feasibility Study Section 1.4) show that household rainwater harvesting can meet normal daily demand (35 lpd) during normal climatic periods. In the last decade however they can only just meet

¹⁹ The Comoros' Institute of Statistics has projected the population to be 1.16 million by 2030.

²⁰ Comoros Intended Nationally Determined Contribution (INDC) to the UNFCCC Sep 2015

²¹ Raphael Tshimanga, 2015. Mutusmudu Watershed Integrated Management Plan. UNEP IWRM SIDS Programme.

²² The majority of population on Grande Comore lives upland and have poor access to water resources..

²³ African Development Bank, Strategy and Programme of PEAPA in the Comoros, Annex 2: Hydrogeologic context

²⁴ Jean-Christophe Comte et. al, Challenges in groundwater resource management in coastal aquifers of East Africa: Investigations and lessons learnt in the Comoros Islands, Kenya and Tanzania, Journal of Hydrology: Regional Studies 5, pp. 179–199 (2016)..

²⁵ Charmoullie, 2012. Ebauche du fonctionnement hydrogeologique de l'île d'Anjouan (Comores)

²⁶ MP2L, 2015. Schema Directeur AEP Mbadjini Est

the extended dry season household demand²⁷ when households reduce consumption to 20 litres per person per day – the calculations show a 15 day period without rainwater supply. The analysis identifies the cause is primarily because the house roof area catchment size areas are insufficient to supply the tanks, the size of the tanks themselves is adequate – necessitating the households to then rely on community tanks or other alternatives. A detailed study in south-east Grand Comore²⁶ confirms rainwater harvesting in the rural villages is highly efficient with high percentage roof area guttering and rainwater capture. The full roof area of these rural communities is therefore already being maximized.

Community rainwater harvesting tanks are also reported²⁶ to be used as a communal water reserve for when household rainwater tanks have gone dry – to cover the 15 days identified in the analysis above. Daily water balance calculations²⁷ demonstrate communal tanks can only provide water supply for a maximum of a week, although this could be stretched to 21 days on minimum (20 lpd) drinking water rations only. As with the household tanks, the community buildings are integrated into the fabric of the village conurbation, the rainwater collection systems are highly efficient out of necessity and there is no opportunity for roof catchment expansion.

The communal tanks therefore provide a useful additional emergency buffer to cover current climate change induced rainfall shortage experience to date (*i.e.* 21 days minus 15 days = 6 days to spare), but are not large enough to provide for the predicted climate change additional drought of a further 48 days water supply.

If the future climate situation remained as it is today, Grand Comore would not need extra investment in water infrastructure. However, with the forecasted increase of dry days, the current water supply will not be able to satisfy the needs of the communities. The most obvious option to respond to the increase of the dry days would be to increase the size of the roof, but there is no opportunity for roof catchment expansion. Indeed, the typical village housing structure on Grand Comore is of an interlinked and interconnected walled fortified mini-urban area.

For these rural communities on Grand Comore the only option to address the predicted increase in drought water scarcity is to access the island's groundwater resources.

Water Supply Coverage

The population accesses water via rainwater harvesting (25%) or from household (40%) and community (25%) fountains fed directly by gravity fed stream intakes and some well supplies in coastal towns. The type of water supply system varies from island to island, with groundwater supplies only being used in Grand Comore and stream intakes only in Anjouan and Moheli. There are no significant dams or reservoirs on Comoros and no major natural lakes.

Groundwater distribution systems are restricted to the coastal plain due to limitations on pumping and electricity, as well as the locations of boreholes and wells being restricted to low ground elevations near the sea. There is minimal water storage within the groundwater distribution systems, with pumps generally operated 20 hours out of 24. The groundwater system is not currently sized to support the rural population currently using rainwater harvesting.

Surface water distribution systems are gravity fed only and are therefore generally restricted to supplying communities and towns within the watershed from which water is abstracted. There is minimal water storage within the surface water distribution systems, with water supply effectively being from instantaneous flow schemes.

Water consumption per capita is estimated to be only 35 litres per day, but is reported to be as low as 11 litres per capita per day for 50% of the population of Mutsamudu in Anjouan²⁸. Water consumption is therefore mostly limited to the strict minimum for daily needs (drinking, cooking and hygiene washing). Excessive water use and water user wastage are not issues in Comoros.

For those not connected to a reliable year round local water supply system there is a thriving private sector road tanker distribution system. A jerrycan of 20 liters of non-potable water (not sterilized) costs 250 to 500 KMF (US\$ 25-50/m³) in Grand Comore against 25 KMF (US\$ 2.5/m³) in Anjouan - the difference in cost reflecting the better water quality in Grand Comore (non-turbid groundwater compared untreated stream water) and the greater poverty in Anjouan.

Irrigation and Livestock

²⁷ Pro Doc Feasibility Study

²⁸ Raphael Tshimanga, 2015. Mutusmudu Watershed Integrated Management Plan. UNEP IWRM SIDS Programme.

The majority of agricultural production is rain fed. On Anjouan and Moheli communities do use stream water for irrigation using hand held containers but there is no larger scale irrigation practiced.

At Sangani on Grand Comore, a volcanic cinder cone has in the past been lined with cement and this used to store rain water, which is used to irrigate an area of 10-14 hectares for cash crops. A daily water balance analysis (see Feasibility Study Section 4.1.3) has shown the Sangani Impluvium is just capable of irrigating cash crops (e.g. tomatoes) through the driest season of the most recent decade (1-2 days of no supply), but it is very dependent on crop type and cropping months, and requires good agricultural practice with respect to soil moisture retention. The predicted increase in future climate change extended drought by 48 days cannot be met at all with the current structure.

Water Governance

Comoros recently updated its National Water Code (2015) to deliver institutional reforms necessary to improve sustainable water planning and management. Unusually, the country has not only national ministries but also island specific water management authorities, which fragments national efforts to manage sectoral planning, and disperse national capacity across different islands and agencies. The revised National Water Code was approved by the President in 2016 but has not yet been ratified by Parliament. Once ratification is obtained, the Code will need to be translated into specific regulations. The Code lays out reforms necessary to improve water resource management, in particular the harmonization of procedures across all islands, extending the authority of MA-MWE to the urban areas within Anjouan and Moheli.

With respect to O&M roles and responsibilities for the various institutions within the water sector, they are outlined in Annex XIII(b) Operations and Maintenance plan and aligned with the future Water Code.

There is no mention of climate change issues in the new Water Code, which is more focused on service provider institutional strengthening and improved cost recovery. Drought and flood planning and management are not currently practiced by the national or island water management agencies.

In the Water Code, GoC has committed to put in place the framework for implementation and full enforcement of a new tariff system in 7 years. Plans for a comprehensive tariff rollout require significant regulatory work prior to implementation. The Code lays out general principles but leaves to future regulation any specific decision on levels. Principles include universal access, preferential treatment for lowest-income households and cost recovery for service providers, differentiated between urban and rural networks (urban tariffs must cover both capex and O&M, rural tariffs must cover at least O&M). Tariffs will include a fixed and a volume-based component and will be revised periodically by a water regulator to reflect changes in costs.

The process to set tariff levels entails the establishment of an independent water regulator (with its own staff and budget), public consultations, affordability studies (undertaken during the project) and pilots on-site to fine-tune tariffs levels. The latter is an important step since the sustainability of a tariff system relies on evidence of improved service quality.

The Code also envisages the creation of a National Fund for the Development of Water and Sewage Infrastructure (FNDIEA) but offers only very generic indications on the sources of funds (budget, borrowings and concessional) as well as management and supervisory structure. Like tariff reform, FNDIEA appears to be little more than a concept at this stage.

Existing Extreme Weather Coping Strategies

Dealing with climate variability and climatic extremes is an ongoing problem over the last decade, with a range of informal coping mechanisms being used. During periods of climatic norm, the above technical sections confirm the population is adequately served (35 lpd) by rainwater harvesting, fresh groundwater and ample flows in streams²⁹. **The critical problem of water supply in Comoros is only during exceptional dry periods and following storm events.** The daily rainfall analysis demonstrates climate change has had a significant impact on rainfall variability in the last decade,

²⁹ Feasibility Study

including the worst drought and worst flood to date. The GCM predictions confirm a further worsening of both drought longevity (+48 days) and storm intensity (+26-34%) and frequency (+45%) over the project lifetime (2045).

At times of drought, when household rainwater tanks become empty, rural Comorian populations travel to the towns to fill up containers from running water sources and/or road tankers from the towns which deliver water at elevated prices to the rural villages. At the village level, water from ablutions, ylang ylang distillery cooling water and washwater are re-used during dry seasons.

With respect to water quality, during periods of strong rainfall (*i.e.* cyclones and tropical storms), high turbidity levels in streams have been shown to cause rheixistasia³⁰. Intakes cannot be sealed to prevent water ingress, and where not blocked entirely by such events, reduced water supply is distributed without any physio-chemical or disinfection treatment. Consequently, typhoid fever and diarrhoea are the leading sicknesses for children between 3 and 5 years.³¹ The impacts of more extreme climatic rainfall variability are therefore already impacting on the Comoros population.

Water Supply Benchmarking to other LDC SIDS

It is useful to compare Comoros to water supply provision in other LDC SIDS. The Pacific Water and Wastewater Association (PWWA) benchmarks the urban water providers in the Pacific Region³². In the 4 LDC SIDS in the Pacific – Kiribati, Tuvalu, Solomon Islands and Vanuatu, the following levels of water provision are provided:

Country	National Population Served (%)	Utility Supply Water Per Person (lpd)	Comments
Comoros	90% (50% in Grand Comore)	35	Provision of water to capital and island coastal towns. Rural areas on Grand Comore rely on rainwater only.
Kiribati	43%	2	Provision of water only to capital town. Rural areas rely on rainwater harvesting
Solomon Islands	19%	130	Provision of water only to capital and provincial towns. Rural areas rely on rainwater harvesting
Tuvalu	50%	5	Public Works tanker distribution of rainwater & desalination water
Vanuatu	17%	300	Supply to capital is by French Utility Concessionaire. Rural areas rely on rainwater harvesting

Table of LDC SIDS Water Utility Performance

In addition, it is noteworthy the PWWA database shows the Marshall Islands to provide 27 lpd, Papua New Guinea 30 lpd and Nauru 16 lpd.

This benchmarking demonstrates Comoros current water provision is entirely consistent with that of other LDC SIDS in terms of water supply coverage, and is typical of average per capita consumption rates (Port Vila in Vanuatu is an

³⁰ [Feasibility Study](#)

³¹ DGEF, UNDP, GEF April 2012. Report on the Vulnerability and the Adaptation of Water Resources at Risk to Climate Change Impacts on Anjouan.

³² <http://www.pwwa.ws/index.php?page=Benchmarking2>

exception being supplied by a French utility company). In fact Comoros is out-performing the other LDC SIDS in terms of percentage population served by a water supply provider.

The level of water supply provision in Comoros is therefore consistent with its level of development and that which is realistically achievable and possible at this time when compared to other LDC SIDS. A daily water consumption of 35 lpd is adequate for the national populace under normal climatic conditions. Ad hoc cultural drought resilience measures are currently routinely deployed to secure water through the existing dry seasons and storm periods despite the increase in their severity observed in the last decade. This demonstrates that the existing water supply is, for an African LDC SIDS, currently fit-for-purpose, if there was no climate change.

However, with the GCM predictions of increasing duration, frequency and intensity of droughts due to climate change, the rainwater harvesting on Grand Comore will fail, placing excessive demand on the existing groundwater supplies beyond the yields they currently provide, unless additional groundwater resources are exploited.

The GCM predictions of increasing duration, frequency and intensity of storm-related flood damage and turbidity will reduce potable water availability in Anjouan and Moheli unless these systems are upgraded to resist these more intense storms, and their treatment capacity be improved.

The vulnerability of the water supply systems across all three islands will reach a point during the project lifetime where safe water supply during climatic extremes will no longer be possible.

Without the GCF project, the existing water supply to 450,000 people will be severely compromised by climate change intensified impacts during dry periods and during and after storm events.

2. Baseline Investments

The Government of Comoros and the international community have over the last decade invested approximately USD 51 million for current and future projects that address climate and non-climate related water supply security through a combination of institutional capacity reinforcement and infrastructure investments.

Climate Change Projects in the Water Sector

The **ACCE** project, *Adapting Water Resource Management in Comoros to Increase Capacities to Cope with Climate Change*, (financed by LDCF and supported by UNDP-UNEP (Feb 2011 to Dec 2016 \$USD 5.14M)) focused on **increasing climate resilience of drinking and irrigation water supplies** on all islands. It also stressed water management, **reforestation, land use planning and the construction of small rural water mobilization infrastructure** (impluviums, cisterns and drip irrigation) all of which contribute to climate change resilience. Specialized training was provided for water management committees to establish a water pricing system and to conduct good community management techniques and practices. The ACCE project failed to mainstream climate resilience into the national bodies, focusing instead on state agencies and water service providers and hence failed to address the institutionalization of climate resilience and its necessary technical and financial resources into sector strategic planning and annual work programmes and budgeting. The national aspect of the project was given to the PEAPA project to develop the draft Water Code, which then neglected to include climate change within the future legislation. Key additional lessons learned to this project were identified by the terminal evaluation of this project included:

1. Ensure a well developed operations and maintenance plan is created to secure sustainability of the infrastructure constructed, including identification of roles and responsibilities.
2. Necessity for inclusion of flow meters within the distribution system to capture long term usage to support the implementation of a tariff structure and detection of leaks to quantify lost water.

The project was completed in 2016, with a budget of US\$ 3.7 Million.

In pilot areas, watershed committees were established and provided training on the **shared uses of water resources** in the context of increasing scarcity due to climate change. The ACCE project conducted training on **collecting and analyzing and downscaling climate models** and reinforced their capacities to acquire and install equipment for measuring hydro-meteorological and agro-meteorological parameters. It has essentially provided the first steps in

establishing an effective climate monitoring system. The ACCE project was only delivered in specific targeted areas, and was not national in outreach. These areas are not within the GCF project target areas.

The **CRCCA** (Strengthening the Climate Change Adaptation and Resilience Capacity of the Sector) project, **Enhancing adaptive capacity for increased resilience to climate change in the agriculture sector in the Union of the Comoros** (UNDP, LDCF) (Jun 2014 to Dec 2018 \$USD 9.39M) is building capacities to **reduce the vulnerability of agricultural systems** to climate change on all islands. The project includes support to agricultural extension services, the development of agro-climate services, and installing climate data collection infrastructure including the interpretation and **use of forecasting and information bulletins on climate change specific to the development of weather and agro-meteorological products**. The CRCCA focused entirely on the agricultural sector and did not address water sector specific resilience.

Similarly, the **GCCA European Commission-funded project** is strengthening the resilience of Comoros to climate change through i) **data management (using satellite and ground data in GIS systems)** and dissemination and ii) **integration of climate change into development strategies**. The **GCCA** project (Aug 2013 to July 2019 \$USD 3.695M) will also implement local pilot projects related to water and sanitation.. The GCCA is essentially a national non-sector specific project, which does not focus on the water sector except at the pilot scale.

Furthermore, two LDCF have recently commenced: 1) the **Building Climate Resilience through Rehabilitated Watersheds, Forests and Adaptive Livelihoods** project (UNEP) (July 2017 to 2021 \$USD 5.14M) is focusing on strengthening technical and institutional capacity for resilient integrated watershed management at the national and local levels, rehabilitating and sustainably managing watersheds and sub-catchments in specific project areas and providing a diversified array of resilient livelihood strategies; and 2) the **Strengthening of the Comoros' resilience to disaster risk linked to climate change and variability** (UNDP and UN-ISDR) project (Feb 2018 to 2023 \$USD 9.482M) plans to strengthen DRM capacities, monitoring of disaster risks and early warning dissemination. This project will develop climate modelling and provide Early Warning System messages for cyclone-related climate disasters. The climate products developed by this Disaster Risk Management project will support the integration of climate risks into all sectors including but not limited to the water sector in Comoros.

It is noteworthy that most of these climate change specific project efforts are not specific to the water supply sector. Only the ACCE project focused on the water supply sector, and even that at a pilot project scale. The ACCE project focused on increasing access to water through watershed and water supply improvements, which helps to increase resilience by developing 'buffering' capacity to climatic shocks, but there was **little focus on pro-active adaptive management and risk management**.

Non-Climate Change Projects in the Water Sector

Water institutional reform and water supply and distribution infrastructure provision has been provided by the **PEAPA** project (AfDB). PEAPA made major investments (US\$ 10 Million) in infrastructure such as intakes and pre-treatment facilities. Through PEAPA, preliminary studies on the hydrology and geohydrology of the three islands were also conducted. This enabled the water-related ministries to understand the potential of groundwater and surface water resources across the country. The project was completed in 2015. Infrastructure support was in areas not being targeted by the GCF project.

PEAPA also detailed the new institutional and regulatory management schemes decentralizing authority to the islands and communities, although these are yet to be enacted. PEAPA did not however address CCA issues, although increasing water supply management capacity clearly contributes towards a more robust and less vulnerable enabling environment.

Other water supply infrastructure has been provided by: the World Bank's **FADC** grant program after the flood of 2011 / 2012 devastated much of the water distribution network; a US\$ 6 Million upgrade to the Domoni town water supply to be completed in 2018; a US\$ 4 Million upgrade to the Djandro water supply completed in 2017; and a US\$ 4 Million upgrade to Sima Pensinsula water supply completed in 2012 . None of these are in the GCF project area. These were all engineering schemes and had no climate resilience considerations.

The Regional *Integrated Water Resources and Wastewater Management Project (IWRM or GIRE in French) in the Atlantic and Indian Ocean SIDS project* (UNDP – UNEP) will be completed in March 2018. It has implemented a watershed protection and integrated management programme in the Mutsamudu River Basin of Anjouan and developed a national IWRM Plan to improve river and watershed management so as to reduce soil and river pollution, to reduce the amount of water-borne diseases and to increase the sustainability of water supplies. Sensitization campaigns were used to encourage upstream users and farmers to modify their practices to reduce pollution and erosion while increasing recharge. GIRE clearly includes CCA as one of its objectives and improved watershed management clearly increases climate change resilience, although the focus of the Mutsamudu work has been on wider IWRM issues. That said the IWRM national plan provides a strong foundation for watershed eco-system based climate adaptation responses. This project has made important progress on establishment of a watershed committee and water resources management committees in pilot municipalities and their training in IWRM under the GIRE project. The project provided a good example in engaging with stakeholders from different sectors. Coordination was mainly made through official correspondences, steering committees participatory planning, technical and dissemination of Information workshops were organized during the implementation of the demo project activities. It has the potential to be national in scope building from the participatory approach, and has valuable lessons in terms of the importance of stakeholder engagement.

The US\$5 Million **GECEAU** project (2012-2018) funded by AFD, **Support to public management of water in a pilot zone in Grande Comore** is also investing in water mobilization infrastructure and has implemented a pilot project on water management. The project is developing boreholes, rehabilitating the water distribution network and creating an impluvium on Grande Comore, using a strategy of water source diversification (rainwater and groundwater) to increase water security.

In addition, recent region-specific water sector projects on Comoros include development of necessary water infrastructure and improvements in institutional capacity:

1. Oichilli - (US\$1.49M) completed in 2011
2. Anjouan and Moheli
 - a. Water Services Strengthening Project (NETWORK)- (US\$0.93M) 2009 to 2011
 - b. Support Project for Water Management - US\$1.5M 2004 to 2009

Whilst all these projects will contribute to national water security through improved institutional operation and management of water supply systems, water supply infrastructure extensions and upgrades and more effective cost recovery, as well as pilot schemes on watershed management and pollution reduction, **they do not specifically consider climate change risk reduction**. The baseline projects did not explicitly address existing water resources and water supply risks from climatic extremes, and have actually increased exploitation of the finite water resources.

Water Sector Climate Resilience Gap Analysis

Despite the considerable efforts of the non-climate specific water supply programmes focusing on broader service provider institutional strengthening, infrastructure upgrades, cost recovery, and wider watershed management, and the strategic vision of a single project focusing on climate resilience in the water sector correctly exploring water resources monitoring, watershed and land use management as well as infrastructure resilience upgrades and climate forecasting, it is evident the water sector currently does not systematically integrate climate risk reduction into water supply provision and as such the sector remains highly vulnerable to climate change risks.

The gap analysis highlights both the **inherent vulnerability to climate variability**, due to the fundamental nature of the small island context – small land masses, with little natural freshwater storage, resources, which respond rapidly to increases and decreases in rainfall and the increasing severity of the climate change impacts to come.

Existing and forecast hydrological variability based on a range of climate change scenarios and its impact on water resources and water supply infrastructure has received almost no attention. There has been no national scale effort on water resources monitoring and no climate focused support to national agencies, policies or legislation.

Water conservation and water loss reduction within the reticulated water supply schemes has received no attention, despite these being obvious demand side climate change adaptation strategies to reduce water shortages.

Perhaps of greatest concern, the fundamental and unavoidable issue of limited and fragile freshwater resources during periods of rainfall variability is not addressed in any way. Whilst currently this issue does not prevent year round water supply from being made available, this will become a critical data shortage with the increase in drought days, reduced aquifer recharge and increased erosion of watersheds.

This is particularly important with the predicted climate change impact of rainwater harvesting collapsing during future droughts resulting in greater demand for public water supply from groundwater when groundwater resources are under greatest pressure. This double-whammy means that climate change impacts necessitate the absolute need for i) assessing and understanding future drought yields from groundwater; and ii) maximizing the contribution of improved watershed management to increasing water retention and thereby increasing groundwater recharge and stream base flows as well as reducing flood flow peaks and erosion. These two approaches will support maximum potable water resources availability during climatic extremes.

The increasing population growth rate (2.4%)³³ and consequential future increases in water demand will obviously increase the number of people at risk of climate vulnerability. Water resources are abundant during periods of non-climatic extremes – rainwater harvesting works well in Grand Comore except during extreme drought periods, and average stream flows in Anjouan and Moheli exceed predicted future demand by usually at least two orders of magnitude³⁴. The critical issue is potable water availability in times of droughts and floods.

What is clearly absent from the water sector interventions, be they climate focused or not, is a **lack of consideration of climate impacts driven by an assessment of risks and how to reduce these risks**. This is all the more important in Comoros because the limited and therefore inadequate levels of human and financial resources means risk reduction assessment needs to be mainstreamed so as to prioritize climate resilience efforts to focus precious human and financial resources into those areas which can have optimum benefit.

3. Barriers to achieving Water Sector Climate Resilience

The Project Outcome is *Increased Resilience of Water Supplies to Climate Risks in the Comoros Islands*. This outcome can only be achieved by ensuring the government agencies, communities and households have the capacity to plan, deliver, operate and maintain appropriately designed and constructed water supplies and protect their water resources from derogation by forecast climate risks.

In order to build this technical adaptation capacity, the Comoros stakeholders require: knowledge, training and information. And in order to acquire these, Comoros has to provide/secure sufficient financial resources and then use these financial resources in an effective and coordinated manner. In order to secure the financial resources and coordinate stakeholders and agencies, Comoros needs to have appropriate prioritization of climate change impacts in its water sector enabling framework. And this requires political will to address climate change risks – which Comoros does have.

These barriers are discussed in more detail below in the order they logically need to be addressed in.

Limited Financial Resources to Deliver Climate Resilience

Water supply service providers are currently undergoing national tariff reforms as part of the on-going water governance reforms mandated by the new water legislation (Water Code) to improve the sustainability of the water sector. However the issue of appropriate tariff setting is only identified within the new Water Code in terms of normal operating conditions. It is not recognized with respect to the level of additional investment required to deliver resilience to climate impacts – which require much greater investment in risk reduction and preparedness planning (CAPEX) as well as risk reduction operation & maintenance (OPEX).

Water resources monitoring and watershed management however appear to receive no current budgetary support at all, given there is no national hydrological monitoring network, no clear understanding of the nation's water resources, and little watershed management. To date the lack of a hydrological monitoring network and protected watersheds have not impacted water supply provision, however climate change rainfall variability necessitates water resources and watershed

³³ <http://hdr.undp.org/en/data>

³⁴ Feasibility Study Section 4.4.3

management as critical strategies to protecting and increasing water resources availability during climatic extremes in SIDS.

Inadequate Stakeholder Coordination and Collaboration to Deliver Climate Resilience

Comoros has national, island and community (water user) governance arrangements in place. The autonomous island/state level specific to Comoros introduces an additional level of institutional mandates and therefore complexity and fragmentation of water resource and water supply mandates, roles and responsibilities, including the planning and budgeting of annual work programmes, than is usually observed.

The new Water Code legislation articulates a more effective and efficient institutional arrangement for delivering more reliable water supplies, but it does not consider climate change risks to water supplies.

There is no formal governance mechanism for water resources and watershed management at the island scale, although the introduction of IWRM to the regional water directorates by the GIRE project has created an entry point for improved coordination. A national IWRM Plan was validated in August 2017, but has yet to be enacted. With periods of climatic extremes this lack of IWRM to date has not been a critical barrier to water security. However the impact of climate change on water resources necessitates the use of watershed management to both prevent water quality derogation and optimizes water quantity availability – through water retention within the catchment. IWRM therefore becomes an essential climate change strategy in countries with fragile and finite water resources. In addition as climate risk reduction is arguably the most important IWRM issue with which to promote IWRM, using climate resilience to secure political and community support to IWRM is highly effective. Climate risk reduction through IWRM can therefore be used to secure wider commitments to improving non-climate related drinking water quality risk reduction to the water sector e.g. pollution.

Climate risks also introduce a much greater requirement for coordination, cooperation and collaboration between agencies and stakeholders, as they require data, decision-making and actions (e.g. preparedness and response) to transfer between organisations more quickly than managing water resources and water supply in normal operating conditions. Whilst the recent Water Code reforms accommodate the requirement for improved coordination, this does not extend to the level required to achieve effective climate risk reduction.

Limited Knowledge and Data to Identify Climate Risks and Develop Adaptation Responses

There is no current on-going monitoring of water resources and water supply infrastructure performance and no monitoring systems in place to allow such monitoring to occur. This lack of knowledge has not been essential to operate the nation's water supply to date, however it is now required in order to understand reliably the nation's fragile water resources, their availability during climatic extremes, and their quality and vulnerability to future climate variability and plan their resilience.

The lack of detailed climate forecasting in-country prevents specific early warning of future climate extreme impacts, other than by evaluating the antecedent conditions experienced in the country in previous hours, days and months. Monitoring requirements to assess climate risks requires even greater monitoring resources as the need for monitoring becomes time critical, especially for flood and erosion risks, but also for drought, as well as understanding how stressed the water resources become during climatic extremes.

Consequently, it is not currently possible to identify climate hazards (beyond the generic), evaluate priorities, determine risk reduction measures and implement adaptation measures to increase water resources and water supply resilience.

Limited Technical Capacity to Assess and Reduce Climate Risks

The technical staff capacity within water-related agencies is limited. There is some capacity as evidenced by local water engineers, geologists and hydrologists being involved in the development of this project, however they are limited in experience to managing and operating the current water supply infrastructure within normal climatic conditions.

There is a complete lack of sectoral technical capacity in understanding climate change risks – beyond that articulated in the NAPA *et al* – with little experience in how to map climatic hazards, determine risks, identify risk reduction and/or avoidance strategies and integrate these into water resources and water supply planning, location selection, design, construction, operation and maintenance.

There is no evidence of Drinking Water Safety and Security Planning being undertaken on any public or private water supply system to assess and mitigation water quantity and quality risks including climate change risks.

None of these climate resilience gaps are being addressed by the project baseline.

4. Adaptation Options

Based upon the analysis above of gaps and barriers, and cross-referenced/checked to a recent UNICEF review of WASH (Water, Sanitation & Hygiene) Resilience³⁵ in LDC Pacific SIDS, the following adaptation actions are considered relevant to addressing the barriers/gaps preventing water sector climate resilience.

Water Resources

- Strengthen meteorological, hydrological and hydrogeological (groundwater) monitoring and investigation to understand climatic impacts and warn of climatic event impact commencement;
- Improve dry season water resources management – especially groundwater resources management to minimize saline impacts and stream low flows to avoid over-abstraction;
- Improve storm event water resources management – to minimize exposure to flood damage;
- Diversify dependence on single water resource types;
- Increase exploitation of sustainable groundwater resources;

Watershed Protection

- Upgrade watersheds to improve water retention and increase dry season flows;
- Upgrade watersheds to improve water retention and reduce flash flood flows;
- Upgrade watersheds to reduce storm erosion to reduce water turbidity;
- Relocate sanitation and other potentially polluting activities away from high risk flood areas;

Climate Risk Knowledge, Forecasting and Preparedness

- Increase national and local government water supply utility capacity to plan, manage and actively adapt to climate change – including incorporating climate forecasts in water resources management;
- Increase community capacity to plan, manage and actively adapt water supply provision to climate change – including responding to climate forecasts.
- Develop a drought warning system effective at the community level;
- Develop a flood warning system effective at the community level;

Climate Resilient Infrastructure

- Design and locate stream water intakes to be resilient to climate change increased flash floods and high turbidity periods;
- Design and locate boreholes to be resilient to climate change increased storm run-off events;
- Design and locate boreholes to be resilient to climate change increased saline intrusion and up-coning;
- Improve storm run-off drainage infrastructure;
- Increase storage available on stream water dependent supply schemes;
- Increase water capture and storage available for irrigation;
- Expand supply networks to communities solely reliant on rainwater harvesting – in Comoros household and community rainwater harvesting cannot withstand drought periods without increasing roof areas;

Water Conservation and Water Demand Reduction

³⁵ UNICEF Pacific, 2016. Review of Pacific WASH Policy Frameworks in 5 Least Developed Country Small Island Developing States. GWP Consultants

- Reduce leakage and unaccounted for water in reticulated water supply schemes to reduce dry season demands;
- Increase efficiency of irrigated water supply schemes to reduce dry season demands;
- Improve farm practices to minimize dry season water demand and reduce storm erosion potential;
- Introduce water conservation public awareness raising;
- Introduce a water tariff that is punitive on high water usage;
- Design sanitation to minimize freshwater usage;
- Advocate and incentivise wastewater re-use.

The Pacific SIDS WASH Resilience Policy Review is noteworthy in that it considered the linkages between WASH, Climate Change Adaptation (CCA) and Disaster Risk Management (DRM) policy frameworks, as well as global WASH Resilience best practice, and concluded that climate change, disaster risk management and sustainable development in SIDS specifically, are so interconnected, and the magnitude of disaster impacts so great, that these can no longer be delivered as separate issues, but need to be addressed together.

The Pacific SIDS WASH Resilience Review also highlighted, as is clearly the case in Comoros as well, that delivery of these adaptation strategies and actions will require considerable investment in:

- increasing technical human resources;
- monitoring of exposure to and forecasting of climate risks;
- strengthening and coordinating institutions and their policy frameworks; and
- ensuring long term financial sustainability of the water supplies – be that at utility, community or household scale,

in order to overcome existing capacity, information, organization and financial inadequacies that are barriers to implementation. These issues are common to all of the adaptation options and are therefore considered to be key barriers.

5. Selected Adaptation Solution

Of the five groups of adaptation actions identified above, four groups: water resources management, watershed protection, resilient infrastructure and climate risk knowledge and forecasting are directly relevant to Comoros water sector climate resilience and are considered fundamental to Comoros water security.

Water conservation is not considered to be such a priority, because water usage per capita is already at minimal levels (35 lpd) and as such water is highly valued and used carefully by communities.

Supply system leakage is however a problem in the reticulated water supply networks (typically 50%) and this is included in the project as part of the approach to reduce water demand during increasingly dry periods. This will reduce borehole pumping rates and hence reduce salinity risks, as well as increase durations that surface water scheme storage tanks can supply water when future storms necessitate intakes to be closed.

Sanitation improvements have not been included as most communities use on-site sanitation, with many households using VIPs and therefore not using freshwater or generating wastewaters. Sanitation improvements are not therefore considered to be a climate change adaptation measure for Comoros.

Drainage infrastructure improvements have not been included as the greater national priority is water supply (as articulated in the NAPA *et al*).

The following Adaptation Solution strategies are proposed to address the barriers preventing climate resilience:

- Long Term Planning, Budgeting and Investment in Climate Resilient Water Supply Infrastructure;
- Climate Informed Water Resources and Watershed Monitoring and Risk Forecasting; and
- Climate Resilient Water Supply Infrastructure Design, Operation and Protection.

Drinking water climate risk reduction will be integrated across all three strategies and will be the primary approach used to identify, predict, forecast and avoid, mitigate and withstand hydrological climate risks.

The first strategy is designed to create a paradigm shift in water governance by mainstreaming climate risk reduction as the priority objective for the sector. This requires appropriate amendments to national water sector policies, legislation

and plans to include mainstreaming of climate change risk reduction approaches into institutional responsibilities, their multi-year targets and goals, and commit the necessary financial resources to strengthening sector climate resilience, year on year.

The second strategy will require investigation and monitoring of the nation's water resources, understanding their vulnerability to climatic (hydrological) risks, protecting and rehabilitating watersheds to minimize the effects of (*i.e.* adapt to) increasingly variable rainfall on these small fragile water resources, and developing simple forecasting and early warning systems for drought and flood.

The third strategy requires island water supply agencies, community supply committees/operators and households to have the capacity to develop and implement preparedness plans to avoid or reduce the hydrological risks to the water supplies, through climate risk reduction preparedness planning in the design, construction, operation and maintenance of their water supplies and their contributing catchments, including responses to forecasts and early warnings, and use of multiple water sources.

For a Small Island Developing State like Comoros **it is essential the project commits to all three strategies** to ensure the programme intervention is sustainable in the long term. Increasing only the resilience of the water supply infrastructure providers (hardware and people) to climate risks is not a solution on its own, as this still relies on the availability of the fragile water resources to exist during climatic extremes so they can be abstracted, hence the essential need for water resources management to inform climate adaptation and supporting warning systems, and for adaptation centric watershed management to protect and augment the water resources.

Similarly upskilling the water service providers in climate risk reduction design & construction and O&M practices is not sustainable without government commitment to ensuring adequate CAPEX and OPEX budgets exist in perpetuity – *i.e.* optimizing the use of tariffs, based on careful review of affordability and willingness to pay, in a way that minimizes the need for future government budget support to the water infrastructure required to achieve climate resilience. Hence water resources, water infrastructure and water governance all require climate risk reduction in order for the other strategies to remain effective. Removing any one and the water sector will not become climate resilient.

6. Project Against Baseline Summary

The project is designed to address critical shortfalls in national capacity and knowledge which currently prevent climate risk reduction being integrated into national and community climate resilient water resources and water supply management, so as to overcome the country's vulnerability to climatic extremes due to its fragile water resources and lack of human and financial resources due to its small population and island isolation.

Specifically the project focuses on: creating a paradigm shift in water governance by mainstreaming climate risk reduction into water sector legislation, institutional arrangements, planning and budgeting; understanding and adapting to the climate risks to the nation's fragile water resources using watershed protection and rehabilitation coupled with hydrological monitoring and forecasting; and the integration of climate risk reduction into water supply scheme design, operation and management including multiple water source exploitation.

With the GCF project, approximately 450,000 people will have a more secure, more resilient and safer drinking water supply, capable of meeting longer drought periods, withstanding more intense storms and supporting food security as well as water security, with 800,000 people to benefit from national level interventions in improvements to climate resilient water governance and water resources protection.

C.3. Project / Programme Description

The key **goal of the project is to strengthen the climate resilience of drinking and irrigation water for 15 of the most vulnerable zones in the Union of Comoros to climate change risks.**

By addressing the critical barriers preventing climate resilience in the water sector – specifically the lack of climate risk reduction centric financial resources, the lack of stakeholder coordination and cooperation, the lack of knowledge and data and the lack of technical capacity (see the Theory of Change diagram) - the project develops, strengthens and

sustains a national level enabling environment that promotes, supports and allows island and national scale roll-out of climate risk reduction practices, as well as enables international knowledge transfer on this adaptation approach. It is this mainstreaming and integration of climate risk reduction into the 'business-as-usual' water sector planning, budgeting and operating environment that will create the paradigm shift needed to address climate change impacts and achieve a climate resilient water sector.



Fund-Level Impact 2.0: Increased resilience of health and well-being, and food and water security

Project Outcome: Ensuring Climate Resilient Water Supplies in the Comoros Islands

Increased water security at household and community level

Increased resilience of rural and peri-urban communities

Output 1: Climate Informed Water Supply Planning and Management

Assumptions: Knowledge transfer, legislation reforms, tariff reforms, guidance tools, planning and technical capacity building, will lead to permanent mainstreaming and integration of climate risk reduction into national and state water sector planning, programming, budgeting and practices.

Risks: Politicians and senior ministry stakeholders have other competing priorities; Political change at national and/or state level; Limited staff capacity to implement climate resilience reforms.

Output 2: Climate Informed Water Resources and Watershed Management Including Forecasting and Early Warnings of Climate Risks

Assumptions: Watershed partnership and changes to catchment activities will lead to permanent improvements in water resources availability and quality during climatic extremes; water resources monitoring will lead to effective early climate hazard warnings and forecasts to increase resilience of catchment water and land users.

Risks: Other sector plans not consistent with watershed resilience plans; Watershed communities not committed to effort required to strengthen capacity in adaptive management; Limited technical staff capacity to implement monitoring and forecasting.

Output 3: Climate Resilient Water Supply Infrastructure

Assumptions: Increasing access to groundwater on Grand Comore, reducing exposure to storm hazards on Anjouan and Moheli, and improving leakage reduction will lead to permanent greater access to potable water supplies during climatic extremes.

Risks: Infrastructure sites are in three islands, in mountainous terrain, limiting construction seasons; Extreme weather conditions and seismic hazards could occur during construction or are beyond design capacity; Lack of political & institutional commitment to water demand reduction strategies.

3.1. Undertake climate risk assessments of existing groundwater abstraction wells to develop risk reduction pumping strategies, and construction of additional boreholes in zones at risk of drought water scarcity in Grande Comore

3.2. Build infrastructure to increase resilience of water supply facilities to extended duration low flow periods, greater intensity flood flow damage and flood flow higher turbidity and bacteria loadings (Grande Comore, Anjouan island and Moheli island)

3.3. Installation of flowmeters to support climate resilient tariff adjustments, and leakage reduction programmes to improve the water pricing and management system taking into account the additional costs associated with climatic hazards

2.1. Establish climate resilience focused IWRM Committees and Watershed Risk Reduction Action Plans in the project intervention areas

2.2. Implement the water protection and risk mitigating measures on the ground/operationalize the risk reduction plans

2.3 Establish water source protection zones and raise public awareness on climate risk reduction benefits of watershed management

2.4. Establish water resource monitoring network and upgrade the existing monitoring infrastructure to enable the collection of the required climate/weather data

2.5 Build the capacities of the meteorological services to analyse and produce drought and flood forecasts for targeted users, including for flood early warning system

2.6 Build the capacity of the key government, local authorities and committees to interpret the climate information and raise awareness of the local population to act upon the forecasts and EWS.

1.6. Strengthen decentralized water resources management capacities to undertake climate risk reduction assessments and develop and deliver awareness campaigns and training programmes to Water Management Committees and users

1.5. Design and conduct trainings on best practices and gender-sensitive techniques of climate change adaptation in the context of water management, health and nutrition among national, regional and local water stakeholders

1.4. Develop planning guidance on source protection and water quality standards in view of climate change, operating procedures during periods of drought/floods; and safety plans

1.3. Develop and apply criteria for assessing socially sensitive water pricing mechanisms ensuring prices take into account the actual costs of production, storage and processing required in view of the projected climate stresses

1.2. Develop water sector climate change risk reduction awareness raising programme for national and state agencies and establish CCA knowledge and information exchange mechanisms

Barriers

Limited technical capacity to ensure quality and supply of water under climate change

Limited knowledge and capacity to fully assess climate change risks to water supply and quality

Lack of expertise, finance and data for generating appropriate legislation/policies in the water sector

Poor levels of service delivery and financial constraints hindering cost recovery and investment in the water sector

In order to deliver and sustain climate resilient water supplies it is necessary, especially in SIDS, to achieve three components. These form the outputs of the project:

- i) a national water planning approach which mainstreams climate resilience into its policies, plans, legislation, budgeting and institutional arrangements including both the regulators and the service providers, to ensure adequate financial and human resources are available year-on-year to sustain climate resilience (this is the objective of Output 1);
- ii) ensure adequate water resources are available during droughts and floods and actively manage the watersheds to both prevent climate induced derogation and where possible augment water resources protection (this is the objective of Output 2), including providing forecasts and warnings of water resources status to enable adaptive management; and
- iii) locate, design, build, operate and maintain water supply infrastructure to explicitly be resilient to climate change increased risks including droughts, floods, storm damage, storm surges, bushfires, power outages, and increased temperature driven water demands (this is the objective of Output 3).

Together these three Outputs deliver climate change resilient water supplies. Failure to address any one of these outputs will result in limited and unsustainable climate resilience.

The target zones are deliberately spread across all three islands to maximize the likelihood of post-project retention and replication in each autonomous state. The zones explicitly cover rainwater, groundwater and surface water reliant communities to ensure each water resource type is included in the project. Northern Grand Comore is the driest area and is included for rainwater harvesting diversification into groundwater. Anjouan and Moheli and southern Grand Comore receive the highest rainfall and are particularly vulnerable to flood damage. The target zones locations are shown in the Feasibility Study in Annex 8

The socio-economic analysis highlighted the current burdens of water supply such as the average time per day women have to search for water (2.5 hours per day) and the non-standardized rates that rural and peri-urban populations have to pay for water of questionable quality and unreliable quantity. Existing and projected water demands up to 2030 were also verified in each of the 15 target zones in the socio-economic analysis. Water demand related to population growth is explicitly excluded from the project. On Grand Comore, population growth requires an increase in the housing stock, which also provides the water catchments for the rainwater harvesting. Hence during normal climatic conditions, the population growth related water demand will be provided by additional rainwater harvesting not part of this project. However, once climatic change impacts increase such that rainwater harvesting fails then this additional population will require access to groundwater resources. This extreme dry season water supply 'backup' is supported by the project for the existing population and the future growth in population but only for the drought period groundwater augmentation.

On Anjouan and Moheli, the proposed upgrades to the water supply schemes to increase their resilience to flood damage and turbidity increases associated with the increased intensity and frequency of storm events due to climate change will not increase the water supply yields from the current schemes. The yields of the stream fed gravity schemes are more than sufficient to already meet the predicted population growth. Most schemes conveyance capacity is an order of magnitude greater than the current community demand estimate. Accordingly the climate resilience upgrades do not augment water supply for population growth but they do provide climate resilience for the existing population and future population growth in terms of minimizing storm damage, turbid water entry and increased treatment requirements to overcome increased turbidity levels.

The 15 target zones on the three islands, comprising 103 villages, have been chosen due to their vulnerability to climate change, their good hydrogeological and hydraulic potential for water storage and capture, limited donor support for water supply in the localities to date and potential collaboration planned with complimentary donor support. The target zones are:

Grande Comore Zones: 1) Bambao et Itsandra et peri-urban Moroni, 2) Ngongwe, 3) Hambou Djoumoipanga, 4) Mboikou, 5) Oichili, 6) Hamanvou;

Anjouan Zones: 7) Hassimpao, 8) Vouani, 9) Vassi, 10) Ankibani, 11) Chitrouni – Saadani, 12) Mjamaoué, 13) Nioumakélé-Bas;

Moheli Zones: 14) Fomboni-Djoiezi, 15) Hoani-Mbatsé.

Output 1: Climate Informed Water Supply Planning and Management

This output intends to improve multi-agency coordination and collaboration, as well as mainstream climate risk reduction into organizational planning, budgeting and programming by strengthening the national water policy and legislation frameworks to require financial and technical resources to be available to undertake climate change informed decision making at all levels on water resources and water supply.

Water is a cross-sectoral issue that is embedded in the mandate of various ministries including the Directions of Energy and water (DGEME), Environment (DGEF), the Ministries of Transport (Meteorology Department) and Health. Within DGEME is the National Water and Sanitation Directorate (DGEA). Because Comoros is a federal republic, each island is a state and each has its own Regional Water and Sanitation Directorate (DREA). The DGEA is responsible for designing and implementing national policy, regulation and sectoral planning framework. The new water code intends to establish autonomous island water utilities but currently only Water user Associations on the outer islands have been established. On Grand Comore, a public service provider MAMWE provides water supply to the wider Moroni capital area. Comoros therefore has a particularly complex and highly fragmented water sector institutional set up for such a small country, with senior regulatory management responsibilities disseminated across 4 organisations.

New Water Management Committees are currently being created within the communes as the new Water Code becomes progressively enacted. There are several different governance models used for water service providers in Comoros. The models to be used at each location have yet to be finalized by Government of Comoros and it is possible various different models may be used. For example, a local supply management organization model has been successful in managing water resources and in supporting payment collection mechanisms in rural districts (e.g., SOGEM in Moheli), whereas another option is the widening of MAMWE's (the capital water provider) remit to cover the other islands as a semi-autonomous water utility. The project will support improved coordination and the transversal integration of climate change risk reduction approaches into the existing framework of institutional reforms on decentralizing water management brought about by the newly revised Water Code. Climate-risk informed professional management of the water and sanitation supply systems is needed to ensure the long-term resilience of the climate-proofed infrastructure to be rehabilitated or installed.

In light of the above, this output will revisit the current on-going water tariff pricing reform process in order to ensure prices reflect actual production, storage and treatment costs required to deliver climate resilient infrastructure. Socially-sensitive billing rates for water tariffs will be determined through comprehensive affordability surveys and will be standardized so that they include locality-specific production and distribution costs required to address climate extremes, so as to supply sufficient, good quality water during and after dry periods and intense rainfall events. The responsibility, timing and regulatory process to review tariffs after the initial setting will also be defined as a part of this output.

The project recognises the need for re-sensitisation and awareness raising of climate risks with the national and state sector stakeholders (given the lack of inclusion of the issue in the Water Code) and includes activities to transfer and advocate existing global climate risk reduction practices for water supplies, and training of national and state regulators in planning for climate resilience in the water sector.

Furthermore, a Water Security Plan will be drafted and the Water Code will be updated in order to detail the requirements and responsibilities for climate risk reduction focused i) protection of sources, ii) operational procedures during dry / wet periods and iii) water quality standards. Best practices for addressing intense rainfall events, prolonged drought and saltwater intrusion will also be outlined.

In addition, drinking water management structures will be strengthened in rural and peri-urban areas through climate risk reduction training for the Water Management Committees (WMCs). Training will detail the existing and predicted impacts of climate change on water resources, required practices for water quality testing and monitoring, best protocols for climate resilient water infrastructure Operation and Management (O&M) and application of water tariff rates to reduce the identified climate risks to the water supplies within a risk based management framework known as Drinking Water Safety and Security Planning (DWSSP). To ensure sustainability of the water management system, user behavior change will be promoted with respect to protection and conservation of water services. Awareness-raising on climate change will be

provided on water laws, water quality issues and water production and supply costs relative to water tariff rates. Users will be informed about the link between water and climate change and the need for water conservation.

The Government of Comoros will provide co-financing for: supporting the development of criteria for assessing socially sensitive water pricing mechanisms, regulatory processes to review tariffs and training related to tariffs; and strengthening water resource management capacities for the WMC's and users

Output 1 activities are as follows:

Activity 1.1. Prepare recommendations and legal guidance on the integration of climate change adaptation into the national (federal) and regional (state) water sector agencies governance frameworks, regulations and operations

Activity 1.2 Develop water sector climate change risk reduction awareness raising programme for national and state agencies and establish CCA knowledge and information exchange mechanisms

Activity 1.3 Develop and apply criteria for assessing socially sensitive water pricing mechanisms ensuring prices take into account the actual costs of production, storage and processing required in view of the projected (GoC funded)

Activity 1.4 Develop planning guidance on source protection and water quality standards in view of climate change, operating procedures during periods of drought/floods; and safety plans

Activity 1.5 Design and conduct trainings on best practices and gender-sensitive techniques of climate change adaptation in the context of water management, health and nutrition among national, regional and local water stakeholders

Activity 1.6 Strengthen decentralized water resources management capacities to undertake climate risk reduction assessments and develop and deliver awareness campaigns and training programmes to Water Management Committees and users

Output 2: Climate Informed Water Resources and Watershed Management Including Forecasting and Early Warnings of Climate Risks

Output 2 intends to optimize the availability of the finite and fragile water resources in each watershed and use watershed based adaptation approaches to reduce the impacts of climate change on dry season aquifer recharge, stream low flows and water quality, storm flood run-off and soil erosion and storm water quality (thereby extending the climate resilience of Output 3 hard infrastructure solutions) through improved inter-sectoral coordination and capacity building in watershed adaptation approaches. This output will support an integrated approach to water management by creating Integrated Water Resources Management committees that will guide water abstraction, use and protection activities at the watershed scale to ensure resilience to climatic hydrological extremes. The revised Water Code has mandated the establishment of the IWRM committees to focus on ensuring groundwater and surface water recharge and watershed protection. The IWRM committees will coordinate with the DGEME (role of water supply) and the DGEF (role of water quality) at the national level, the DREA at the regional level and will work with the communes at the local level. Best practices will be used from a recent IWRM (GIRE) project that used a national IWRM Plan and national steering committee as an entry point to promote the IWRM watershed committees to focus initially on the priorities of drought and flood resilience planning and adaptation, as part of broader watershed management planning.

The IWRM committees will be responsible for generating watershed specific Climate Risk Reduction Action Plans based on initial Climate Risk Vulnerability Assessments (CRVAs) to be conducted. The CRVAs will provide a hazard mapping inventory of areas exposed to climate risks. The IWRM Climate Risk Reduction Action Plans will detail the most appropriate adaptation measures to ensure the conservation, protection and sustainability of water resources. In collaboration with UNEP-GEF's watershed project the development of these plans will be community-focused, gender-sensitive and elaborated in a participatory manner. These plans will detail feasible watershed management and restoration climate adaptation activities including zoning of source water areas and delineation of buffer zones, re-vegetation in riparian zones and reforestation of river basins, improving farming practices and other land use practices which de-stabilize soils eg road construction and drainage, so that water resources become more resilient to climate

change through improved dry season water retention, flood flow attenuation, soil stabilization and water quality clarification and oxidation. The zoning areas will be established by decree under the Water Code.

Furthermore, output 2 will also address the lack of knowledge and data on the climate vulnerability of the water resources of the country and to develop the technical capacity to both forecast climatic risks and create sector specific hydrological products to inform and upskill the adaptation capacity of water users – be they water supply, agricultural or other sectors. The collection, analysis, interpretation and dissemination of water resources data for all aspects of the water cycle (rainwater, surface water, groundwater, evapo-transpiration) allows the impacts on the water resources due to climate change to be observed and predicted. In Comoros, where rainwater harvesting is already restricted, much groundwater is almost brackish, and surface water already highly variable in flows and turbid in quality, then small further increases in rainfall shortages or rainfall intensities can have very significant consequences in terms of water resource availability during periods of climatic extremes. In this situation, effective water resource monitoring enables agencies/ organizations to develop effective adaptation strategies.

This output will contribute to improving climatic extreme water resource data management through the procurement and installation of 10 hydrological (surface water) gauging stations in Anjouan and Moheli and the procurement and installation of 30 groundwater piezometers on Grande Comore and 13 across Anjouan and Moheli. All water related monitoring stations will be coupled with the existing synoptic stations (4), automatic weather stations (10) or with the 90 rain gauges already installed. Technicians will be trained on monitoring system operation and management. Data will be captured, processed, stored and analyzed, as well as distributed and shared, by the Directorate of Meteorology and not the watershed basin committees. This data will contribute directly and immediately to final engineering upgrade designs to be delivered in Output 3 – such as optimal locations and pumping rates of new production boreholes.

This output will also focus on developing simple but customized water-related forecasts for targeted user agencies (including water service providers, farmer associations) and local community end-users, by staff in key government departments and local authorities who will have been trained in interpreting water-related climate and weather forecasts to support sector specific decision-making tools, eg agricultural drought warnings and response/preparedness advice s.

Furthermore, a simple flood alert early warning system using total rainfall and rainfall intensity from selected automated rainfall gauges and telemetry links to the volcanic observatory (which uses a similar real time seismic monitoring system) will be developed to provide a flood risk warning system for each island. This will be connected to the mobile phone system and provide SMS flood warning messages. The technical capacities of staff in the Directorate of Meteorology will be improved in-house using near-by available trainings (e.g., WMO Hydrology and Water Resources Programme (HWRP), the WMO Regional Training Centres in Madagascar and Kenya, WMO distance learning courses, RSMC Réunion,) and by exploring existing university knowledge. The University of Comoros is working with the MAMWE to monitor groundwater salinity levels in pumping wells brought about by saltwater intrusion. The national University hosts the national water quality laboratory and represents the main groundwater knowledge base in the country. This output will also support mainstreaming water resources and water supply climate risk reduction into relevant existing curriculums of existing graduate level classes and programmes within the university.

The Government of Comoros will provide co-financing for the following: supporting the establishment and formalization of climate risk centric IWRM committees; Establish water resource monitoring network and upgrade the existing monitoring infrastructure; facilitate building capacity of key stakeholders to interpret the climate information and raise awareness of the local population to act upon the forecasts and EWS

Output 2 activities are:

Activity 2.1. Establish climate resilience focused IWRM Committees and Watershed Climate Risk Reduction Action Plans in the project intervention areas

Activity 2.2 Implement water protection and risk mitigating measures (eg re-vegetation, reforestation, best practice low-erosion farming) on the ground/operationalize the risk reduction plans

Activity 2.3 Establish water source protection and buffer zones and raise public awareness on climate risk reduction benefits of watershed management

Activity 2.4 Establish water resource monitoring network and upgrade the existing monitoring infrastructure to enable the collection of the required climate/weather data

Activity 2.5 Build the capacities of the meteorological services to analyze and produce drought and flood forecasts for targeted users, including for flood early warning system

Activity 2.6 Build the capacity of the key government, local authorities and committees to interpret the climate information and raise awareness of the local population to act upon the forecasts and EWS.

Output 3: Climate Resilient Water Supply Infrastructure

Output 3 aims to deliver a programme of risk-prioritized evidence-based climate risk reduction infrastructure upgrades to the existing water supplies on each island - an approach known as Drinking Water Safety and Security Planning (DWSSP) and promoted by UNICEF - as the design process. This will deliberately include a range of groundwater and surface water fed potable water supply schemes and rainfed and surface water fed irrigation supply schemes, to ensure maximum relevance nationally both on sustaining these schemes against climate change and later replication throughout the country. These infrastructure upgrades address drought vulnerability and water shortages for rainwater reliant communities on Grand Comore and storm vulnerability and storm damage and high turbidity risks to surface water intakes on Anjouan and Moheli.

In specific, this output includes hard infrastructure measures to scale-up and improve water abstraction and storage during climatic extremes. The GCF project will focus on 1) increasing access and exploitation of groundwater when rainwater harvesting fails during extended dry periods, , 2) climate-proofing water sources and infrastructure (elevating pump stations, using protective covers for reservoirs), against flood risks, 3) treating and filtering water in a cost-effective manner to account for climate change water quality deterioration (especially in BOD, turbidity) and 4) expanding agricultural water storage mechanisms for dry periods and higher crop demand associated with increasing high temperatures. Taking into account the physical characteristics of watersheds and the types of risks they are exposed to, this output will upgrade surface water and groundwater abstraction systems and develop and / or rehabilitate rainwater irrigation systems. Rather than opt for a single type of solution which will increase susceptibility to climate change impacts, the output will enable each island to have more diversified sources of water supply. Details on the activities in each target zone are provided below.

Output 3 specifically focuses on increasing the reliability of water supply during future climate change increased extreme rainfall variability events (*i.e.* droughts and storm floods) However, this output will not aim to provide additional water supply to serve the increasing demands of the growing population. In Grand Comore, communities will meet their normal water demand through their own household and community rainwater harvesting – which is not funded by the project. Only when this rainwater harvesting fails, due to climate change induced impacts and extreme events, will these communities need to access the groundwater scheme upgrades being provided by the project.

The Arab Fund for Social and Economic Development (FADES) will provide co-finance for the safe borehole salinity yield assessments in Zone I of Grand Comore where the largest and most productive boreholes are known to exist. The Chinese Government through one of its international construction companies will fund extension of the groundwater distribution network to 18 villages in Zone 1 of Grand Comore and will co-fund borehole pumps and treatment for 19 villages in Zone 2 of Grand Comore.

On Anjouan and Moheli, the existing surface water schemes have sufficient excess capacity to already meet the predicted population growth – most schemes are small and the pipework and intakes can transmit an order of magnitude more water than the current demand requires. This output will fund the upgrades that will protect these schemes during storm flood events and minimize the entry and use of storm waters into the systems – including intake closure systems, storm flow diverters and short term storage tanks. This output will also fund for the increased treatment requirements for the potable water supply schemes necessary to overcome the climate change increases in turbidity due to higher rainfall intensity.

Furthermore, as stated above, most of the surface water schemes transmit greater flows than are needed for water supply. Therefore, this output will also support agricultural storage infrastructure to hold some of these flows to secure food production during the dry season. On Grand Comore a similar approach will use rainwater schemes. These irrigation storage schemes will only support existing agricultural production and will not support new or proposed expansions of farming areas. All schemes on Anjouan, Moheli and Grand Comore will be funded through GCF funds.

Studies have indicated that natural water resources are sufficient to ensure a continuous water supply if managed sustainably. Water balance estimates demonstrate that Grand Comore has a renewable water volume of 1.2 billion m³ per year, of which 1.1 billion m³ are from groundwater.³⁶ Current exploitation of groundwater and of surface water and RWH is 0.5% and 1.9% of the total renewable volume.³⁷

Across the 6 target zones in Grand Comore, the water demand is estimated to be 18,235 m³/d. When rainwater harvesting fails due to future climate change, this entire demand will need to be met for up to consecutive 44 days from the groundwater supplies. The potable groundwater boreholes on the island can only provide an estimated 11,088 m³/d, even if they could get this water to the rural communities, which they currently cannot. The shortfall of 7,147 m³/d (82 l/s) i.e. 40% of the normal water demand, will be able to be supplied by 5 new boreholes – 1 in each of 5 target zones. But to meet the total demand, this also requires existing groundwater schemes with spare pumping capacity to be have their distribution systems and pumping stations expanded into the rural upland areas.

In Anjouan and Moheli, the climate change risks do not relate to water quantity but to water quality derogation, both during the storm events themselves and then the increased turbidity due to storm damage to the catchments. Here the project approach is to provide sufficient storage tanks as upgrades to the schemes to enable the intakes to be closed during the storm periods, and then increase treatment capacity of the schemes to deal with the increased turbidity of the raw water after the storm has passed. Great amounts of fresh groundwater resources are also thought to exist in Anjouan and Moheli on perched aquifers for instance. Due to a lack of groundwater studies, this source has never been exploited.³⁸ Groundwater resources analysis will be undertaken in Output 2 in advance of and to inform locations for new groundwater abstraction boreholes on Grand Comore.

Output 3 activities are:

Activity 3.1. Undertake climate risk assessments of existing groundwater abstraction wells to develop risk reduction pumping strategies, and construction of additional boreholes in zones at risk of drought water scarcity in Grande Comore

Activity 3.2. Build infrastructure to increase resilience of water supply facilities to extended duration low flow periods, greater intensity flood flow damage and flood flow higher turbidity and bacteria loadings (Grande Comore, Anjouan island and Moheli island)

Activity 3.3. Installation of flowmeters to support climate resilient tariff adjustments, and leakage reduction programmes to improve the water pricing and management system taking into account the additional costs associated with climatic hazards This activity is fully funded by the Government of Comoros.

Activities include conducting simple rapid (1 day long) groundwater saline up-coming capacity tests on existing production wells to optimize existing pumping regimes (altering pumping rates and pump depths) to reduce the risks of saltwater intrusion during drought periods when recharge is low and groundwater demand and therefore pumping rates increase

³⁶ African Development Bank, Strategy and Programme of PEAPA in the Comoros, Annex 1: Hydrologic and water resources balance

³⁷ African Development Bank, Strategy and Programme of PEAPA in the Comoros, Annex 1: Hydrologic and water resources balance, Annex 2: Hydrogeologic context, Annex 3: Socio-economic context, Annex 5: Water resources monitoring

³⁸ DGEF, UNDP, GEF April 2012. Report on the Vulnerability and the Adaptation of Water Resources at Risk to Climate Change Impacts on Anjouan.

due to rainwater harvesting failure. FADES will fund these tests in Zone 1 of Comoros. The other zones will be funded by GCF.

In areas on Grand Comore where roof surface areas are minimal and where boreholes can be constructed in shallower, productive groundwater supply areas, boreholes will be upgraded and new boreholes constructed where groundwater investigations (see Output 2) confirm fresh groundwater to exist.

Expansion of existing coastal groundwater fed water supply networks into the high interior rural villages is an appropriate adaptation strategy in Grand Comore, into those areas where rainwater harvesting is inadequate (and cannot be expanded) and where depth to groundwater prevents direct exploitation of the underlying aquifers. The Chinese Government is funding the expansion of the groundwater distribution system to 18 villages in Zone 1 and co-funding the system to 19 villages in Zone 2. The other zones will be funded by GCF.

In contrast, on Anjouan and Moheli, climate-proof river intakes will be constructed to ensure they are not damaged by flood events. This will be fully funded by GCF. Rainfall catchment water balances undertaken as part of the Feasibility Study (see Feasibility Study Section 4.4.3) suggest dry season flows will be more than sufficient to meet public water supply requirements (including for population growth), however additional surface water monitoring (see Output 2) will be installed to confirm dry season flows do exist in each basin prior to intake upgrading.

All storage tanks attached to surface water and groundwater distribution systems will be equipped with the appropriate water disinfection and filtration systems – to address the increased turbidity and therefore water quality derogation expected due to climate change. The tanks will also provide greater flexibility in supply scheme operation to enable the intakes to be shut down during storm events, and will provide a few days of storage during the dry season.

Reservoirs, impluviums and water troughs will also be constructed for non-potable uses including agriculture. Impluviums, or natural crater collecting rainwater, will be used for agriculture due to their tradition in the Comorian culture. The impluviums are located above the villages enabling gravity feed pipelines to be constructed. This will increase flood security during the dry seasons and reduce use of precious rainwater necessary for drinking water in the upland households of Grand Comore. These will be fully funded by GCF.

Finally, water meters will be installed to both promote water demand management (allowing service providers to identify areas of water leakage and/or theft in each supply zone) of water supply schemes thus minimizing groundwater and stream abstraction rates during climatic extremes and reducing saline up-coning and water shortage risks, but also demonstrate water delivery and therefore inform the tariffs required to sustain climate risk reduction operation and maintenance levels. This entire activity will be fully funded by the Government of Comoros. These hard infrastructure measures will be accompanied by a capacity building and strengthening of institutions in charge of managing the water provision infrastructure (see Output 1) to use the data to manage water leakage and reduce lost water, as well as how to use the data to inform tariff structures.

Table. Proposed Water Supply Infrastructure per Target Zone

Zone	Existing Borehole Protection	New Borehole Drilling	Protecting Stream Intakes	Expanding Pipeline Network	Protecting Water Treatment	Potable Water Storage	Irrigation & Livestock Storage
1 Moroni	3 Bores	1	n/a	Rural	Yes	10 tanks	No
2 Ngongwe	1 Bore	0	n/a	Rural	Yes	1 tank	No
3 Hambou	n/a	2	n/a	Rural	Yes	2 tanks	No
4 Mboikou	1 Bore	1	n/a	Rural	Yes	2 tanks	No
5 Oichili	1 Bore	1	n/a	Rural	Yes	2 tanks	No
6 Hamanvou	n/a	n/a	n/a	Rural from impluvium	No	4 crater impluvium	No

7 Hassimpao	n/a	n/a	1	Rural	Yes	2 tanks	16 basins & 8 troughs
8 Vouani	n/a	n/a	1	Irrigation only	Yes	2 tanks	20 basins & 12 troughs
9 Vassi	n/a	n/a	1	Irrigation only	Yes	1 tank	12 basins & 6 troughs
10 Ankibani	n/a	n/a	1	Rural	Yes	4 tanks	30 basins & 24 troughs
11 Chitrouni	n/a	n/a	1	Rural	Yes	1 tank	12 basins & 6 troughs
12 Mjamaoue	n/a	n/a	1	Rural	Yes	1 tank	12 basins & 6 troughs
13 Niomakele	n/a	n/a	3	Rural	Yes	3 tanks	44 basins & 35 troughs
14 Fomboni	n/a	n/a	1	Rural	Yes	12 basins	6 troughs
15 Hoani	n/a	n/a	1	Rural	Yes	16 basins	6 troughs

C.4. Background Information on Project / Programme Sponsor (Executing Entity)

Operating Experience

The Ministry of Energy, Agriculture, Fisheries, Environment, Country Planning and Urbanism (MEAPEATU) is the sponsor (or activity coordination unit) for the Proposed Project. MEAPEATU will be the main focal point for the GCF project and will coordinate with island coordinators during project implementation.

The Vice-Presidency in charge of the Ministry of Environment has the primary responsibility for environmental management. The organizational framework includes an Administrative and Financial Branch supported by six departments: (i) Planning, Monitoring and Evaluation, (ii) Public Relations, (iii) Training and Development, (iv) Communication, (v) Legal Support and Cooperation, and geographic information system (GIS).

The Vice-Presidency is structured into six national directorates and a research institute: Environment and Forestry, Fisheries Resources, Agricultural Strategies, Energy, Water and Sanitation, Industry, Crafts and the National Research Institute for Agriculture, Fisheries and the environment (INRAPE). The department has 98 positions across all branches.

The primary function of the MEAPEATU is to protect the environment, in particular: regulation, control, education, public awareness, sustainable conservation and management of natural resources, including water resources, in the Comoros, and management of protected areas and other areas of environmental interest. The MEAPEATU also has dual responsibilities for managing and implementing the national environmental policy and the national climate change agenda.

The Vice-Presidency in charge of Ministry of Energy, Agriculture, Fisheries, Environment, Country Planning and Urbanism (MEAPEATU) as the national executing entity - also referred to as the national 'Implementing Partner' in UNDP terminology - will implement the project in compliance with UNDP rules and regulations, policies and procedures, including the NIM Guidelines. These include relevant requirements on fiduciary, procurement, environmental and social safeguards, and other performance standards. In legal terms, this is ensured through the national government's signature of the Special Agreement concerning Technical Assistance between UN organizations and the Government of Comoros (signed by both Parties on 27 January 1976) and the Agreement between UN Special Fund and the GOCI concerning Assistance from the Special Fund (signed on 27 January 1976/20 October 1959), together with a UNDP project document which will be signed by the Implementing Partner to govern the use of the fund

MEAPEATU has significant operating experience in Comoros on climate change-related project coordination through its long experience with implementing LDCF, EU and AFD projects (e.g., ACCE, PACEAU). Currently, the Ministry of Environment is the focal point of many multilateral environmental agreements and works in close collaboration with all government institutions at both the federal and island levels.

Financial Status and Management

The Comoros Country Office (CO) of the United Nations Development Programme (UNDP) will oversee and monitor the implementation of the project and will provide a financial oversight and quality assurance role. The Comoros UNDP CO will engage with all three islands to ensure GCF resources are allocated on an equitable basis.

MEAPEATU will provide the UNDP CO with certified periodic financial statements, and with an annual audit of the financial statements relating to the status of GCF funds according to any established procedures. The Audit will be conducted annually by the legally recognized auditor of the MEAPEATU or by an independent commercial auditor engaged by the GCF. The periodic financial statements will cover information in relation to the whole funded activity (as required by clause 17.02 of the AMA) rather than just that relating to the status of GCF funds.

C.5. Market Overview (if applicable)

Comoros is one of the poorest countries in the world, with a gross national income per capita of US\$ 840 and an annual GDP growth of 3.5% in 2013, dropping to 1% in 2015.³⁹ Furthermore, the Comoros has high levels of poverty (42.4%) and a chronic economic deficit, and is considered a highly indebted poor country.⁴⁰

The Comorian economy is largely an inward-looking and poorly diversified subsistence economy, with 34.5% of GDP generated in the agriculture sector, 54.4% in the tertiary sector, and the secondary sector accounting for 12% (2014).

The budget of Comoros has been in significant deficit in recent years and will continue to be so at least until 2020 (last year of IMF projections). The budget deficit was 14.8% in 2016 and is expected to increase to 15.6% in 2017, settling at 11.5-12% in 2018-20. The economic slowdown in 2015/16 and related decrease in tax revenues accentuated deficits and lead to large arrears in the payment of public sector wages.

Populations pay by volume in the Moroni region whereas in Anjouan and Mohéli, the *Union des Comités de d'eau à Anjouan/Mohéli* charges a fixed tariff at 1,000 KMF per month for water that is not potable.⁴¹ This was a similar rate set by AFD's RESEAU pilot projects in the villages of Ongoni and Mjimandra. This project implemented individual connections but the consumption was limited to 10 m³/household/month (i.e. approximately 35 lpd at US\$ 0.2/m³).

The average household water bill in Moroni is around US\$4-5 per month, or approx. 8% of the average wage – a high percentage, especially for lower-income households.

In the dry season rural villagers procure water from the town water supplies. A container of 20 liters of non-potable water (not sterilized) costs 250 to 500 KMF (US\$ 25-50/m³) in Grande Comore against 25 KMF (US\$ 2.5/m³) in Anjouan.

C.6. Regulation, Taxation and Insurance (if applicable)

The proposed project will be governed by the Ministry of Energy, Agriculture, Fisheries, Environment, Country Planning and Urbanism (MEAPEATU) Guidelines for Integrated Water Resource Management Projects. The guidelines set the design standards for such investments, and the minimum water supply targets that such systems must achieve. The proposed project will fully align with MEAPEATU guidelines as well as offer greater efficiency and optimization of the multiple sources of water so that it adheres to the MEAPEATU requirements that such system "shall be designed in a manner to provide sustained potable water to the consumers within the supply area on a continuous basis". The Guidelines set specific standards and requirements for conveyance system, water harvesting, storage standards, water transmission, desalinization, power supply, water treatment, disinfection, and groundwater recharge.

The MEAPEATU has the power to control and monitor water resources and to issue authorizations for exploitation and mobilization. It has the power and the duty to control the conditions of exploitation and the conformity of this one with the stipulations of the act of authorization on all the plans. Authorization is required for all cases of mobilization and

³⁹ <http://www.worldbank.org/en/country/comoros/overview>

⁴⁰ National Progress Report on MGGs (2012)

⁴¹ Rapport sur la vulnérabilité et l'adaptation sur les ressources en eau face aux changements climatiques sur l'île d'Anjouan

exploitation of water. It is issued for a maximum of five years, renewable once. The refusal of the authorization must be motivated and communicated to the person concerned within a period not exceeding three months from the date of receipt of the duly constituted file. The following are subject to the permit system :

- Temporary occupation of the public hydraulic domain (DPH) ;
- The construction of wells or boreholes for the abstraction of groundwater and the construction of any other water catchment works ;
- water intakes established on rivers and reservoirs of low surface water capacity (lakes or dams) ;
- The establishment in the DPH of all kinds of facilities, access or exits on the dykes and freeboards ;
- the construction, reconstruction or repair of works that are established between the fringes of rivers, lakes, pipelines

To obtain the permit, the project activities need to comply with site specific EIA requirements for the installation of such systems. The project builds on the ESIs conducted for prior projects (ACCE). For the purpose of this specific investment ESIs will be conducted in all target islands as required by MEAPEATU regulations. Additionally, the project will follow the government's tariff setting policies. The request for permits for the project activities should attach:

- The SESP and any related assessments and management plans.
- Detailed terms of reference for key project positions and the project board.
- Standard agreements such as the project cooperation agreement when a civil society. organization is implementing and cost-sharing agreements.
- Standard letter of agreement between the implementing partner and responsible parties for the implementation of the project.
- Procurement plan.
- Project monitoring schedule plan.
- Resource mobilization and communications plan.
- List of foreseeable knowledge products, if relevant.

Additionally, UNDP's Programme and Operations Policies and Procedures which apply to the Project are publicly available here: <https://info.undp.org/global/popp/Pages/default.aspx> Furthermore, Section 7 of the Convention on the Privileges and Immunities of the United Nations provides, inter alia, that the United Nations, including its subsidiary organs, is exempt from all direct taxes, except charges for utilities services, and is exempt from customs duties and charges of a similar nature in respect of articles imported or exported for its official use. Also, the goods and services procured directly by GoC implementing partners, are exempt from customs duties and will be tax free. All the project procurement/inputs will be tax free. There are no specific insurance policies relevant for the project activities.

C.7. Institutional / Implementation Arrangements

The GCF Project's National Implementation Management (NIM) arrangements will be consistent with arrangements successfully implemented via other adaptation projects in Comoros supported by UNDP.

The project will be implemented following UNDP's National Implementation Modality (NIM), according to the Special Agreement concerning Technical Assistance between UN organizations and the Government of Comoros (signed by both Parties on 27 January 1976) and the Agreement between UN Special Fund and the GOC concerning Assistance from the Special Fund (signed on 27 January 1976), and according to policies and procedures outlined in the UNDP POPP (see: <https://popp.undp.org/SitePages/POPPSubject.aspx?SBJID=245&Menu=BusinessUnit>).

The national executing entity - also referred to as the national 'Implementing Partner' in UNDP terminology - is required to implement the project in compliance with UNDP rules and regulations, policies and procedures, including the NIM Guidelines. These include relevant requirements on fiduciary, procurement, environmental and social safeguards, and other performance standards. In legal terms, this is ensured through the national government's signature of the Special Agreement concerning Technical Assistance between UN organizations and the Government of Comoros and the Agreement between UN Special Fund and the Government of Comoros concerning Assistance from the Special Fund together with a UNDP project document which will be signed by the Implementing Partner to govern the use of the funds.

The Implementing Partner (IP) for this project will be the Ministry of Energy, Agriculture, Fisheries, Environment, Country Planning and Urbanism (MEAPEATU), which is accountable to UNDP for managing the project, including the monitoring and evaluation of project interventions, achieving project outcomes, and for the effective use of UNDP resources. The MEAPEATU will have project ownership and will appoint a Project Manager (PM), paid for by the project, to coordinate project operations. The MEAPEATU is also the National Designated Authority of the Green Climate Fund and all the national level coordination mechanisms will be under the aegis of the ministry.

The DGEF (General Directorate for Environment and Forests) and the DGEME (General Directorate for Energy, Mines and Water) will be the main technical departments within the MEAPEATU in charge of the implementation of the project.

The DGEME is the main state institution responsible for the sovereign missions in the water sector namely:

- Collect, create, maintain and manage a sectoral database for energy, water, mining and sanitation;
- Control and order the rational exploitation of energy resources in water and mineral, all over the territory;
- Implement programs and activities aimed at the optimal implementation of these same resources, their integrated management as well as good governance within the framework of the strategy defined by the Government;
- Develop, in conjunction with the General Planning Commissariat (CGP), the Public Investment Program (PIP) and the Public Expenditure Program
- To co-operate with other administrations and institutions, including the Directorates-General of Agriculture and Rural Development, Environment, Health, Economy, Customs, Comorian Society, hydrocarbons and the national electricity and water company, as well as private professional operators and NGOs AND development association, a dashboard on resources and their use
- Provide technical oversight of structures involved in these sectors, including community and regional organizations, involved in the development, integrated management, sustainable development, protection and exploitation of resources in these sectors
- Assure, on behalf of the Government, the project management of sectoral projects within the remit and competencies of the Ministry of Energy, Water, Sanitation and Mineral Resources.
- Develop and conduct comprehensive and partial technical and financial studies as part of the sectoral development program
- Define areas and different areas of protection of resources
- Collaborate with national, regional and international, bilateral, multilateral and non-governmental organizations for the coordination and orientation of activities, in accordance with the national sectoral development plan, in accordance with the various international conventions signed by the country

The DGEME will be responsible in the framework of the implementation of the project of the following tasks

Will serve as an executing Partner for the implementation of some of demonstration projects

- Will be a member of the PSC.
- Will be responsible for reviewing existing policies to ensure the incorporation of climate change considerations.
- Will facilitate the sharing of lessons and experiences at a national level as resources permit.
- Will serve as a resource institution during the project or policy-related issues.
- Will serve as a resource institution for water resources monitoring, and managements.
- Will contribute in the M&E.

The DGEF is responsible among other for the following water resources management related tasks:

- Preserving the quality of water resources, according to the uses
- Increasing the availability of water resources while avoiding wastage
- The prior authorization of the ministers responsible for the management of water resources and the environment, on all works relating to rivers, their beds and banks, likely to modify the water regime, their chemical composition, their temperature or their fishing suitability or to compromise, in general, the ecological balance
- Definition of bacteriological, biological, physical and chemical standards for drinking water and water for irrigation and watering of cropland

- The establishment of protective perimeters (immediate, close and remote protection) on the water sampling points, intended for human consumption, according to the needs noted for each individual case
- Prohibition of all activities likely to affect water quality
- The prior authorization of the Minister of the Environment in relation to the operation of agricultural or industrial enterprises whose wastewater is discharged into watercourses and devices to be taken for the purification of wastewater.
- The classification of forests by decree in Council of Ministers, on the joint proposal of the Minister for Rural Development and the Environment when soil protection, watercourses regime, conservation of a threatened species of fauna or the ecological balance require it

In this framework , the DGEF will be responsible in the project implementation of the following tasks:

- Will serve as an Executing Partner and will, therefore, be responsible for executing the project.
- Will chair the Project Steering Committee (PSC).
- Will delegate implementation responsibilities to other national directorates such as, Directorate of Water and Energy, associations, unions, NGO's and others, which are still to be identified.
- A National Project Director will be appointed from within the Directorate.
- Will house the Project Support Team (PST: project manager and project support).
- Will implement project activities through its extension network.
- Responsible for M&E.

Subsidiary agreements will be signed between UNDP, the Government of Comoros and Ma-MWE, and ANACM (National Agency for Civil Aviation) respectively.

The MA-MWE is a separate entity with a legal personality but under the technical supervision of the MEAPEATU.

The MA-MWE will be carrying out the following activities:

3.1. Undertake climate risk assessments of existing groundwater abstraction wells to develop risk reduction pumping strategies, and construction of additional boreholes in zones at risk of drought water scarcity in Grande Comore

3.2. Build infrastructure to increase resilience of water supply facilities to extended duration low flow periods, greater intensity flood flow damage and flood flow higher turbidity and bacteria loadings (Grande Comore, Anjouan island and Moheli island)

3.3. Installation of flowmeters to support climate resilient tariff adjustments, and leakage reduction programmes to improve the water pricing and management system taking into account the additional costs associated with climatic hazards

ANACM is also a separate entity with a legal personality but under the technical supervision of the Ministry of Transport, Tourism, Poste and Telecommunication and Information and Communication Technologies. ANACM will carry out the following activities:

2.4 Establish water resource monitoring network and upgrade the existing monitoring infrastructure to enable the collection of the required climate/weather data

2.5. Build the capacities of the meteorological services to analyze and produce drought and flood forecasts for targeted users, including for flood early warning system

The UNDP country office provides support services for the implementation of project activities, including assistance in reporting and direct payment. In doing so, it must ensure that the capacity of the Government (the Implementing Partner) is strengthened.

In addition, the UNDP country office may provide, at the request of the Implementing Partner, the following support services for the implementation of the project activities:

- (a) Identification and / or recruitment of staff to be assigned to the project;
- (b) Management of service contracts and institutional / professional contracts;
- (c) Management of fixed-term contracts, temporary contracts;
- (d) The management of specialized expertise contracts;
- (e) Definition and facilitation of training activities;
- (f) Complex procurement processes including civil works;
- (g) Acquisitions of specialized equipment to be ordered using open international tenders;
- (h) Local acquisitions exceeding the ceiling of USD50,000.00;
- (i) International Travel Management;
- (j) Financial and administrative management of the project;
- (k) General Service Management;
- (l) Management of information technology and communications services

Contracts and flow of funds

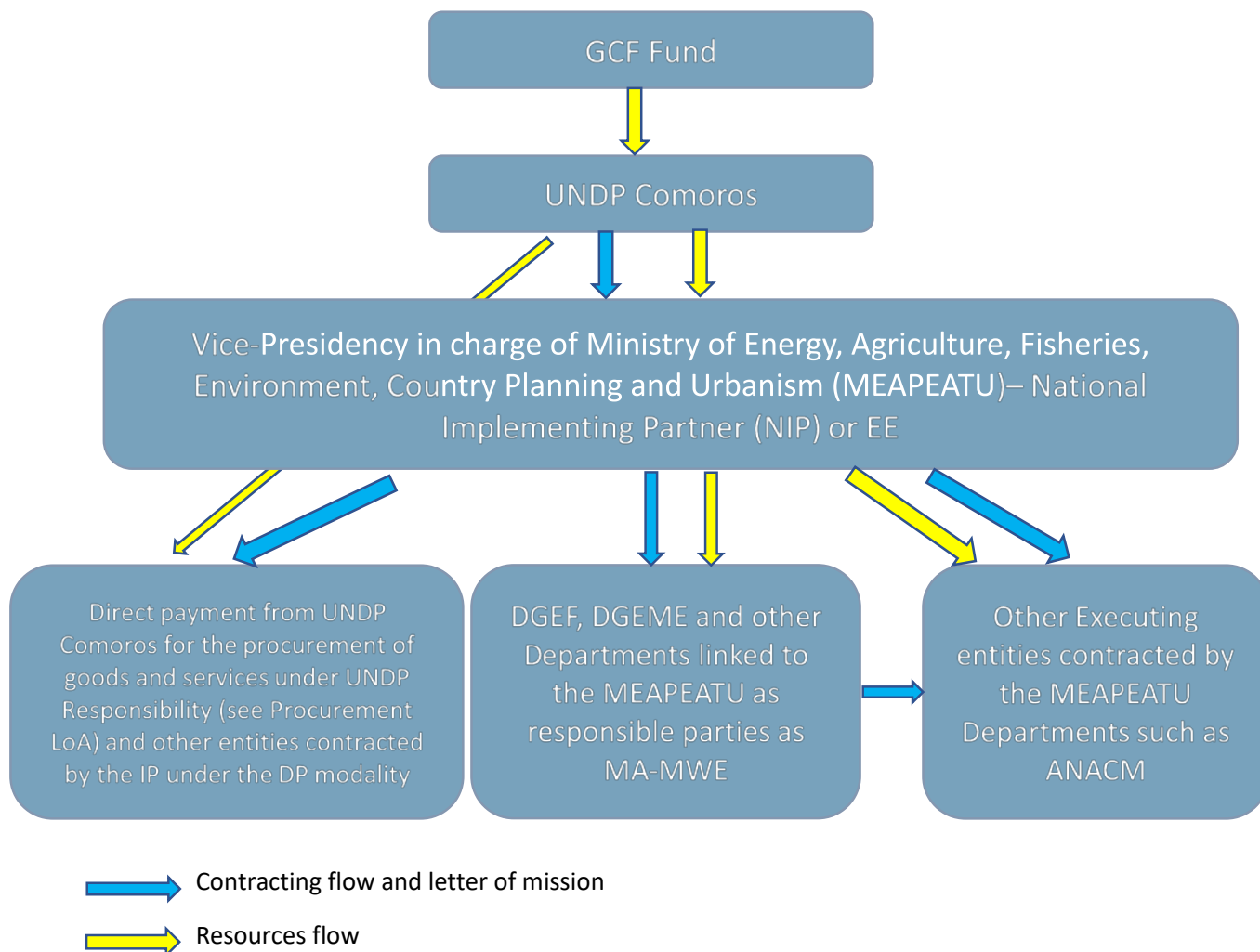
Under the National Implementation Modality, guided by the Special Agreement concerning Technical Assistance between UN organizations and the GOC (signed by both Parties on 27 January 1976) and the Agreement between UN Special Fund and the GOI concerning Assistance from the Special Fund (signed on 27 January 1976), and in terms of the Project Document to be signed between UNDP and GOC (Subsidiary Agreement under the FAA), UNDP will advance cash funds on a quarterly basis to the MEAPEATU as EE. The major agreements to be entered into are those between the MEAPEATU and:

- the General Direction for energy, Mines and Water (DGEME)
- the General Direction of Environment and Forest (DGEF)
- the Water and Electricity Enterprise of Comores (MA-MWE)
- the National Agency for Civil Aviation and Meteorology (ANACM)

As the AE, UNDP will disburse funding (received from the GCF according to the FAA disbursement schedule), to the MEAPEATU, as the Executing Entity, for the purposes of undertaking the project. The MEAPEATU will conclude agreements with the DGEME, DGEF, MA-MWE and ANACM and these Letters of Agreement (LOAs) will be attached to the Project Document. In terms of these LOAs, these 3 Divisions of the MEAPEATU will be made Responsible Parties (RPs), responsible for delivering particular projects outputs as set out in detailed terms of reference attached to the agreements.

The main beneficiaries of this project will be the MEAPEATU (and its 2 key Divisions: DGEF, DGEME) the Water and Electricity of Comoros Agency (MA-MWE), the local Water Committees of Anjouan and Moheli (UCEA) and (UCEM), Electricity of Anjouan (EDA), the Direction of Meteorology as well as the local Water User Associations and CBOs

Contracting structure and flow of funds



The Project Steering Committee, led by MEAPEATU, will be responsible for approving program activities. Based on the approved activities, the Project Management Unit (PMU) will ensure the provision of funds to all institutions/organizations for their respective activities. All executing agencies will be responsible for managing tasks allocated to their institution/organization.

The UNDP Comoros Programme Officer will be responsible for **Project Assurance**. In addition, the Government of Comoros has requested UNDP to provide direct project services for this project. The UNDP and Government of Comoros acknowledge and agree that those services are not mandatory and will be provided only upon Government request and specified in the Letter of Agreement. The letter of agreement is annexed to the project. The direct project services will follow UNDP policies on the recovery of direct project costs relating to GCF funded projects. The UNDP country office provides support services for the implementation of project activities, including assistance in reporting and direct payment. In doing so, it must ensure that the capacity of the Government (the Implementing Partner) is strengthened.

UNDP country office will provide, the following support services for the implementation of the project activities:

- (a) Identification and / or recruitment of staff to be assigned to the project;
- (b) Management of service contracts and institutional / professional contracts;
- (c) Management of fixed-term contracts, temporary contracts;
- (d) The management of specialized expertise contracts;
- (e) Definition and facilitation of training activities;

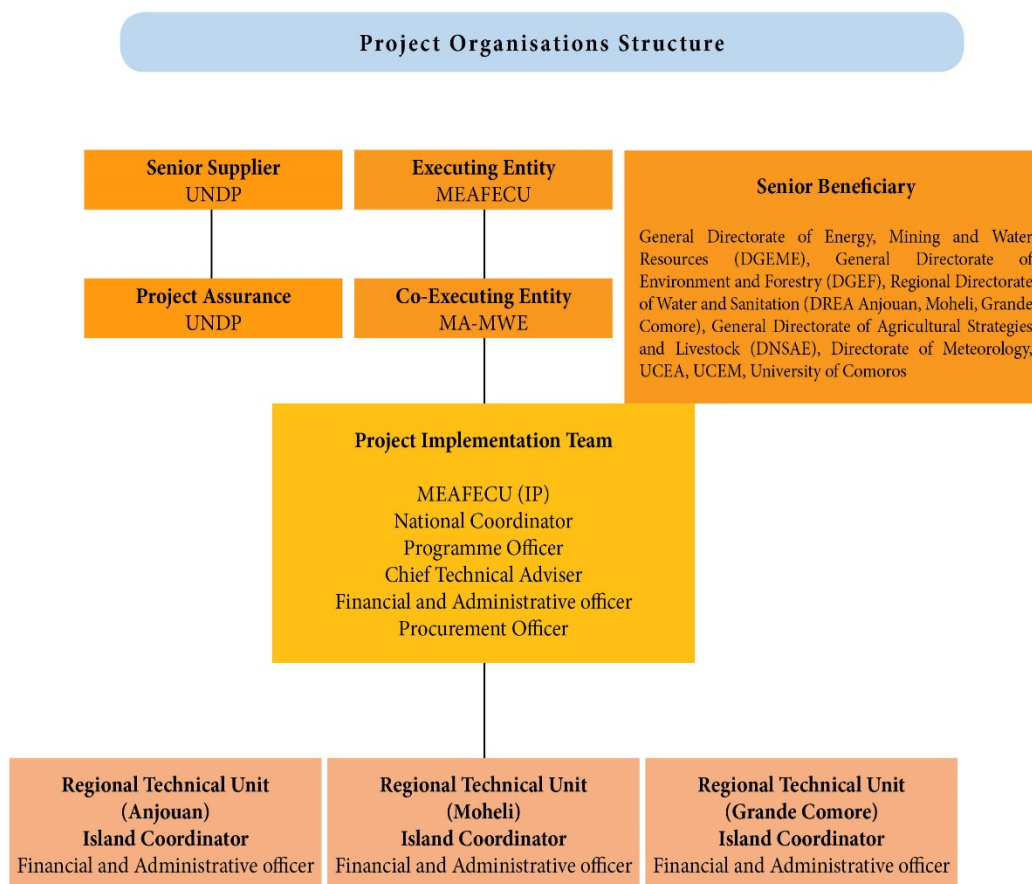
- (f) Complex procurement processes including civil works;
- (g) Acquisitions of specialized equipment to be ordered using open international tenders;
- (h) Local acquisitions exceeding the ceiling of USD50,000.00;
- (i) International Travel Management;
- (j) Financial and administrative management of the project;
- (k) General Service Management;
- (l) Management of information technology and communications services

The **Project Steering Committee (PSC)** established by a Ministerial Order will be directed by MEAPEATU and will be responsible for approving reports and activities. It will also provide guidance for proper implementation of the project. Members of the Project Steering Committee will include UNDP, representatives from the Ministries of Environment, Economy, Transport, Health and Interior as well as MA-MWE, UCEA, UCEM and EDA. The PSC will be responsible for making management decisions for the project, in particular when guidance is required by the PMU. The PSC plays a critical role in project monitoring and evaluation by quality-assuring processes and products and using evaluations for performance improvement, accountability and learning. The Committee will convene 2 times per year. Representatives from other institutions/organizations such as local Water User Associations can be included in the PSC as appropriate.

The **Project Manager** has the authority to run the project on a day-to-day basis. The PM is accountable to UNDP, the IP and the PSC for the quality, timeliness and effectiveness of the activities carried out, as well as for the use of funds. He/she will also be responsible for coordinating budgets and work plans at the island level with the **Island Coordinators**. The PM will be assisted by a Chief Technical Advisor, a Procurement Officer and a Financial and Administrative Assistant.

Three technical committees, one from each island will provide financial and technical support to the PMU.

UNDP provides a three – tier oversight and quality assurance role involving UNDP staff in Country Offices and at regional and headquarters levels. The quality assurance role supports the Project Board by carrying out objective and independent project oversight and monitoring functions. This role ensures appropriate project management milestones are managed and completed. Project Assurance must be independent of the Project Management function; the Project Board cannot delegate any of its quality assurance responsibilities to the Project Manager. The project assurance role is covered by the accredited entity fee provided by the GCF. As an Accredited Entity to the GCF, UNDP is required to deliver GCF-specific oversight and quality assurance services including: (i) Day-to-day oversight supervision, (ii) Oversight of project completion, (iii) Oversight of project reporting. “The ‘senior supplier’ role of UNDP is to represent the interests of the parties, which provide funding and/or technical expertise to the project (designing, developing, facilitating, procuring, implementing). The senior supplier’s primary function within the Board is to provide guidance regarding the technical feasibility of the project.



C.8. Timetable of Project/Programme Implementation

For Timetable of Project/Programme Implementation, please refer to Annex X of the Funding Proposal.

Assumptions:

1) Output 1:

- a. project will begin with integrating CC risks into institutional reforms brought about by the Water Code;
- b. technical and operational capacities of DGEME will happen in year 2
- c. trainings and participative workshops on water management will happen each year for the first 5 years
- d. the development of a Water Security Plan will occur towards the end of year 2
- e. all climate risk training and workshops on the Output 3 infrastructure will take place in the complimentary wet seasons when Output 3 construction is placed on hold

2) Output 2:

- a. Activity 2.1, reforestation / revegetation and implementing soil/water conservation measures will take place in the catchment areas during the dry season
 - b. All training and workshops on IWRM will take place in the complimentary wet seasons
 - c. All monitoring equipment installations will take place during the dry season
 - d. All training and workshops on monitoring equipment will take place in the complimentary months to the training on bulletin and forecast preparation
- 3) Output 3:
- a. Delineation of the protection zones and production capacities will occur in Year 1 and early Year 2
 - b. For Activity 3.2, all rehabilitation and construction of the water supply storage and collection infrastructure will take place during each dry season
 - c. Meters will be installed on select water networks during the dry season

D.1. Value Added for GCF Involvement

The water resources and water supplies of Comoros, a Small Island Developing State and Least Developed Country, are inherently vulnerable to climate change and specifically vulnerable to increased climate variability, given the small land masses, finite and fragile water resources and limited existing technical and financial capacity to adapt to and manage climate change risks. Existing public and private water resources and water supplies are capable of meeting national water demand during normal operating hydrological conditions, but are progressively being impacted upon by climate change aggravated rainfall shortages (droughts) and rainfall intensities (flood damage and erosion). Recent baseline water supply projects have failed to target or address water resources and water supply management in these climatic extremes, except at the pilot scale, and the new national Water Code does not refer to climate change adaptation – resulting in the current reform of the water sector enabling environment not recognizing, let alone prioritizing, addressing climate risk impacts on what is a fragile and vulnerable water supply sector, currently only capable of dealing with normal operational water supply needs.

The GCF involvement is therefore critical to Comoros because it provides the financial resources to address the additionality caused by climate change impacts by taking a functional existing national water supply regime and increasing its resilience to climatic extremes. The support provided by the project for each Output is described below.

Output 1 focuses on achieving a paradigm shift in changing the national water policy and legislative enabling environment to focus on and mainstream climatic risks in water management – a paradigm shift in approach from the current business-as-usual philosophy. Existing non-climate change water sector reforms are on-going. The GCF will specifically pay for the inclusion, mainstreaming and prioritization of climate risk reduction planning, preparedness and management, from national to local level, into the water sector, using a combination of legislative, regulatory, institutional, capacity building and tariff adjustment approaches to ensure not only adequate climate risk reduction technical and financial resources exist but are also prioritized in the water sector. By using a climate risk reduction approach added to more conventional drinking water safety planning (WHO-DWSP) philosophy the water sector will be using climate risk reduction to guide water sector design, operation, maintenance and reinvestment. The contribution of GCF support is very specifically to add climate resilience onto existing water sector governance reform.

Output 2 focuses on protecting the nation's water resources from climatic extremes (droughts and floods) through watershed ecosystem based adaptation management improvements, and using monitoring of the water resources condition to both inform watershed adaptation approaches and provide forecasts and early warnings of drought and flood risks. Without protection of the nation's water resources, water supply infrastructure will always remain vulnerable to climate change. The support brought by the GCF is to focus and prioritise watershed management partnerships, plans and investment onto activities which will reduce climatic extreme impacts on the basins surface water and groundwater resources – using a range of watershed based adaptation approaches - with the specific objective of increasing dry season flows and groundwater recharge, reducing storm event peak flows and minimizing storm event soil erosion and pollution mobilization, to improve water availability and quality.

In addition the GCF support is to fund water resources monitoring which is specifically focused on monitoring and tracking climate impacts on water resources e.g. groundwater salinity, stream flow quantity and turbidity, and use this information not only to inform and monitor ecosystem adaptation efforts but also be used for forecasting encroaching drought risks or impending flood risks.

Output 3 focuses on delivering, operating and maintaining climate resilient water supplies to 50% of the national population. GCF funds are addressing the climate change additionality by specifically increasing and protecting

the quantity and quality of safe drinking water available during climate change extended severe drought and flood climatic extremes.

GCF funds are not being used to improve day-to-day normal operational condition water supplies.

Specifically, GCF funds will be used to address the following climate change additionality:

- a) dry season water supply access to rural and peri-urban households in the island of Grand Comore (who currently rely on only rainwater harvesting), by expanding and constructing new groundwater fed water supply systems, which can supply water when the household rainwater tanks fail during drought periods;
- b) dry season water supply access to town and per-urban households in Grand Comore, whose borehole water supplies currently become non-potable due to increasing salinity during dry periods, by improving borehole pumping regimes to reduce salinity, increasing water storage to enable pumping rates to be reduced (which requires improving water treatment to allow water to be stored and remain potable) and by drilling new boreholes to reduce individual borehole yields and target areas of fresher groundwater;
- c) protection of stream intakes (and boreholes) from storm flood damage, through flood protection upgrades (e.g. flood walls, overflows), addition of water supply network storage to allow intakes to be closed during flood events, and improvements to treatment plants to treat increasingly turbid stream waters; and
- d) installation of flow meters to measure climate resilient improvements in drought water supply provision so as to inform correct setting of tariffs to sustain continued climate risk reduction practices, and for use to reduce water leakage to reduce water demand during drought periods.

These GCF infrastructure improvements all focus on protecting and improving drinking water supply during climatic extremes, to provide the same quantities and qualities of water per capita during climatic extremes as are currently provided during normal operating weather conditions. They are not designed to increase water supply during normal operational periods per capita or to larger populations, other than during periods of climatic extremes. The GCF is therefore solely funding climate risk reduction improvements which are required to overcome the climate change impact additionality that is above and beyond existing water supply provision during normal operating conditions.

The project targets 450,000 direct beneficiaries with climate resilient upgrade infrastructure support. This consists of approximately 367,400 people across Grand Comore, Anjouan and Moheli by Year 8 (2026), the final year of project implementation. After project completion, the national population is projected to increase. In Grand Comore this population growth will be almost entirely in the rural and peri-urban areas, who will be reliant on new housing supplied by rainwater harvesting tanks. These tanks are expected to fail during droughts due to the size of the house roofs. Accordingly the GCF support extends to the groundwater fed water supply scheme upgrades in Grand Comore being installed with the capacity to support drought water supply provision to these future vulnerable communities. Hence the number of beneficiaries receiving drought water supplies (but not normal operating condition water supply) will increase to 450,000 by 2043.

Without the GCF project, the nation's water resources will be derogated by poorly informed water supply improvements, water supplies will fail more frequently and more severely during dry seasons and be damaged during storm events, and the water security of the Comoros population will decrease year on year, resulting in greater illness, death and poverty.

For the GCF this project provides an opportunity to support one of the mostly highly vulnerable countries in the world (being one of only 6 countries classified as a SIDS and LDC), to have a direct impact on 50% of a country's population, an indirect impact on the entire nation, and hence be a truly transformative impact in the country.

D.2. Exit Strategy

The core strength of the proposed interventions in terms of sustainability is creating an enabling environment for water supply management, implementing concrete water mobilization works and in protecting water sources.

Institutional Sustainability – Legal Requirements, Capacity and Tariffs

Climate change risk reduction will be included in the national water legislation which will require evaluation and minimization of climate risks in all agency and organization planning and design. More robust climate proofed construction methods shall be employed and a comprehensive operation & maintenance has been initiated with budgets identified of all water supply schemes in Comoros. Through the update and use of a Water Security Plan improved operation and maintenance of all infrastructure works will be realized.

Financial Sustainability – Tariffs and Legal Requirements

In the medium term a water tariff will be introduced that provides adequate cost recovery to enable water supply providers to have sufficient budgets for not only addressing operation & maintenance during climatic extremes but also where necessary capital re-investment of infrastructure. Projected operations and maintenance costs have been developed to better budget (and plan for long term sustainable infrastructure. In the short term all infrastructure will have sufficient budget to cover long-term operation and maintenance costs for at least the next 25 years) through co-financing support. The O&M costs and related government co-financing support are detailed in Annex XIIIb.

Non-infrastructure agencies e.g. water resources and watershed management, will have budgets clearly identified through climate risk mandated work programmes – and prioritized for funding through the water code legislation.

Environmental Sustainability – Water and Land Resources Protected Exploitation

The GCF project will contribute to a greater physical understanding of water resources. A hydrogeological measurement campaign will be conducted on all islands to quantify groundwater and surface water exploitation potential. Such measurements will support climate-informed water management to ensure sources will not be impacted by saltwater intrusion or sedimentation caused by intense rainfall events.

Additionally, the project will support the development of site-specific, community-based Integrated Water Resources Management Plans of Action that will facilitate watershed protection and increase subsistence agricultural productivity due to enhanced water resources available for irrigation.

Technical Sustainability – Capacity Building

The Project includes technical training for Water Management Committees and IWRM Committees on climate-informed water and watershed management practices respectively.

Eleven Comorian institutions (DGEF / 3DREF, DGEME / 3DREA, UCEA / UCEM, Mamwe) trained each island on the pricing of socially sensitive water that takes into account the impacts of climate change.

Similarly, staff from the Meteorological Department (ANACM) will be trained on hydro-meteorological data collection, analysis and presentation. Users such as agriculture unions, Women's Groups, local Water User Associations will be trained in interpreting water-related bulletins to assist their planning and budgeting.

Female trainers will be trained and empowered to ensure that women's specific vulnerabilities to climate change and water management are addressed.

As a result, the users will be able to take timely and in some cases urgent actions, to ensure water supply is of good quality and has sufficient capacity. The forecasts will also enable the agencies to take preparedness measures against natural hazard events such as flooding or extended dry periods.

The capacity to implement climate resilient drinking water solutions, helps safeguard the project investments as climate risks evolve. This ensures that the project interventions do not remain short-term responses rather can be sustained and scaled beyond the project lifetime. The ToT approaches to technical training further institutionalize these skills so as to build long-term, continued capacities across the agencies. Moreover, broadening the capacity building to include sub-national and national level government staff as well as institutional partners creates synergistic approaches to support for coastal communities that can also ensure that impacts last beyond the project.

Social Sustainability - Participative Approach

The project, through enforcement of the reformed Water Code and the Decentralization Strategy of the Comoros will ensure participation with existing Water User Associations and Women's Groups to create sustainable and fair water tariff scheme and equitable water supply schemes.

The Integrated Water Resources Management Committees will have a minimum of 30% women representation in addition to cross-sectoral representatives to determine the most effective and equitable water supply mechanisms for subsistence agriculture.

To ensure sustainability of the water management system, user behavior change will be promoted with respect to protection and conservation of water services. Users will be informed about the link between water and health, and the need for water cost recovery.

In this section, the accredited entity is expected to provide a brief description of the expected performance of the proposed project/programme against each of the Fund's six investment criteria. Activity-specific sub-criteria and indicative assessment factors, which can be found in the Fund's [Investment Framework](#), should be addressed where relevant and applicable. This section should tie into any request for concessionality made in [section B.2](#).

E.1. Impact Potential			
Potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas			
E.1.1. Mitigation / adaptation impact potential			
The project will contribute to achieving Fund Level Impact, increased resilience of health and wellbeing, and food and water security; The climate-impact potential of the Project introduces an integrated, holistic, participative approach to reduce hydrological (drought and flood) risks and increase the resilience of reliable and safe water supply accounting for climate stresses. All water resources, watershed and infrastructure designs and management protocols will account for expected climate change impacts (drought, extended dry periods, intense rainfall events). Increased resilience of health and well-being, and food and water security (Fund-Level Impact 2.0) The project will directly benefit 450,000 people (of which 229,500 are women) in the urban and peri-urban towns of each island capital (Moroni, Mutsamudu and Fomboni) and 100 rural villages of Comoros (see Feasibility Study Annex 5) by improving drinking water supply climate resilience to droughts, flood damage and poor storm run-off water quality. This is 64% of Comoros's current population.⁴² The project will also reduce and potentially eliminate the number of people severely impacted by floods due to water disruptions (e.g., 84,000 people had interrupted water supply after the 2012 flood) and by water-borne diseases (approximately 4,000 annually).^{43 44}			
E.1.2. Key impact potential indicator			
GCF core indicators	Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided (Mitigation only)	Annual	N/A
		Lifetime	N/A

⁴² World Bank statistics

⁴³ OCHA: Floods in April 2012 affected approximately 50,000 people, including more than 9,000 who were forced to leave their homes.

⁴⁴ Comoros Second National Communication to the UNFCCC (2012)

	- Expected total number of direct and indirect beneficiaries, disaggregated by gender (reduced vulnerability or increased resilience); - Number of beneficiaries relative to total population, disaggregated by gender (adaptation only)	Total	Direct Beneficiaries 450,000 people (229,500 women) will benefit from potable water supply climate risk resilience improvements, watershed climate risk resilience improvements and irrigation water supply climate risk resilience improvements Indirect Beneficiaries 800,000 (375,000 women) will benefit from national and island scale climate risk resilience water governance reforms, national scale water resources monitoring and management during climate extremes, national scale climate risk forecasting and adaptation product dissemination, and national scale climate risk reduction public awareness raising
		Percentage (%)	Direct – see above 64% (51% women) Indirect – see above 100% (51% women)
Other relevant indicators	5.1 Mainstreaming of climate change into national sector planning & coordination in information sharing and project implementation 5.2 Number and level of effective coordination mechanisms 6.2 Perception of population & emergency response agencies on timeliness, content and reach of Early Warning Systems 7.1 Extent of use by public sector and communities of improved tools, strategies & activities to respond to climate variability and climate change 7.2 Number of males and females reached by climate -related early warning systems and other risk reduction measures established/strengthened 8.1 Percentage of target population aware of potential impacts of climate change & range of possible responses		
The methodology for calculating the above core indicators is as follows: The total direct beneficiaries calculation methodology is as follows: i) The population (2018) for the 103 villages and urban areas receiving water supply infrastructure in the 15 project target zones has been estimated from the 2003 Census and official population growth rates (UN Stats); ii) The total beneficiaries for the project is estimated based upon the 2018 estimated population plus population growth to 2042 in these target populations using approved population growth rates (UN			

Stats). The target date of 2042 refers to the 25 years of Government of Comoros tariff support and hence is the end date for project co-financing benefit;

- The indirect beneficiaries are defined as those benefiting from national governance reforms and nationally available climate information and products eg radio/SMS storm flood early warning alerts and drought advisories.
- The indirect beneficiaries calculation methodology uses the national population (2018) calculated using the 2003 Census and official population growth rates (UN Stats).

E.2. Paradigm Shift Potential

Degree to which the proposed activity can catalyze impact beyond a one-off project/programme investment

E.2.1. Potential for scaling up and replication (Provide a numerical multiple and supporting rationale)

National

By addressing the critical barriers preventing climate resilience in the water sector – specifically the lack of climate risk reduction centric financial resources, the lack of stakeholder coordination and cooperation, the lack of knowledge and data and the lack of technical capacity (see the Theory of Change diagram) - the project develops, strengthens and sustains a national level enabling environment that promotes, supports and allows island and national scale roll-out of climate risk reduction practices, as well as enables international knowledge transfer on this adaptation approach. It is this mainstreaming and integration of climate risk reduction into the pre-project 'business-as-usual' water sector planning, budgeting and operating environment that is the paradigm shift the project is delivering to address climate change impacts and achieve a climate resilient water sector.

A national water code re-focused on climate risk reduction will provide the legal direction and requirement for national and island institutions to plan, budget and programme their water supply, water resources and watershed activities beyond that of just the project target zones. By mainstreaming climate risk reduction into institutional responsibilities, annual plans, performance indicators, staffing units and budgets, there will be a sustained planning and management paradigm shift in improving climate resilience.

The introduction of water tariffs designed to achieve sufficient cost recovery to maintain and invest in climate resilience infrastructure upgrades ensures water service providers will have the financial resources to maintain both capital and operational investment of climate resilience. Service providers, whose remit extends beyond the project intervention zones, will be able to expand this approach to the rest of the country, as well as provide a model for take-up by communities not directly supported by the project.

Climate forecasting, adaptation tool development and guidance, early warning systems and public awareness raising, will be national in availability and extent increasing the capacity, understanding and interest in climate risk reduction across the country as a whole.

Within each island there is considerable potential to replicate the infrastructure, water resources and watershed activities to the 40% of the people and 55% of the areas not directly benefiting from the project.

Groundwater and surface water monitoring whilst national in extent will not be in every watershed (11 only) and can be increased to have a much greater coverage than the project will deliver – ideally every catchment (60).

Groundwater investigation in Anjouan and Moheli will be preliminary and exploratory in character and can be increased substantially to give a similar coverage to that proposed for Grand Comore.

Even within the project target zones there are opportunities to transfer best practice from one island to another.

Specifically, Grand Comore exploits groundwater whereas the other islands do not. The project will investigate the occurrence of groundwater in Anjouan and Moheli, and where located, groundwater abstraction technologies can be introduced to these other islands.

Similarly, successful rainwater harvesting approaches currently used in Grand Comore could be replicated in the other islands, increasing resilience to flood events.

Watershed activities will primarily focus on those catchments or sub-catchments supporting or about to support stream water supply intakes, however watershed activities can be replicated in those catchments which do not, especially those which have groundwater abstraction boreholes which will benefit from increased groundwater recharge and/or have populations living in marginal land along flood route corridors.

Whilst climate hydrological forecasting will be national in scale the climate impacts will be watershed and aquifer specific and hence the preparedness planning and management responses can be expanded to a greater number of areas than the project target zones.

And although capacity building will focus within the project on the agencies and communities receiving the benefits of resilient water supply infrastructure upgrades and watershed improvements, those agencies and communities not receiving such targeted training in climate risk identification and risk reduction, will still require this post project.

Equally the climate adaptation hydrological products – which will focus on water supply and food security – can be expanded and applied to other sectors.

International

Comoros is but one of 4 Small Island Developing States in the Indian Ocean, along with Mauritius, Seychelles and the Maldives, with additional overseas territories in the region (e.g. Reunion, Mayotte) and larger coastal countries (e.g. Madagascar).

Whilst not all are steep volcanic terrains (Maldives most obviously), replication of many of the lessons learned from the project, including successful GCF application, will be relevant to these states – most obviously Seychelles and Mauritius in terms of water resources investigation & monitoring, watershed management, but for all countries and territories with respect to climate forecasting and product development, capacity building materials and national legislative codes and institutional reforms. The UNDP/UNEP GEF funded AIOSIDS IWRM programme has created a web forum <http://aio-iwr.org/> for knowledge transfer which this project can use for knowledge transfer to support replication.

Indeed, the 56 recognized Pacific, Atlantic and Caribbean SIDS other than Comoros, all have potential replication value of using this GCF project as a guide and blueprint, especially given the difficulties SIDS have in successfully accessing global funding mechanisms.

E.2.2. Potential for knowledge and learning

Knowledge and learning are critical to building adaptation capacity, both within government and in the wider general public.

Knowledge and learning are therefore embedded across all four components of the project design as approaches to increase resource and infrastructure sustainability, build capacity, promote partnership, ensure ownership, secure political support, raise awareness and deliver evidence based risk reduction planning and management of the nation's water resources and water supply schemes.

These include both formal and informal knowledge strengthening and learning activities, including:

- Climate change risk knowledge and information sharing integrated into the Water Code;
- Assessment, guidance and standards for tariffs required to fund climate adaptation design, operation & maintenance of water supply infrastructure;
- Inclusion of risk mapping, risk reduction actions, source protection, climatic extreme operation & maintenance best practice and water quality standards in the Water Code;
- Training of national and island government agencies in implementation of the revised climate risk reduction Water Code;
- Promotion, development, training and use of Water Safety and Security Planning during climatic extremes for government and water management committee stakeholders;
- Water committee and community training on water conservation and efficient production of potable water;

- Training in IWRM governance to IWRM committees;
- Knowledge generation from watershed characterization including climate hazard and risk mapping;
- Training of IWRM committees and watershed communities in climate risk adaptation strategy options planning and budgeting, management and required monitoring;
- Development of evidence-based watershed adaptation action plans and knowledge exchange with island and national agencies;
- Implementing watershed adaptation plans involving community and other stakeholder awareness raising – including soil and water conservation, source protection and pollution prevention and groundwater recharge techniques;
- Assessment and guideline development on optimum well pumping techniques to minimize saline up-coning, and training of water supply service providers;
- Knowledge transfer of UNICEF Pacific Drinking Water Safety and Security Planning (DWSSP) approaches, guidance and materials
- Development of Comoros best practice DWSSP documents for water supply service providers and communities – including location selection, design, construction, operation & maintenance, emergency procedures and drought preparedness responses;
- Training in water supply system performance monitoring during climatic extremes;
- Investigation, assessment and characterization and monitoring of groundwater, stream and rain water resources, reliable yields and vulnerabilities to climate risks;
- Training of Meteorological service in climate forecasting, drought forecasting and floods forecasting;
- Development of standard procedures for data collection, processing and analysis and training in water resource monitoring network maintenance
- Training on application of rainfall predictions to water resources balances and risks of water supply failures;
- Training of different sector water users on water resources forecasts and information advisories;
- Training on forecast dissemination best practice – communication networks and media; and
- Inclusion of water sector climate risk reduction modules in local university curricula.

E.2.3. Contribution to the creation of an enabling environment

The Proposed Project will contribute to the creation of an enabling environment by delivering political, institutional and financial sustainability for climate resilient water resources, watershed and water supply management.

This is primarily achieved through the project inclusion of climate risk adaptation into the new Water Code. This will ensure the necessary legislative political and institutional commitment to ensuring climate risk reduction is included in ministerial and departmental programmes, with appropriate planning, recruitment and budgeting.

With improvements in quantities of treated water quality delivery the public have confirmed they will pay for reliable safe water supplies. Once improved secure water supplies are established then the tariff system will be set to be self-sustaining.

Decentralized Water Management Committees will be trained on decentralized water supply and management solutions including setting of water tariff structures to support operation and maintenance protocols focusing on preventative actions required to adapt to dry period and intense rainfall conditions.

Once financially sustainable water supply schemes become established, private sector interest will increase in the water sector and based on successful experiences by the semi-private Water User Association, SOGEM, the project could potentially catalyze PPPs to support with water management.

IWRM committees founded in accordance with the reformed Water Code with the objective of ensuring watershed protection, will use the human resources within the watersheds to be actively involved in climate risk reduction measures and to recognize the cost savings they benefit from reduce climate impacts and hence value of water achieved through climate risk reduction.

In order to plan for integrated water management, the water cycle in each target watershed will be monitored. Safe yields from wells that do not permit saltwater intrusion will be documented. Also, the water resource data will be forecasted in order to provide targeted water conservation advisories to subsistence farmers, etc. An MOU will be established with various regional forecasting groups and the WMO in order to have more accurate forecasts. Also, due to the experience of the university with regards to hydrogeological measurements (e.g., piezometers), the water agencies have set up collaborative agreements to enhance groundwater monitoring.

Increasing the water resources and water supply technical management capacity through using the human resources available within the watershed communities to be directly involved in participatory management reduces both national and island institutional governmental costs whilst accessing the monetarized and non-monetarized benefits received by other sectors, notably cash cropping, and attributed to be delivered through eco-system services, hence sustaining the watershed managed improvements and eco-system based adaptation measures.

Lastly the improved dry season water provision will remove the market demand which is currently supporting the excessive and extreme private trader water prices that occur during dry seasons, providing rural and peri-urban communities with incentives to use water conservatively and manage their water supply systems effectively.

E.2.4. Contribution to regulatory framework and policies

The Project will contribute to the strengthening of the national water sector reforms, by ensuring climate risk reduction is a central component and integral to the new Water Code, across water resources, watershed and water supply infrastructure management at the national, inland and community management scales. Without the GCF project climate risk reduction will not be included in the new Water Code and there will be no policy driver or legislative requirement to undertake climate risk reduction at all in the water sector, let alone mainstream it into work programmes and institutionalize it agency mandates and capacities.

The project then follows up with supporting implementation of revised climate risk reduction enhanced Water Code through training, advocacy and capacity building of the main institutional stakeholders, as well as strengthens the financial resources available to deliver climate risk reduction through improved tariff structures.

This is a paradigm shift in the approach to water resources, watershed and water supply infrastructure management by using a risk-based approach to identify, avoid and reduce exposure to climate risks, introducing both efficiency and collective effectiveness in improving dry season water flows and wet season water quality as well as reductions in soil erosion and flood damage to infrastructure.

By actively engaging with the watershed stakeholders and water supply customers the project also reduces the institutional regulatory compliance burden by advocating and raising awareness with the public and hence effectively creating a self-regulatory culture which requires much reduced enforcement monitoring.

In doing so, the project also contributes to both the National Strategy for Accelerated Growth and Sustainable Development (*Stratégie de croissance accélérée et de développement* (SCA2D) 2015-2019), but also to the National Action Programme of Adaptation (NAPA, 2006) to Climate Change and the Sustainable Development Goals.

E.3. Sustainable Development Potential

Wider benefits and priorities

E.3.1. Environmental, social and economic co-benefits, including gender-sensitive development impact

The following co-benefits are recognized as being realized by the project.

Environmental Co-Benefits

Improved and enhanced water resources in terms of quality and quantity

The project will reduce groundwater salinity through improved understanding of the available groundwater resources (sustainable yields), their vulnerability to drought (drought yields) and optimizing how best to exploit the groundwater without resulting in saline up-coning.

The watershed rehabilitation activities in 32 catchments will contribute to groundwater (and surface water) quality and quantity improvements by slowing runoff and increasing groundwater recharge. They will also reduce pollution risks to both water resource types.

Water conservation measures – supply system leakage reduction, user conservation, and more efficient irrigation will reduce water abstraction contributing to fresher groundwaters and greater stream base flows.

Improved soil quality and reduced erosion

The watershed improvement component includes soil erosion reduction measures, through both improved farming practices and reforestation and revegetation efforts with native, drought-tolerant species. The reduction in stream peak flood flows and storm run-off will also reduce direct erosion of soils and land.

Improved biodiversity

The watershed improvement component directly supports watershed reforestation and re-vegetation of habitats.

The improved water quantities (greater baseflows, reduced flood flows) as well as reduced stream turbidity will all contribute to healthier aquatic eco-systems and those wider eco-systems that in turn are dependent upon the riverine eco-system corridor health e.g. the nearshore marine environment and wider watersheds.

Increased carbon sequestration

Although not an explicit objective of the project, the increased reforestation and re-vegetation of the watersheds will increase carbon sequestration in the country.

Social Co-Benefits

Improved Health

The project will improve drinking water quantity and quality to 450,000 people throughout the year, including in periods of droughts and during and after cyclones and associated flooding events

Reduced Lost Time – Access to Education and Work

Currently rural communities (especially women) spend considerable periods of time fetching and carrying water, especially in the dry season, but for some communities this is year round. This saving in lost time provides an opportunity to increase school attendance and/or work on household income activities.

Improved Safety in Climatic and non-Climatic Disasters

The climate forecasting linked to storm event flash flood forecasting will provide an early warning system for extreme flooding events. This will enable vulnerable communities and those working in or near streams to safely evacuate to higher ground.

Community Empowerment

The project focuses on increasing the capacity of rural communities to manage and protect their water supply schemes. This development of adaptation management capacity will strengthen the accountability of wider

community governance generally and can be applied to other sector/issue resilience more specifically e.g. food security.

Economic Co-Benefits

Reduced Costs of Climatic Damage

The timely forecasting and prediction of flood events will reduce the economic damage to all sectors from flood water inundation, as well as improve event response and recovery performance. The April 2012 floods were estimated (by COSEP) to cost US\$20 Million, 5% of Comoros GDP.

Water supply utilities will design, locate, construct and operate and maintain water supply schemes which will be less exposed to drought and flood risks and therefore will have reduced event associated damage and repair costs.

The provision of a more regular water supply during droughts and floods will reduce losses associated with business/trade/manufacturing operation disruption.

Increased Employment Opportunities due to Improved Health

The 450,000 people receiving improved water supplies will have better health and therefore can be more productive in their household incoming earning activities – either informal or formal in nature.

More Productive Agricultural Sector

The agricultural sector has very little irrigation infrastructure with which to reduce exposure to the dry seasons. The project will increase irrigation water storage using impluvium and stream water intakes to increase dry season cash crop productivity.

Direct Employment opportunities

Adaptation interventions such as impluvium rehabilitation, water supply construction upgrades and even watershed improvements will all require construction workers, field workers, material suppliers and importers, which will create employment opportunities in the country.

The wider project enabling environment will require a permanent increase in technical, management and financial capacity across a wide breadth of different ministries, agencies and organisations, including meteorologists, hydrologists, hydrogeologists, water risk reduction engineers and so on.

Gender-Sensitive Development Impacts

In Comoros, women and girls are traditionally in charge of collecting water, and are additionally affected by increasingly scarce water supplies. Consequently, women and girls in Comoros have to walk approximately 195 meters to the closest water source and distances to safe water resources are predicted to increase in rural areas due to the projected drying up of rivers and decrease of wells yields that will likely be led by the increased rainfall patterns and temperatures. To get the minimum amount of necessary daily water, women and girls walk this distance, back and forth, up to five times per day; which results in about 2 kilometers per day per household.⁴⁵ Women spend an average of 2 hours per day collecting water. Water collection is increasingly demanding and represents an opportunity cost in terms of time and labor.

Women are responsible for the majority of household food production in the Comoros. Agriculture, including food production, is suffering the effects of climate variability and the lack of water with an increase in food scarcity. Children's malnutrition and slow development rates are increasing (25.8 per cent and 44 per cent respectively in 2004) with more recent statistics showing that more than 42% of Comorian children age five and under suffer from chronic malnutrition.⁴⁶

To address the issues raised above, the project will reduce salinity risks and will improve water treatment which will have a direct impact on women and youth through reduced high salt water intakes, improved health, wellbeing and nutrition. Moreover, women's skills on operation and maintenance of local water management systems, and

⁴⁵ Ibid.

⁴⁶ Ibid.

knowledge on water resource management will be improved as a result of this project. A Gender Action plan has been developed (see Annex XIII: Gender assessment and budgeted action plan). The action plan provides suggested entry points for gender-responsive actions to be taken during project implementation.

E.4. Needs of the Recipient

Vulnerability and financing needs of the beneficiary country and population

E.4.1. Vulnerability of country and beneficiary groups (Adaptation only)

Country Vulnerability

The Union of the Comoros is classified as a Least Developed Country and a Small Island Developing State (SIDS) by the UN System. It is one of not only the poorest countries in the world with an estimated 80% of the rural population considered poverty-stricken and 46% of the population living in absolute poverty (<\$1.25/person/day), but also one of the most vulnerable to climate change impacts, with a land mass area of approximately 1,600 km² with no land >7km from the coast.

The global position on SIDS vulnerabilities is articulated in the SIDS Accelerated Modalities of Action (SAMOA) Pathway⁴⁷ (2014). This position statement (previously defined in the Barbados Programme of Action and later the Mauritius Strategy) clearly identifies the unique and particular vulnerabilities of SIDS, including isolation, small land masses (and therefore water resources), small populations and distance to global markets, as well as greater exposure to natural hazards (cyclones, droughts, volcanic and seismic hazards), which not only hinder sustainable development but also climate resilience.

The IPCC Fifth Assessment Report (AR5, 2014) Chapter 29⁴⁸ specifically identifies SIDS as having high vulnerability and low adaptation capacity to climate change. It explicitly states addressing critical economic, social and environmental issues will increase human and environmental resilience to climate change, whilst also recognizing substantial changes to human settlement patterns within SIDS may mask climate change effects. Land and water management in particular are highlighted as complimentary climate change and development strategies.

In addition to limited technical, financial and human resource barriers to climate adaptation, constraints from weak political and legal frameworks and difficulties in translating national mainstreaming into over-coming rural and especially outer island existing social and cultural practices are also cited as SIDS specific in the AR5 assessment.

National capacity to adapt to climate change risks in Comoros is extremely limited, as it is for many SIDS. At least 14.3% of the population is unemployed. The unemployment rate among those aged 15-24 is very high at 50.5%.⁴⁹

Between 70-80% of the Comorian population are small-scale farmers that are dependent on rain-fed water resources for subsistence agriculture. National food security is therefore closely linked to water security and to climate change impacts and their successful adaptation.

More widely, poverty issues and limited employment opportunities are severely hindering the country from self-sustaining economic growth.⁵⁰

⁴⁷ United Nations Resolution RES/69/15 SIDS Accelerated Modalities of Action (SAMOA) Pathway

⁴⁸ Nurse, L. *et al* (2014) IPCC Fifth Assessment Report: Chapter 29 – Small Islands

⁴⁹ Programme Pays pour le Travail Décent 2015-2019

⁵⁰ Comoros' Country Work Programme (2015-2019)

National Distribution of Climate Vulnerability

A national consultation process on climate variability within Comoros was undertaken as part of this project design process. Facilitated by UNDP, the participatory consultation was undertaken between November 2015 and November 2016 using an expert stakeholder group involving senior national experts from the key line ministries and island agencies in a series of workshops. The consultation process included a review of water supply vulnerability based upon group evaluation of existing water supply feasibility studies as well as national climate vulnerability and disaster risk management assessments, followed by expert discussions. Consultation documentation is enclosed in Annex XIII (f & g).

The country was divided into 26 zones. The vulnerability of these zones were then assessed using 34 metrics covering 8 criteria groups: climatic impacts, type and condition of water resources, land degradation, role of women in water management, socio-economic vulnerability, on-going governmental and non-governmental programmes, and likely ease of project implementation (e.g. accessibility). The zones were then ranked based upon a summation of climatic and socio-economic vulnerability and an overall ranking derived which included other on-going similar project initiatives and ease of project implementation.

Using these criteria, the most vulnerable zones to climate risks which were not receiving on-going or planned climate adaptation support were identified. Of the 19 most vulnerable zones identified, 4 already had other development partners working in them, resulting in 15 zones being selected as targets for the GCF project.

These 15 zones deliberately covered all three islands (6 on Grande Comore, 7 on Anjouan and 2 on Moheli), which enables the zones to cover not only the different governance arrangements on each island but also the different hydrological regimes (i.e. rainwater, groundwater and surface water) and water supply types.

Community Consultations

Consultations were held with zone communities, where those communities were responsible or partially responsible for water provision. 14 Memoranda of Understanding were signed with these communities securing their agreement and commitment to the project, including the type of intervention, land access and involvement in future training and scheme management (where applicable). The Memoranda of Understanding are available in the Annex XIII (g).

E.4.2. Financial, economic, social and institutional needs

Financial resource constraints

As a SIDS, the overall cost of the economic impacts related to climate risks identified for the case of Comoros has been projected to amount to USD 506 million in 2020, (192 billion KMF) representing 112% of real GDP in 2020. In 2050, the economic impacts of change climate represent 1.5 times the GDP in 2050.⁵¹

As indicated above, the population of the Comoros experiences severe difficulties in accessing water of good quality. Most citizens consume a limited quantity of water, and a large share of the consumed water is non-potable. In 2014, the total cost of imports of medicine directly associated with the treatment of gastro-intestinal diseases in the 15 zones targeted by the project reached approximately \$6.4 million. This cost increased to \$7.7 million in 2015, and reached \$9.5 million in 2016, representing an average annual cost of approximately \$7.8 million over the period 2014-2016. In these zones targeted by the project, the average cost per person for the treatment of gastro-intestinal diseases increased from an estimated \$20.60 in 2014, to \$24.00 in 2015, and reached \$28.80 in 2016 for an average cost of \$24.50 per capita over these 3 years⁵². Climate change is projected to significantly alter this situation and to make it even more challenging. However, Comoros has limited financial resources that prevent it from adequately addressing climate change adaptation needs. Classified as a Least Developed Country by the UN System, the Comoros ranks 169 on the Human Development Index (out of 187 countries), with US\$ 840 GNI per capita, and 46% of the population

⁵¹ Comoros's Second National Communication to the UNFCCC 2012

⁵² These calculations are shown in the row 3 to 16 worksheet "Benefits" of the economic analysis in Annex XI.

living in absolute poverty with less than US\$ 1.25 per day. Anjouan and Mohéli are the poorest islands. (REF: Anjouan Vul). A large proportion of the population relies on remittances from the Comorian diaspora (mainly in France), which was estimated to have sent in US\$ 35.4 million in 2004. (REF: UNEP)

In spite of donor interventions, there is a lack of financial resources for climate-informed water management. With the current scheme, pilot projects are contributing to improving the water storage and supply system without any cohesion and using limited integration of climate change projections. A National Water and Sanitation Infrastructure Fund is to be created under revisions to the Water Code. However, the fund is not currently active. As a result, only 35% of the country has access to sanitation and watersheds are continually polluted by release of wastewater.⁵³

The rural and peri-urban populations have a willingness to pay water tariffs. However, the lack of standardization has caused these marginalized populations to pay as much as 4 times the price in the capital for water that is of poor quality.

Capacity for addressing climate change adaptation

As a SIDS which is highly susceptible to climate variability, natural hazards and climate change impacts, one of the most significant challenges for Comoros is water security. Inadequate management of water production, supply and treatment is expected to become more challenging in the future due to sporadic and intense rainfall events as well as an increase in natural hazards. AfDB, AFD, the EU, UNEP and UNDP have tried to improve water services by building the capacities of DGEME and other water-related organizations but have not considered climate change impacts to any extent. Water-sector coordination is limited between ministries and water infrastructure has not been designed to account for the existing and projected impacts of climate change. To do so, the appropriate ministries and Water Management Committees to be formed under the revised Water Code require capacity reinforcement and practical demonstrations of how to best manage, operate and maintain water resources in a systemic manner that ensures cost recovery in the context of expected climate change and recurring natural hazards.

The project includes institutional strengthening of the water sector to include tariff setting at such a level as to sustain and maintain climate risk adaptive management across the water supply, water resources and watershed management agencies and stakeholders.

The project recognizes communities and customers will not immediately support tariffs until such time as reliable climate resilient water supplies can be demonstrated to be sustained. Consequently, the Government of Comoros is supporting operation, maintenance and management costs for 25 years, during which time the reliability of the water sector will be demonstrated and the tariff system progressively introduced. This approach allows for sustainable financing of climate resilience post-project intervention.

E.5. Country Ownership

Beneficiary country (ies) ownership of, and capacity to implement, a funded project or programme

E.5.1. Existence of a national climate strategy and coherence with existing plans and policies, including NAMAs, NAPAs and NAPs

The Proposed project is highly aligned with Comoros's climate change adaptation commitments and priorities.

The Government of Comoros ratified the **United Nations Framework Convention on Climate Change** in 1992 and submitted its **National Adaptation Plan of Action (NAPA)** in 2006. NAPA priorities are aimed at adaptation in agriculture and water. The NAPA identifies loss of water bodies, drought and low river flows, and climate-related storms as major threats and hazards to Comoros. It also identifies water/groundwater availability, food security and income generation as the main issues vulnerable to climate change. Some of the priority adaptation projects identified in the NAPA are i) increase water supply and increase its quality; and ii) fight against soil erosion and promote restoration

⁵³ AfDB Country Report 2015

and reconstitution of basin slopes.

The NAPA identified 19 climate change impacts of which the most acute (1st ranked) was that of acute drought, food security was ranked 5th. Heavy rains, cyclones and sea level rise were also identified as concerns.

Water resources was ranked the most vulnerable environmental service, with agriculture and small farmers most likely to be impacted. The NAPA prioritized ranking for adaptation options was:

- Drought resilient crops;
- Increased water supplies;
- Improved water quality;
- Soil and land restoration.

The GCF financed project is also in agreement with the *Initial and Second National Communications (INC (2003) and 2NC (2012))* respectively) to the UNFCCC⁵⁴ which identify Comoros's principal environmental problem to be water scarcity and the ephemerality of surface water requiring the following strategies:

- strengthening water resources management and environmental monitoring;
- improving groundwater management and preservation;
- expanding hydrological and meteorological monitoring infrastructure;
- protecting ecosystems and regulating stream flow;
- increasing storm run-off collection and discharge infrastructure; and
- integrating local populations into water resources management.

Furthermore, the project is entirely aligned with Comoros's recently drafted *Accelerated Growth and Sustainable Development Strategy (2015 – 2019) (Stratégie de Croissance Accélérée de Développement Durable-SCA2D)*. The SCA2D's aim is to make the Comoros an emerging country by 2040.⁵⁵ Two of the four objectives of the SCA2D are i) to improve the population's living conditions and ensure equity in the access to basic social services; and ii) to promote the Comorian natural and cultural heritage and the optimal use of natural resources.

Similarly, the Project supports i) the *Water Act* (1994) that indicates water resources management should be led by MA-MWE (Autonomous Agency for Water and Energy Distribution) in urban areas and by the Ministry of Production in peri-urban and rural areas and ii) the 2011 decree on the *Decentralization Process* that stipulates that water and sanitation management is attributed to communities on the three islands.

The Project also takes steps to advance Comoros's *National Adaptation Plan* process. To date, Comoros has mobilized many government, private and civil society organizations to provide awareness on adaptation, water security and DRM.

It also fully supports one of the pillars of the *United Nations Development Assistance Framework 2015-2019 (UNDAF)*: improvement of the population's living conditions and equity in the access to basic social services.

Finally, the Project supports Comoros's aim in achieving the *Sustainable Development Goals (SDGs)*. The project will address SDG 13 – climate change action, and contribute to SDG 2 – food security, SDG 6 – sustainable water management, SDG 14 – sustainable land management and SDG 11 – making cities more resilient.

E.5.2. Capacity of accredited entities and executing entities to deliver

⁵⁴ Comoros's Initial National Communication to the UNFCCC 2001

⁵⁵ Stratégie de croissance accélérée et de développement durable 2015-2019 (SCA2D) (May 2014)

Executing entities

The Executing Entity will be the Ministry of Energy, Agriculture, Fisheries, Environment, Country Planning and Urbanism (MEAPEATU). MEAPEATU has long experience in managing water and energy-related projects such as the recent ACCE and all AFD water projects. MEAPEATU works in close collaboration with all government institutions at the federal, island and community levels. The DGEF and the DGEME will be the key technical departments of the MEAPEATU involved in the implementation of the project.

The DGEF has 15 permanent staff members, most of whom have more than 20 years of experience in project management. Since then, the DGEF has been implementing development projects related to environmental management and climate change financed by various development partners. To this end, she has a very good experience in the management and implementation of projects.

Proven experience will enable it to monitor project implementation, from the achievement of project outputs, to full responsibility for achieving the goals and objectives of government projects, the United Nations Framework Convention on Climate Change (UNCCD), UNCCD, and UNFCCC.

The DGEF, in accordance with the tasks assigned to it, provides strategic guidance for the implementation of projects and ensures their supervision, administrative and financial management in conjunction with the administrative and financial management of the Vice-Presidency and monitoring and evaluation in collaboration with the monitoring / evaluation of the Vice-Presidency.

The DGEME is the main state institution responsible for the sovereign missions in the water sector. It has 20 permanent staff members, most of whom have more than 20 years of experience in water related project management.

The MA-MWE and ANACM will be co-executing entities and will be in charge respectively of the activities 3.1, 3.2, 3.3 and 2.4, 2.5. They have also a long experience with executing projects activities in line with the activities they will be in charge in the framework of this project.

A list of the projects managed and completed by these organizations over the last decade is included in the Feasibility Study.

The Accredited Entity is the United Nations Development Programme (UNDP). The UNDP has a long history in promoting sustainable development in Comoros and has been closely involved in climate change adaptation activities as evidenced by presently supporting four adaptation projects (GIRE, CRCCA, DRM, and Geothermal).

E.5.3. Engagement with NDAs, civil society organizations and other relevant stakeholders

The overall goal of stakeholder consultations has been to identify relevant agencies involved with water and watershed management and climate information production. Consultations have ensured the proposed project is grounded in local realities whilst being aligned with national policies and able to support the most vulnerable communities, particularly female-headed households who are responsible for water collection and subsistence farming.

Engagement with NDA

The project design process has fully involved the NDA (MEAPEATU) since the start of the concept genesis in late 2015 (see Annex XIII (f)). The Concept Note design (January to June 2016) involved the partnership of MAPEIA and UNDP in preparing a Pre-Feasibility Study (April 2016) and Concept Note (June 2016), formally endorsed by the NDA.

Full project design commenced in July 2016, with a project design overview workshop between the NDA, UNDP, water sector agencies and consultant design team, during which the project concept design was reviewed and the design engagement process agreed with the NDA. The minutes are in Annex XIII (f).

A series of key stakeholder engagement and project design meetings were held between September and December 2016, including the direct participation of MEAPEATU where relevant. MEAPEATU were kept fully informed of progress

during later design missions in March and April 2017 by the UNDP. In November 2016, the President of Comoros met the GCF Executive Director at COP22 to demonstrate government commitment to the proposal and further secure GCF support (see Annex XIII (f)).

Proposal Consultation Process

A series of tri-partite full project document design and consultation meetings has been undertaken since July 2016 between UNDP, government stakeholders and the consultant design team. These included a review of the Concept Note design in July 2016, discussions on the roles of different water-related agencies and a series of national workshops on the target zones to be selected involving more than 30 agency representatives.

Meeting minutes and agency attendance records are held in Annex XIII (f), confirming the involvement and engagement of the country's water related stakeholders in the design process.

In early 2017, environmental impact assessment and socio-economic impact assessment of the detailed design were undertaken and included additional consultation with the proposed national and island agencies to be involved with implementation as well as consultations with beneficiary communities (see Annex XIII (g)).

In May 2017, community visits were undertaken by the national design consultants on all three islands and discussions held with the local communities on likely requirements for land access, land occupation and land disturbance and potential vegetation loss.

All communities were asked to confirm their agreement to the project activities being undertaken on land within the community ownership and control. The signed agreements are held in Annex XIII (g) with photographic records of these community consultations in Annex XIII (g).

Future Stakeholder Engagement during Project Delivery

At the implementation level, there are several types of stakeholders that will be engaged during the implementation of the Proposed Project. Details of each stakeholder, their role and responsibilities with respect to project implementation and details of their engagement modality are shown in Annex XIII (e).

The implementing national and island governmental water organisations are well aware of the project and have been fully involved in its genesis.

To ensure local ownership of the Project, island coordinators will be responsible for local project activities. Training will be provided to all Water Management Committee trainers (implicating at least 50% women and making sure some of the trainers are female) on the integration of climate-informed practices into water management (e.g., storage) and on the operation and maintenance of climate-proof water supply systems.

Water User Associations at community levels will play a crucial role in monitoring the functioning of water resources in Comoros. Having a large women representation, these Associations focus on vulnerable groups such as women and children. Similarly, Women's Groups will be heavily implicated. At the moment, a focal point in each ministry is focusing on gender, however, with the current project, specific Women's Groups will be implicated in project decision-making.

Finally, the University which is the sole institution with experience in monitoring the groundwater will work in collaboration with the national and regional water management groups as well as the National Meteorological Service (ANACM).

E.6. Efficiency and Effectiveness

Economic and, if appropriate, financial soundness of the project/programme

E.6.1. Cost-effectiveness and efficiency

A grant is deemed as the most suitable form of GCF financing for two reasons: (i) GoC's ability to service a GCF loan from budget or capital market sources is severely constrained and (ii) a nationwide water tariff scheme is not yet in place and, when it will be with the project's help, tariffs affordable to Comorian water users are unlikely to allow full capex and O&M recovery.

GoC Budget Considerations

1. Widespread poverty and poor economic prospects significantly constrain Comoros' ability to support additional borrowing. As highlighted in Section C.1, Comoros is one of only 6 countries to be classified as both a Least Developed Country and a Small Island Developing State. It is one of the poorest countries in the world, with an estimated 80% of the rural population considered poverty-stricken and 46% living in absolute poverty (<\$1.25/person/day). In recent years, structural constraints (e.g. in the electricity sector) and slower-than-expected implementation of the public investment program have resulted in economic stagnation, which is particularly bad considering a population growth rate of 2.5%.
2. Large budget and current account deficits make the payment of interest on sovereign debt, let alone repayment of amortization tranches, prohibitive. The budget of Comoros has been in significant deficit in recent years and will continue to be so at least until 2020 (last year of IMF projections). The budget deficit was 14.8% in 2016 and is expected to increase to 15.6% in 2017, settling at 11.5-12% in 2018-20. The economic slowdown in 2015/16 and related decrease in tax revenues accentuated deficits and lead to large arrears in the payment of public sector wages. In 2015 Saudi Arabia came to the rescue with a large grant that plugged the budget hole for the year and for part of 2016. As a result, a 12.2% deficit in 2015 turned into a 2.9% surplus and a 14.8% deficit in 2016 was reduced to 6.4%. The IMF assumes the continuation of grants from various sources in the future, but still projects overall budget deficits (7.3% in 2017 and approx. 4% thereafter). Current account deficits (excluding foreign grants) are steadily above 10%, in part due to imports stemming from investment projects in power and telecom and stagnating exports. As a result, Comoros is heavily reliant on external sources of funds to support its domestic growth and development.
3. With constant budget deficits, debt sustainability is already stretched, as proven by a debt relief negotiated in 2013. Unable to service its debt pile, in 2013 Comoros entered into agreements with all but one of its bilateral creditors, receiving irrevocable debt relief that cut nominal external debt from 40.5% of GDP in 2012 to 18.5% at year-end 2013. Since then, Comoros only contracted one new loan at very concessional terms with India for the construction of a power generation plant. Due to the accumulation of budget deficits, however, debt has risen already to over 25% and will continue to do so. The IMF projects debt/GDP ratios of 30% in 2026 and 39.2% in 2036, highlighting the continuing challenges in terms of debt sustainability. As a result, the IMF explicitly recommends that the country only seek external support on concessional terms.
4. Comoros has a very shallow financial system and no capital markets, limiting the domestic sources of finance for both government and private projects. Eight financial institutions operate in Comoros: four commercial banks, three micro-finance institutions and the postal bank. Their combined loans were approx. 25% of GDP in 2015 – higher than the average Sub-Saharan low-income country, but lower than the average SSA frontier country. Most loans are to the private sector – lending to the public sector is very limited.
5. Despite its severe fiscal constraints, GoC has committed to co-finance part of the initial project capex, all O&M costs and replacement capex over the 25-year project period and part of the project's soft components. Over the project period, the total financial burden for GoC would be US\$25.2 million. As previously discussed, introducing tariffs outside of the current limited MA-MWE user base in Moroni and determining their structure and affordable levels will require significant technical and regulatory work, helped by the project. Should tariff implementation prove difficult, GoC may have to finance O&M and replacement capex entirely from budget – a very significant commitment in light of the debt sustainability issues discussed above.

Project-Specific Considerations

The project involves both hard (water infrastructure and related equipment) and soft (technical assistance) components. Only the former are suitable, in principle, to capex and O&M recovery via tariffs. Hard components involve

total capex of US\$52.9 million, mostly for Output 3 and to a lesser extent Output 2. The capex plan will be implemented over the first 8 project years, with the vast majority falling in the first 5 years. Approximately US\$2.5 million in replacement capex will be needed cumulatively over 25 years. O&M costs will total US\$9.5 million over the 25-year project period. The total financial commitment required by the hard components of the project over 25 years will therefore be US\$65 million (all forecasts assume zero inflation). Other non-capex activities of Output 1 and 2, representing a total financial commitment of US\$5.46 million, involve mostly technical assistance activities that are not prone to revenue generation via tariffs or service fees.

As discussed in Section C.2 (Water Governance), the new Water Code envisages the design and implementation of nationwide water tariffs aimed at ensuring the financial sustainability of the water supply and distribution system. GoC has committed to implementing tariffs in 7 years – roughly coinciding with the time required to implement the project's capex plan. The project will support, through GoC co-finance, the establishment of the appropriate tariff system, making sure that climate risks are taken into consideration.

Under the Water Code, the principle underlying tariff structure and levels is full cost recovery, with costs defined as capex and O&M for urban water infrastructure and at least O&M for rural infrastructure. However, at this stage evidence of tariff affordability is limited to a very small portion of the urban population of Grande Comore. Willingness to pay for the rest of population of Grande Comore and the other islands will need to be tested through comprehensive affordability surveys supported by the project. Given the limited track record of charging and collecting water tariffs and widespread poverty in the three islands, it is doubtful that tariffs can be set immediately in year 7 at levels that ensure recovery of both capex and O&M. Introducing full tariffs upfront would likely encounter significant opposition, resulting in poor collection levels and undermining the long-term success of the tariff scheme. It is more likely that tariffs in the first years will be sufficient to cover for O&M, and only progressively over the 25-year project period they can be raised to levels that may also cover capex.

Specifically, on tariff affordability it is worth highlighting the following issues:

Only 4,225 households (30% of total) in the capital Moroni are connected to a water distribution network providing treated, drinkable water. This represents less than 10% of the total population of Comoros. The Moroni water network is managed by national utility MA-MWE, which charges tariffs on a per cubic meter basis. The average household water bill is around US\$4-5 per month, or approx. 8% of the average wage – a high percentage, especially for lower-income households. Only 34% of water volume is metered and tariff collection rate is a low 64%. The Moroni water distribution network is under-invested and poorly maintained, with leakage as high as 50%.

Only 1,275 households in Anjouan and 470 in Moheli get treated water. An additional 5,590 households in Anjouan and 2,650 in Moheli have access to untreated water through infrastructure managed and maintained by two water management committees, UCEA in Anjouan and UCEM in Moheli. These committees were set up by a French NGO in 1997 and subsequently reorganized under an AFD-sponsored project. Tariffs were introduced to recoup capex and O&M costs and make the committees financially self-sustaining, but with little success. UCEM set tariffs at a fixed €3/month/household, regardless of volumes used, and targeted 2,500 paying households. It only managed to connect 1,700 households and is therefore unable to cover operating costs. This in turn results in poor maintenance and service quality, which reduces willingness to pay – 3 out of 4 households occasionally disconnect from the network. UCEA started a small network that distributes treated water in the Sima area of Anjouan three years ago. Tariffs were initially set at a level that would allow for capex and O&M recovery, but users refused to pay. One year ago tariffs were lowered to €2/cubic meter. In other areas of Anjouan, ACEA distributes untreated water for free.

The remaining 90% of the population of Comoros, including the entire rural area of Grande Comore, uses free, untreated water from various sources, including: 25% collects rainwater through cisterns (especially in Grande Comore where surface water is very scarce), 40% uses house fountains and 25% communal fountains. Some of this water is distributed through small, independent community networks.

During the dry season, a minority of households resorts to purchasing water by the jerry can or having water delivered by truck. The former are mostly rural households that commute to Moroni to purchase treated water. Jerry cans are very expensive. The high cost of water purchased by the jerry can or delivered by truck is not indicative of widespread

capacity to pay for water distribution. Should a household fulfill its entire water needs for the dry season with jerry cans, at the recommended per capita consumption of 35 l/day, the bill could run up to US\$200/year. Only 7% of households in Comoros can afford to buy jerry cans and, when they do, they purchase highly rationed volumes, well short of optimal daily consumption. Water delivery by truck is even more expensive (US\$24 per cubic meter in Moroni) and affordable only to the wealthiest 3% Comorians.

In the Comorian economy, there are no large, commercial water users with greater capacity to pay for water. 70-80% of the Comorian population consists of smallholder, subsistence farmers. Comoros exports three agricultural commodities – ylang-ylang, cloves and vanilla. None of these, however, is grown in large commercial farms (there are large exporters but they purchase the commodities from many small farmers). The tourism industry is underdeveloped: only one large hotel is present in Moroni. With such fragmented demand, there are no obvious user segments that could afford to pay high tariffs in support of the planned infrastructure investment. Water tariffs are currently the same for households and commercial users. Only some ylang-ylang distilleries in Anjouan pay higher tariffs since, for environmental reasons, separate distribution networks had to be set up for them; however, they represent only 3% of total water consumption in Comoros.

Tariff and financial IRR simulations

In the absence of nationwide affordability data, the project design team has run simulations based on two scenarios: (i) the current water tariff of US\$0.53/m³ charged by MA-MWE in Moroni is applied to the entire user base of the project from year 7 (low likelihood scenario) and (ii) tariffs start in year 7 at two thirds of the current Moroni level and subsequently increase every 5 years by 20% (higher likelihood scenario). If GoC were to fund the entire project – initial capex, replacement capex, O&M – on its own, without any concessionality, it would realize a negative IRR of -8.5% (full tariffs scenario) to -8.7% (progressive tariffs scenario). In other words, the government would have to step in and support the project with additional budget resources. Since the IRR is negative, lending to the project – even at GCF's very concessional terms – would only widen the funding gap by adding interest expenditure.

This funding proposal envisages a GCF grant of US\$37 million dedicated to the implementation of the capex plan in years 1-8. FADES and China Geo-Engineering will provide additional grant and in-kind co-financing for the amounts of US\$293,363 and US\$ 1.9 million, respectively, also dedicated to capex funding in years 1-8. GoC will take charge of the remaining US\$11.4 million capex, as well as US\$1.1 million in replacement capex and US\$10.9 million in O&M over the 25-year period (assuming no inflation) – a total funding contribution to the hard components of the project of US\$23.4 million. With the GCF grant the IRR would be just above breakeven: 2.0% in the full tariffs scenario and 0.2% in the progressive tariff scenario.

From a sustainability standpoint, most of the infrastructure installed has a lifetime exceeding 25 years. As a result, over the long-term, tariffs will need to cover mostly O&M and limited capital replacement. This should ensure the long-term financial sustainability of the project with limited need for support from the central government's budget or new concessional sources.

Please refer to Section E.6.3 for a detailed discussion of the financial IRR methodology.

E.6.2. Co-financing, leveraging and mobilized long-term investments (mitigation only)

N/A

E.6.3. Financial viability

Please specify the expected economic and financial rate of return with and without the Fund's support, based on the analysis conducted in [F.1](#).

Financial rate of return

Financial IRR calculations are based on the following assumptions:

- Capex, replacement capex and O&M schedules for the 25-year project period as presented in Section I Annex III.b
- The number of water users served by the new infrastructure increases progressively from 136,000 to approx. 450,000 by year 25 (~60,000 households)
- Each individual consumes 30 liters of treated water per day (short of the 35 liters recommended amount, to reflect existing water sources such as rooftop tanks)
- Comorians take advantage of treated water distribution only during the dry period, currently lasting six months but expected to increase to seven months by year 10 due to climate change. During the wet season, Comorians use free untreated water from rooftop tanks or fountains
- In the full tariffs scenario, all users, rural and urban and in all three islands, are charged US\$0.53/m³ starting in year 7. This is the same tariff currently charged by MW-MWE for the fraction of the population served in the capital city of Moroni. As discussed in Section E.6.1, this scenario is considered unlikely due to water affordability issues
- In the progressive tariffs scenario, all users, rural and urban and in all three islands, are charged US\$0.35/m³ for the first five years of the new tariff regime starting in year 7. This is equivalent to two thirds of the current Moroni tariff and would allow coverage of O&M over those years. Subsequently, tariffs will increase by 20% every five years (since the model assumes zero inflation, this increase is in effect above inflation). This scenario is considered more likely from a water affordability standpoint. Until tariffs catch up and ultimately surpass the level of US\$0.53/m³ assumed in the full tariffs scenario, GoC would realize lower cash flows than in the previous scenario

Without concessionality, GoC would fund the entire project – initial capex, replacement capex, O&M – on its own, for a total investment of US\$63.6 million over 25 years. It would realize a negative IRR of -8.5% (full tariffs scenario) to -8.7% (progressive tariffs scenario).

With concessionality, GoC would fund part of the initial capex (US\$11.4 million out of US\$51.5 million) and all replacement capex and O&M, for a total investment of US\$23.4 million over 25 years. The remainder of the initial capex in years 1-8 would be covered by the GCF grant (US\$37 million), FADES grant (US\$293,363) and China Geo-Engineering in-kind (US\$1.9 million). The IRR on the GoC investment would be just above breakeven: 2.0% in the full tariffs scenario and 0.2% in the progressive tariff scenario.

In a more optimistic scenario where Comorians take advantage of treated water distribution for the entire year (not just in the dry season), generating higher tariff revenues, the IRR without grants would range between -2.9% (full tariffs) and -3.6% (progressive tariffs) – again highlighting the need for concessionality. The GCF/FADES/China Geo-Engineering grants and in-kind co-financing would bring the IRR into positive territory: 11.0% with full tariffs and 7.9% with progressive tariffs.

E.6.4. Application of best practices

Best practice climate resilience approaches have been considered and applied to the Comoros water sector, as well as recent SIDS specific Resilient WASH guidance recently developed in the Pacific Region by UNICEF. These approaches have then been tailored to the specific Comoros context, including recognizing and incorporating some specific indigenous adaptation approaches.

Water supply side approaches include increasing access to drought water supply through increased abstraction, as well as increasing water storage to allow for dry season shortfalls, system storm outages and flexibility to use the water

supply infrastructure from multiple sources to supply multiple supply zones. This flexibility includes the opportunity to access different water sources and resources at different times, *i.e.* to have multiple resource schemes *i.e.* available to exploit groundwater and surface water.

Water demand side approaches to reduce and/or suppress water demand, include leakage reduction, water conservation awareness campaigns, and tariff structures that penalize increasing usage.

In addition, it is more recently (and not so widely) recognized that climate resilience requires an understanding of the water resources vulnerability to climate extremes, including likelihood of low stream flows, saline intrusion of groundwater, reducing borehole yields, and increasing raw water turbidity during and after storm events – often with higher pollution levels. To manage these issues requires investigation of the water resources to understand them and then monitor their condition before and during climatic extremes. The IPCC itself⁴⁷ recognizes this as an under-appreciated issue. These are included in the proposed project.

Hazard-based (*i.e.* avoiding or reducing exposure to the hazard *e.g.* flood flow), vulnerability-based (*i.e.* protecting the asset *e.g.* stream intake), adaptive-capacity (*e.g.* forecasting and utility preparedness) and policy-based CCA approaches are all included in the project design.

The proposed project includes adaptive management and water resources and watershed approaches, recognizing insufficient data exists for scenario-based approaches and that water supply infrastructure improvements alone will not resolve water shortages.

In summary, it is clear from the above that the Project is following best practice of combining water supply side and water demand side improvements, investing in water resources management, adopting policy, institutional and technical scale approaches, including watershed adaptation, whilst building adaptation capacity at national (*e.g.* meteorological), institutional (*e.g.* infrastructure resilient planning) and community levels (*e.g.* rationing and agricultural water use efficiency).

The Pacific Region is arguably more cohesive as a SIDS region than the Indian Ocean, and as such develops regional approaches to addressing some of these knowledge and capacity shortfalls. UNICEF Pacific commissioned a WASH Resilience Policy Review⁵⁶ of 5 Pacific SIDS in 2015. This serves as a useful synopsis of current and best practice in 3 LDC SIDS in the region – countries with very similar challenges to Comoros – another LDC SIDS.

This work recognized the limited technical, institutional and financial barriers to sustaining climate resilient WASH in these countries, and focused on identifying optimum approaches to assessing and reducing water sector climate risks in these data scarce environments further restricted by little adaptation capacity. The work identified Drinking Water Safety and Security Planning (DWSSP) as the most effective approach, using a risk-based approach to prioritize the deployment of scarce technical, management and financial resources to address the most significant risks.

This approach is an adaptive management approach, which includes IWRM when expanded to include the catchment areas.

The above SIDS water sector resilience best practice has been applied to the specific context of the Union of Comoros and in particular tailored to the different water resources occurrence on each island, and their vulnerability to climate risks.

At the national scale, the importance of ensuring a strong policy level enabling environment is evident in the proposal design, which for the Comoros context means focusing on integrating and mainstreaming climate risk reduction into the recent legislative reforms articulated within the Water Code. The proposal goes beyond this however recognizing the lack of institutional adaptation capacity to be a risk to Code implementation, and therefore explicitly supports institutionalizing of the future climate risk components of the Code within national and island scale institutions (water

⁵⁶ GWP Consultants, 2016. Pacific WASH Climate Resilience Review. UNICEF Pacific.

sector planning), as well as water user associations/community management entities (water sector operation & maintenance) – the institutional aspects of DWSSP.

The proposal also recognizes the SIDS vulnerability of the water resources and watersheds in Comoros to drought, flooding and storm event erosion, and specifically focuses project resources on watershed scale IWRM, watershed community adaptation capacity building and eco-system based adaptation as a means to protect dry season flows, reduce flood risk and both reduce turbidity and increase natural clarification.

In parallel, the proposal provides specific resources to the investigation of the nation's groundwater and surface water resources, and the creation of a national climate and rainfall forecasting capability, the development of early warning systems/tools, and developing sector specific water & climate guidelines and tools to support and advise on adaptation approaches and responses. This work will allow existing water resources to be exploited more carefully during climatic extremes, not used when too vulnerable, and help plan further water resource exploitation where unused, under exploited and non-vulnerable water resources are identified.

The last component discussed here is the water supply infrastructure upgrade part of the proposal. This was designed based upon preliminary water balance analysis of rainwater, groundwater and surface water resources availability and knowledge, to inform sustainable risk resilient water supply optioneering and design approaches.

Household and community rainwater harvesting was evaluated and found to be drought vulnerable due to the limiting roof sizes of the houses, not because of tank size. Without a national property construction programme, further investment in household rainwater harvesting would not to improve drought resilience.

In Grand Comore, the fresh groundwater resource is not uniform nor well understood, but years of successful operating in some locations of existing boreholes with high yields without increasing salinity confirms groundwater is resilient to drought periods at least in some locations. Existing boreholes with acceptable salinity will be evaluated to assess their risk to saline up-coning through controlled pumping trials and their pumping regime altered so as to provide maximum resilience.

New production boreholes will be drilled once exploration drilling confirms fresh groundwater to exist, and used to replace or blend with more saline groundwater and/or used to reach high elevation villages with currently no access to groundwater and sole reliance on rainwater.

Groundwater monitoring equipment (e.g., multi-level piezometers and conductivity equipment) will be installed to measure the freshwater lens (a technique used across the Pacific SIDS) to ensure that groundwater is extracted sustainably and that aquifers are effectively recharging after rainy seasons. Preliminary groundwater investigation will also be undertaken on Anjouan and Moheli to confirm for the first time the presence of viable groundwater resources on these islands.

The groundwater abstraction water supply networks will be fitted with water tanks. These tanks will hold 1-2 days of water demand. These are used to ensure the boreholes can pump at the lowest uniform flow rate – which minimizes the risk of saline up-coning from having to pump at peak water demand rates (peak flow rates are typically 3-4 times average daily demand rates).

Storing the water for 1-2 days does however require the water to be treated, hence the water treatment systems are only required in order to achieve deliver the adaptation response to drought periods. The treatment is also necessary due to the risk of storm water run-off entering the existing pumping wells – as happened in April 2012 for the capital Moroni water supply wells – and hence is also a climate resilience measure to storm periods.

On Anjouan and Moheli the groundwater resources are not exploited although they are believed to support the dry season flows in the watersheds. Consequently, nearly all water supplies currently use stream intakes. These intakes are highly vulnerable to storm event erosion associated turbidity and its associated pollution, as well as dry season low and no flow periods.

Additional sedimentation tanks will be placed near the river intakes to reduce turbidity during and after intense rainfall events, as well as valves and diverters to prevent storm flows entering the intake pipes.

Filtration and disinfection systems will be upgraded to deal with the increased turbidity and BOD loading associated with the increased flow and turbidity associated with increased frequency and severity of storm events. These are therefore an adaptation response.

As with the borehole supplies, the stream intakes will have 2-3 days of water storage within the supply system. This allows for the intakes to be closed for a day or two if the stream water is excessively turbid, as well as provides backup storage, should an intake or transmission pipeline be damaged by storm flows or wind damage.

To provide an evidence base for supply scheme performance and checking for water leakage, as well as promoting water conservation, particularly during extended dry seasons, water flowmeters will be placed in key points throughout the distribution networks.

Finally, on Grand Comore the traditional technique of creating rainwater impluviums from lining volcanic cinder cones will be used to collect rainfall for non-potable uses – specifically irrigation for food crops. This removes the pressure of irrigation from potable water supplies whilst allowing rainwater to be captured without having to consider treatment. In addition, more conventional livestock troughs will be provided across the country to reduce the risk of livestock mortality. These agricultural sector interventions reduce climate risks to food security, whilst promoting water efficient farming approaches to a wider agricultural stakeholder community.

These water supply infrastructure upgrades, informed as they will be by both the earlier capacity building and institutional strengthening of the water supply organizations, as well as the water resources investigations and monitoring networks, provide the means to operationalize the climate risk reduction assessment and planning procedures of DWSSP into operation and maintenance tasks, staffing and monitoring review processes, thereby demonstrating improved reliance in supplied water quantity and quality thereby justifying sustainable levels of tariffs to allow on-going water sector climate risk resilience and water security.

E.6.5. Key efficiency and effectiveness indicators

GCF core indicators	Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation only)
	N/A
	Expected volume of finance to be leveraged by the proposed project/programme and as a result of the Fund's financing, disaggregated by public and private sources (mitigation only)
	N/A

Other relevant indicators (e.g. estimated cost per co-benefit generated as a result of the project/programme)	
---	--

F.1. Economic and Financial Analysis

Approach and Methodology

The economic analysis of the proposed project was carried out in accordance with the *Guidelines for the Economic Analysis of Projects of United Nations Development Program*.⁵⁷ The economic efficiency of the investment was determined by computing the economic net present value (NPV) with an assumed 10% discount rate, and the economic internal rate of return (EIRR). For consistency purposes, all proposals developed with the support of UNDP have thus far opted to use a 10% discount rate, in line with the existing practice of multilateral development banks.

Economic values (costs and benefits) are all measured in real terms of 2017. Economic costs of the project are net of taxes, duties, and price contingencies. Furthermore, the analysis assumes a shadow wage rate of 1.00 for unskilled and semi-skilled labor in Comoros. Provided that the economic cost of labor in Comoros is expected to be lower than the market wage rate (financial cost), we expect this assumption leads to significantly over-estimating the economic cost of the project, and under-estimating the true net economic value of the project. For example, in a recent (2016) cost-benefit analysis of an energy efficiency project in Comoros, the World Bank used a shadow wage rate factor of 0.75 for unskilled labor, reflecting a judgment that the labor component of the project had a lower opportunity cost than is implied by the financial labor cost (as a result of the existence of surplus unskilled and semi-skilled labor in Comoros).⁵⁸ The above assumption allows the use of financial cost as a measure of the economic cost of the project (once again noting that in doing so, the economic cost of the project is over-estimated, and the net present value of the investment is then under-estimated).

As is common when undertaking the economic analysis of investment projects, numerous assumptions were used to delineate the “with project scenario” from the “without project scenario”. These assumptions are presented and discussed below. Assumptions were made so as to under-estimate the true net economic value of the proposed investment project. The analysis covers a period of 25 years, from 2018 to 2042.

Economic Analysis of Proposed Investment

The economic costs of the project are provided in the project budget and summarized in the table below.

Total Cost of the Project per Output per Year

Output	Total	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
Output 1	1,800,163	261,702	421,002	682,788	301,002	60,117	60,117	6,717	6,717
Output 2	5,662,532	363,219	1,151,315	1,673,262	1,033,015	648,115	503,852	246,752	43,002
Output 3	50,056,574	2,919,517	9,203,679	13,620,537	10,100,702	7,997,549	4,268,513	1,788,500	157,575
Mgment	3,232,226	398,006	473,678	473,678	401,966	405,966	347,606	347,606	383,718
Total	60,751,495	3,942,446	11,249,675	16,450,264	11,836,686	9,111,748	5,180,088	2,389,576	591,013

At least two important benefits are included in the economic analysis.

⁵⁷ UNDP. 2015. Guidance on the conduct and reporting of the Economic and Financial Analysis of Climate Change Adaptation and Mitigation Projects and Programmes. UNDP.

⁵⁸ The economic analysis is available at <https://www.adb.org/sites/default/files/linked-documents/48224-002-ea.pdf>. The shadow wage rate is presented in paragraph 10.

First, Comoros imports large quantities of medicine directly associated with the treatment of gastro-intestinal diseases. According to the sole importer of such medicines in the country (Ministry of Health), Comoros imported approximately \$8 million of medicines for the treatment of gastro-intestinal diseases in the 15 target zones of the project representing an average cost of approximately \$24.50 per capita. The mitigation of future treatment costs is a significant benefit of the project.

A second important benefit is estimated by the economic willingness-to-pay for water. In the times of dry period, households have access to limited supply of water, including bottled water and truck water, with truck water being the least expensive of these sources. The price of truck water varies across project zones but on average is estimated to be 11,000 CF or approximately \$25 per cubic meter.

In the base case (with a 20% reduction in the number of cases of gastro-intestinal diseases and a WTP of \$8 per m3), the NPV of the project is estimated to reach \$9.8 million, with an IRR of 13.2% (see worksheet "Benefits", row 40 to 47 thus indicating the economic desirability of the proposed investment project.

In this base case, a 20% increase in total costs would reduce the NPV and IRR to \$1.2 million and 10.3% respectively. The NPV of the project would become negative and the IRR fall below the required 10% (discount rate) if the project were to experience an increase in cost of 23% (trigger value for cost increase assuming no change in benefits).

As indicated earlier, a sensitivity analysis was also conducted using alternative values for the assumed percentage reduction in the number of gastro-intestinal diseases (ranging from 15% to 25%) and for the assumed WTP (ranging from \$6 to \$10 per m3). The table below presents the NPV and IRR for the various possible combinations of these values. As shown in the table, the economic efficiency of the project remains robust and the NPV positive except in the only scenario where both mitigation of health costs and the WTP are both assumed simultaneously to be at their lowest values. Given that all of the remaining assumptions in this economic analysis lead to under-estimating the true economic value of the proposed project, there is a very high degree of confidence that the proposed project is economically efficient.

NPV and IRR for Different Scenarios of Mitigation and WTP

	Mitigation of health costs		
Willingness to pay	15% mitigation	20% mitigation (base case)	25% mitigation
\$10 per m3	NPV = 15.6million IRR = 15.1%	NPV = 19.6 million IRR = 16.2%	NPV = 23.2million IRR = 17.3%
\$8 per m3 (base case)	NPV = 6.3 million IRR = 12.0%	NPV = 9.8 million IRR = 13.2%	NPV = 13.5 million IRR = 14.3%
\$6 per m3	NPV = -3.4 million IRR = 8.8%	NPV = 0.2 million IRR = 10.1%	NPV = 11.2 million IRR = 11.2%

F.2. Technical Evaluation

The proposal takes a SIDS specific approach to strengthening the climate resilience of the water sector of Comoros.

This approach recognizes both the inherent natural vulnerability of SIDS due to their small land masses, fragile freshwater resources, lack of natural water storage and direct exposure to climatic risks (cyclonic flooding, storm surges et al), as well as their vulnerability resulting from isolation, small population base, and consequential limited technical

expertise, high costs of transaction, limited access to global markets and hence weak economies, which reduces their capacity to adapt.

This lack of hydrological, technical, financial and institutional resources and capacity creates particular challenges to managing and adapting to climate change and reducing the risk associated with these climatic hazards.

The approach focuses on a systematic risk-based and evidence-based approach to water supply risk reduction by maximizing risk reduction based on prioritized ranking of risks to ensure the optimum resilience outcome is achieved with the scarce national resources available. This approach is currently being promoted by UNICEF Pacific and is an extension on the WHO Water Safety Planning (WSP) approach. UNICEF Pacific refers to this approach as Drinking Water Safety and Security Planning (DWSSP).

DWSSP promotes delivery of demonstrable improvements in water (supply, resource and catchment) management, to increase climate resilience of existing water supply systems, before introducing infrastructure resilience upgrades. This both provides an incentive to the beneficiaries to deliver the initial improvements and increases the likely resilience of the later infrastructure investment having already institutionalised climate risk reduction practices.

In order to address the technical capacity and financial resource challenges, it is essential the proposal includes a national enabling environment component to ensure mainstreaming and integration of climate risk reduction into water supply, water resources and watershed management institutions, work programmes, budgeting and staffing capacity. In Comoros this will occur by first integrating climate adaptation into the national water legislation reform process on-going in the country, and then secondly committing project resources to ensure the water supply providers and associated line ministries and island agencies implement it through a combination of technical capacity building on risk reduction planning, water supply and water demand side implementation, monitoring and delivery to increased human resources, as well as supporting tariff reforms designed to support water supply provision during climatic extremes.

Of particular importance to SIDS, the project recognizes and focuses on the need to protect and sustain the fragile water resources on the islands, which respond so rapidly to extended dry periods and storm events. This includes both water resources investigation, and monitoring of all water resources types available – groundwater, surface water and rainwater – including those resources not currently exploited e.g. groundwater in Anjouan and Moheli – enabling multiple source infrastructure schemes to be considered and developed. Groundwater in particular is considered to be less vulnerable to climatic extremes.

The project also includes IWRM watershed management commitments to reduce watershed derogation so as to retain dry season flows, attenuate flood flows, reduce erosion and turbidity and improve eco-system based clarification of the raw waters used for water supply.

The IWRM watershed approach also recognizes the close link to climate-risk induced food security, and the project explicitly supports improvements in dry season food production and water use efficiency as well as anti-soil erosion measures. The IWRM approach is particularly important non-infrastructure approach in human resources poor environments, encouraging both the use of the public's capacity as well as their awareness and support to reduce the need for regulatory enforcement approaches.

The IWRM approach also provides a coordinated mechanism to try to manage floods, erosion and droughts in a more holistic approach, e.g. use of retention structures to retain storm run-off and encourage infiltration into the underlying aquifers.

Given the rapid response of the water resources and watershed to climatic extremes, climate adaptation through risk reduction requires forecasting and early warnings of floods and droughts, which themselves require monitoring of water resources to inform the forecasts. The project is committing resources to develop a national forecasting capacity, link it to the regional Indian Ocean cyclone tracking agency, as well as developing risk reduction planning and response guidance for each sector.

Lastly the project supports water supply scheme resilience upgrades including: providing communities with water supply schemes fed by multiple sources, that are protected from flood damage, have greater flexibility, avoid unnecessary climatic risks (e.g. avoiding pipelines crossing streams and peak daily demand pumping of wells), distinguish between potable and non-potable uses, and are capable of treating water to overcome increasing levels of turbidity, but does so as part of a systematic risk reduction process for each water supply scheme, so as to ensure maximum resilience for the project and management effort committed to achieve it.

Notably, careful use of daily water balances has revealed upgrades to household rainwater tanks will not improve drought resilience, as the controlling factor is roof size not tank size – hence household rainwater harvesting improvements are not included in the project approach.

F.3. Environmental, Social Assessment, including Gender Considerations

The project has been screened against UNDP's Social and Environmental Standards Procedure (SESP) and deemed to be a moderate risk (Category B) project. A detailed discussion of this assessment is provided in the ESMF in Annex VI.

The ESMF identified 4 Moderate Impact risks and 11 Low Impact risks related to project implementation, after risk mitigation measures have been included. These are identified below:

Medium Impact Risks identified are:

- Disturbance to watersheds and livelihoods due to management changes;
- Water supply construction phase water flow and quality disturbance and derogation;
- Water supply construction phase land degradation and erosion;
- Over-abstraction of water resources due to increased water supply networks;

Low Impact Risks identified are:

- Limited water utility human resources for and insufficient gender representation in capacity building;
- Damage to aquatic habitat during watershed habitat enhancement activities;
- Accidental spillage of water supply treatment dosing chemicals;
- Seismic, volcanic and landslide events damaging infrastructure;
- Generation of construction waste;
- Greenhouse gas production due to increased use of water supply pumping;
- Failure of power supply systems;
- Climatic risks – in particular flood and storm damage;
- Construction impacts on threatened species;
- Poor data collection and poor equipment selection;
- Inappropriate meteorological training, staff retention and inadequate gender representation

The ESMF identifies impact a suite of mitigation measures to address each of the above potential impact risks.

ESMF compliance will be monitored using indicators on: Surface Water; Groundwater; Ecology; Erosion, Drainage & Sediment Control; Social Management; Waste Management; Noise & Vibration; Air Quality; and Emergency Management Measures.

The ESMF makes the following conclusions and recommendations:

- i) None of the interventions will require displacement of people;
- ii) None of the interventions will be conducted in protected areas or sensitive locations;
- iii) Appropriate erosion and sediment control will need to be undertaken during all stages of the project; and
- iv) There must be no release of pollution and/or chemicals as a result of the project

The EMSF will be implemented by the implementing agency DGEF.

Gender Considerations

The project development has included Gender, Equity and Social Inclusion (GESI) approaches. A GESI Assessment has been completed for the proposed project and a GESI Action Plan prepared (see Annex XIIIc).

The Assessment identified the primary role of women in provision and management of household water supply and water usage, as well as the constraints upon gender equity in the country more generally. The Assessment identified lack of water availability impacted on women and girls specifically through time taken to collect water (2 hours per day in rural areas), loss of income and education opportunities, as well as gender specific sanitary needs.

The Assessment found the project will therefore benefit women at the holistic and water supply specific levels as follows:

In this project, the expected outcomes in relation to gender include the following:

Global outcomes:

- 1) Women's reduced vulnerability to the effects of climate change and better resilience to handle these impacts.
- 2) Increased awareness of the impacts of climate change as well as the adaptation and mitigation of climate change.
- 3) Increased awareness of the importance of gender equality and women's involvement in climate-related projects and activities.
- 4) Improvements in health and wellbeing.
- 5) Improved livelihoods.
- 6) Increased empowerment of women in the concerned communities.
- 7) Improved response to women's needs and priorities.
- 8) Contribution to the promotion of gender mainstreaming in the society.

Specific outcomes:

- 1) Reduced time for water collection for women and girls as compared to before the project.
- 2) Reduced incidence of waterborne diseases.
- 3) Increased number of women who will receive training related to water collection and management.
- 4) Improved involvement of women in community decision-making as a result of women's participation in Water Management Committees and Integrated Water Resource Management Committees.
- 5) Women's increased access to water planning information to be able to anticipate water supply in relation to subsistence agriculture.
- 6) As a result of improved water supplies, possibilities to improve sanitation issues in communities and schools.
- 7) Increased awareness and capacity to handle local water management systems.
- 8) Increased awareness of sustainable water management practices.
- 9) Increased implication of local women's organizations in water and climate-related projects.

The following gender specific inclusions have been adopted by the project design:

- Women included in national, island and service provider scale Water Code CCA consultations, information exchange and awareness raising through the help of women's associations and representatives
- Gender specific water CCA issues included in revised national water code
- Ensure women fully involved in water supply management committees including training and women specific issues included in the Water Security Plans
- Ensure women fully represented on IWRM committees, receive equitable training and have women prioritized issues included in the watershed adaptation management plans

- Ensure women equitably involved in IWRM plan implementation, monitoring and evaluation
- Include gender issues in IWRM public awareness campaigns
- Ensure female staff included in hydrogeological investigations of water resources
- Include women equitably in the water supply upgrade design process
- Ensure women fully represented and empowered on water management committees and water user associations
- Ensure women equitably involved in national water resources monitoring programme
- Ensure women fully represented on climate forecasting training and operationalisation
- Ensure women fully involved in design of sectoral water CCA forecast preparedness & response products and tools
- Ensure women fully involved in workshops and public engagement & awareness raising on CCA preparedness

The GESI Plan includes monitoring indicators and targets, which will be used to track and if necessary re-dress gender equity during the project implementation.

F.4. Financial Management and Procurement

The financial management and procurement of this project will be guided by UNDP financial rules and regulations available here: https://info.undp.org/global/documents/frm/Financial-Rules-and-Regulations_E.pdf. Further guidance is outlined in the financial resources management section of the UNDP Programme and Operations Policies and Procedures available at <https://info.undp.org/global/popp/frm/Pages/introduction.aspx>.

UNDP has comprehensive procurement policies in place as outlined in the 'Contracts and Procurement' section of UNDP's Programme and Operations Policies and Procedures (POPP). The policies outline formal procurement standards and guidelines across each phase of the procurement process, and they apply to all procurements in UNDP. See here: <https://info.undp.org/global/popp/cap/Pages/Introduction.aspx>.

The project will be implemented following the National Implementation Modality (NIM) following NIM guidelines available here: http://www.undp.org/content/dam/undp/library/corporate/Programme%20and%20Operations%20Policies%20and%20Procedures/NIM_for_Government_english.pdf.

UNDP will ascertain the national capacities of the implementing partner by undertaking an evaluation of capacity following the Framework for Cash Transfers to Implementing Partners (part of the Harmonized Approach to Cash Transfers - HACT). All projects will be audited following the UNDP financial rules and regulations noted above and applicable audit guidelines and policies. The NIM Guidelines were corporately developed and adopted by UNDP, and are fully compliant with UNDP's procurement and financial management rules and regulations.

The national executing entity is DGEME- also referred to as the national 'Implementing Partner' in UNDP terminology. In legal terms, this is ensured through the national Government's signature of the UNDP Standard Basic Assistance Agreement (SBAA), together with a UNDP project document which will be signed by the Implementing Partner to govern the use of the funds. Both documents require compliance. Prior to signature of the project document, all national Implementing Partners need to have undergone a Harmonized Approach to Cash Transfer (HACT) assessment by UNDP to assess capacities to implement the project. During implementation, UNDP will provide oversight and quality assurance in accordance with its policies and procedures, and any specific requirements in the Accreditation Master Agreement (AMA) and project confirmation to be agreed with the GCF. This may include, but is not limited to, monitoring missions, spot checks, facilitation and participation in project board meetings, quarterly progress and annual implementation reviews, and audits at project level or at implementing partner level on the resources received from UNDP.

The Harmonized Approach to Cash Transfer (HACT) framework consists of four processes: (1) macro assessments; (2) micro assessments; (3) cash transfers and disbursements; and (4) assurance activities. Assurance activities include planning, periodic on-site reviews (spot checks), programmatic monitoring, scheduled audits and special audits. During micro-assessment, there can weaknesses identified for which actions are required to addresses the gaps. When a spot check finds that the gaps are not addressed it will mean that the level of assurance activities must

remain higher and modalities of engaging with that implementing partner must be reviewed if necessary. All details are available here: <https://undg.org/wp-content/uploads/2015/02/2014-UNDG-HACT-Framework-English-FINAL.pdf>. All GCF resources will be made available to the Implementing Partner under National Implementation Modality, less any agreed cost recovery amount. Nevertheless, in this particular project there will be no advance or cash transfer to the government but disbursement will be done through direct payment requests. Any additional requirements will be as in accordance with the AMA as and when it is agreed.

The project will be audited in accordance with UNDP policies and procedures on audits, informed by and together with any specific requirements agreed in the AMA. Per the current audit policies, UNDP will be appointing the auditors. UNDP scheduled audits are performed during the programme cycle as per UNDP assurance/audit plans, based on the implementing partner's risk rating and UNDP's guidelines. A scheduled audit can consist of a financial audit or an internal control audit. A draft procurement plan will be provided (which will be further discussed and revised prior to UNDP Project Document signature).

G.1. Risk Assessment Summary

The Environmental and Social Risk Assessment discussed earlier has highlighted risks associated with the impact of the project on the environment and communities. In addition, there are risks to the project implementation itself. These need careful evaluation, monitoring and mitigation due to the inherent vulnerabilities of Small Island Developing States.

In summary these include risks magnified by Comoros being a SIDS including: Remoteness of Project Country, Remoteness of Project Sites, Limited Qualified Local Staff; and Exposure to Extreme Weather Events; and well as more generic project risks of Community Ownership, Political Commitment, and Land Disputes (although the latter is also magnified in SIDS where land resources are limited, population pressures high and land ownership often traditional.

G.2. Risk Factors and Mitigation Measures

Selected Risk Factor 1: Remoteness of Beneficiary Country

Description	Risk category	Level of impact	Probability of risk occurring
International supply chains are long, expensive and prone to shipping delays. Country staff have little/negligible experience of international procurement procedures (ICB)	Technical and operational	High (>20% of project value)	Medium

Mitigation Measure(s)

The international logistical and procurement issues will be mitigated through a combination of the provision of appropriate logistics and procurement technical assistance to the PMU/PCU to ensure supply chains and times are well understood, delivery contingency plans are prepared, and detailed procurement tasks and timeframes are incorporated and integrated into the project work plans.

Risks can be further mitigated through use of carefully prepared pre-qualification criteria for suppliers, requiring demonstrable international experience of trans-Indian Ocean product & material delivery and incentivizing suppliers to use reputable shipping agents, as well as scheduling deliveries to coincide with more convenient times of year (away from holiday periods and monsoon seasons).

Selected Risk Factor 2: Remoteness of Project Sites

Description	Risk category	Level of impact	Probability of risk occurring
The project sites are located across three separate islands as well as some at remote upland locations within these islands, dependent on inter-island transport as well as off-road vehicles and well maintained access tracks.	Technical and operational	High (>20% of project value)	Medium

Mitigation Measure(s)			
The national logistical and access issues will be mitigated through a combination of ensuring the PMU/PCU has adequate access to national logistics expertise, the use of contingency planning, integration of logistical tasks and timeframes into the work plan and sympathetic scheduling of shipment arrivals (i.e. during the dry season). In addition, the project resources will ensure adequate dedicated off-road vehicles are available to project staff, and where local construction contractors are used they will be required to demonstrate experience and resources to cope with difficult terrains as part of the tendering process. If government departments are used evidence of similar experience will be required Where ground conditions are difficult, the project will allow for phasing of construction of access roads and construction platforms in advance of actual infrastructure activities. If necessary, these will be completed the dry season before infrastructure installation.			
Selected Risk Factor 3: Limited Qualified Staff			
Description	Risk category	Level of impact	Probability of risk occurring
Limited in-country technical, managerial and procurement expertise in critical project areas Inappropriate or inadequate delegates at training workshops Conflicting demands on key governmental and non-governmental staff	Technical and operational	Medium (5.1-20% of project value)	Medium
Mitigation Measure(s)			
The risk relates across the project stakeholder groups including governmental departments, non-governmental organisations and even the UNDP office – and reflects the limited on-island population size and exposure to alternative approaches other than business-as-usual. The risks will be mitigated through the following actions: - Careful selection and deployment of appropriately experienced international technical advisors to support knowledge transfer - Early political advocacy and public awareness raising to ensure ministerial, institutional and community support to project objectives - Careful pre-selection and vetting of training opportunities nominees - Use of Training-the-Trainer approaches to increase capacity outreach - Backstopping and buddying of local staff with national and international experts - Careful monitoring and assessment of project delivery - Development of locally delivered and institutionalised training programmes Adopting and monitoring the success of these approaches is expected to reduce the residual risk to low.			
Selected Risk Factor 4: Extreme Weather & Seismic Events			
Description	Risk category	Level of impact	Probability of risk occurring
Extreme weather events restrict monitoring deployment, infrastructure construction and watershed	Social and environmental	High (>20% of project value)	Low

land use activity periods and delay/limit inter-island access. Natural hazards damage and/or destroy pre-existing and or project activities, and create short-term response & recovery priorities for government and communities.			
Mitigation Measure(s)			
The scheduling of field and construction activities has allowed for concentrating activities outside of the wet season and into the dry season – including where necessary pre-construction dry season preparatory works e.g. access road building and material shipping to minimize later impacts on construction during wet periods, with non-field based activities during the wet seasons e.g. training. Early deployment of hydrological monitoring equipment will allow value of hazard and risk mapping & monitoring to be observed during the hazard events – with the project demonstrating the role of adaptation and resilience during extreme events – e.g. use of early warning systems. Contingency planning will ensure option exists for project to focus on Grand Comore alone during periods of rough seas, whilst can focus on Anjouan and Moheli should volcanic activity on Grand Comore become problematic e.g. ash falls.			
Selected Risk Factor 5: Community Ownership			
Description	Risk category	Level of impact	Probability of risk occurring
Communities not committed to effort required to strengthen capacity in adaptive management, preferring government hand-outs and infrastructure gifts	Social and environmental	Medium (5.1-20% of project value)	Medium
Mitigation Measure(s)			
Targeted community involvement already within the project design phase, including signing of agreements to support project. Early scheduling of public awareness campaigns to reach urban and rural population. Deliberate engagement and involvement in IWRM committees and Water User Associations Intentional targeting and involvement of communities in resilience planning, participatory hydrological monitoring, climate adaptation product/tool development. Community involvement in simple construction tasks and training/upskilling on more complex tasks. The residual risk can be reduced to low with careful and sustained engagement and communication with direct and indirect beneficiary communities.			
Selected Risk Factor 6: Political Commitment			
Description	Risk category	Level of impact	Probability of risk occurring
Senior national, island and community political key stakeholders distracted by other priorities, projects and parliamentary cycles as well as shock events.	Other	High (>20% of project value)	Low

Disengagement or hostility to project.			
Mitigation Measure(s)			
Careful engagement with senior government representatives during the project design process. Promotion of project as an inter-sectoral multi-stakeholder approach with equity and equality in its objectives. Transparent and accountable target community selection not influenced by local political preferences. Transparent and accountable project delivery decisions. Strong communication with government through well established and respected UNDP CO staff. Recognition of parliament annual cycle and necessity of timely inputs to governmental decision making bodies.			
Other Potential Risks in the Horizon			
Community and individual ownership of land has potential to delay and impact on project, especially as ransom strips. This issue has been largely mitigated through identification of land ownership, concept design of interventions to maximize use of government land, and signing of Memorandum of Understanding with communities to secure project intervention in their areas.			

H.1. Logic Framework.

Please specify the logic framework in accordance with the GCF's [Performance Measurement Framework](#) under the [Results Management Framework](#).

H.1.1. Paradigm Shift Objectives and Impacts at the Fund level⁵⁹

Paradigm shift objectives

<i>Increased climate-resilient sustainable development</i>	The project aims to increase the resilience of reliable and safe water supply and mainstream climate change risk reduction into water management in 15 of the most vulnerable zones in the Union of Comoros to climate change risks. All water resources, watershed and infrastructure designs and management protocols will account for expected climate change impacts (drought, extended dry periods, intense rainfall events). It is this mainstreaming and integration of climate risk reduction into the 'business-as-usual' water sector planning, budgeting and operating environment that will create the paradigm shift needed to address climate change impacts and achieve a climate resilient water sector.					
Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	

Fund-level impacts

<i>A2.0 Increased resilience of health and well-being, and food and water security</i>	2.3. Number of males and females with year-round access to reliable and safe water supply despite climate shocks and stresses.	Household (HH) surveys.	0 residents in the target zones	150,000 residents in the target zones 76,500 Females 73,500 Males	450,000 residents in the target zones 229,500 Females 220,500 Males	Sufficient rainfall, groundwater and surface water can be mobilized to help achieve water security.
--	--	-------------------------	---------------------------------	---	---	---

⁵⁹ Information on the Fund's expected results and indicators can be found in its Performance Measurement Frameworks available at the following link (Please note that some indicators are under refinement):

[http://www.greenclimate.fund/documents/20182/239759/5.3 -
_Performance_Measurement_Frameworks_PMF_.pdf/60941cef-7c87-475f-809e-4ebf1acbb3f4](http://www.greenclimate.fund/documents/20182/239759/5.3_-_Performance_Measurement_Frameworks_PMF_.pdf/60941cef-7c87-475f-809e-4ebf1acbb3f4)

H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Programme level

Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	
Project/programme outcomes	Outcomes that contribute to Fund-level impacts					
A5.0 Strengthened institutional and regulatory systems for climate-responsive planning and development	5.1: Institutional and regulatory systems that improve incentives for climate resilience and their effective implementation	Review of work plans, annual reports and meeting minutes for institutions forming IWRM Committee	No national and island agencies incentivizing climate resilience and its implementation under the current water sector planning framework	2 National agencies (DGEME and MEAPEA TU) and 2 Island Agencies (DREAs) annual workplan and multiyear investment plans include adaptation measures	4 National agencies (DGEME, CSRH, DGM, DGEF) and 3 Island Agencies (DREAs) annual workplan and multiyear investment plans include adaptation measures	Political commitments remain high for the period of project implementation.
	5.2: Number and level ⁶⁰ of effective coordination mechanisms	Annual reports of IWRM committee, meeting minutes, work plans, APRs, Interim Evaluation Report and Terminal Evaluation Report	Fragmented water governance - no effective coordination mechanism in place Level=1	1 National IWRM coordination mechanism/ committee Level=2	1 National IWRM coordination mechanism/ committee Level=4 3 Island IWRM coordination mechanism/	Political commitments for coordination remain high for the period of project implementation.

⁶⁰ In this context, the "level" of coordination mechanism is expressed in a scale of 1 to 4 with the scale referring to a particular level of effectiveness. (1 = no coordination mechanism; 2= coordination mechanism in place and meeting consistently; 3 = coordination mechanism in place, meeting regularly with appropriate representation (gender and decision making authorities); 4 =coordination mechanism in place, meeting regularly, with appropriate representation, with appropriate information flows and monitoring of action items/issues raised)

					committees Level=4	
A6.0 Increased generation and use of climate information in decision-making	6.2: Use of climate information products/services in decision-making in climate-sensitive sectors	Agency annual reports and surveys, scorecards, staff questionnaires	0%	20 % of the officials using climate information to advise watershed risk reduction plans and drinking water safety and security planning	100% of the officials using climate information to advise watershed risk reduction plans and drinking water safety and security planning	The fluctuation of ANACM staff (and number of staff) does not increase significantly
A7.0 Strengthened adaptive capacity and reduced exposure to climate risks	7.1: Use by vulnerable households, communities, businesses and public sector services of Fund-supported tools (climate products), instruments, strategies and activities to respond to climate change and variability	Agency annual reports, household surveys on climate sound water use practices	0	150,000 beneficiaries using climate information to increase drought resilience and reduce vulnerability to flooding 76,500 Females 73,500 Males	335,000 beneficiaries (using climate information to increase drought resilience and reduce vulnerability to flooding) 170,850 Females 164,150 Males	Targeted population has the necessary access to the communication channels (e.g. mobiles, media)
	7.2: Number of males and females reached by climate-related early	Annual reports of ANACM and water management committees,	0	150,000 residents total in target areas	335,000 residents total in target areas	Disasters to not destroy installed data collection equipment.

	warning systems and other risk reduction measures established/strengthened	Community Questionnaires		76,500 Females 73,500 Males	170,850 Females 164,150 Males	
A8.0 Strengthened awareness of climate threats and risk-reduction processes	8.1: Number of males and females made aware of climate threats and related appropriate responses	Field survey reports/scorecards; Annual reports of ANACM and water management committees	0	150,000 residents total in target areas 76,500 Females 73,500 Males	450,000 residents total in target areas 229,550 Females 220,500 Males	The targeted population has the necessary access to the communication channels (e.g. mobiles, media)
Project outputs	Outputs that contribute to outcomes					
1. Climate informed water supply planning and management	1.1 Number of primary and secondary ⁶¹ water related legislations and regulations mainstreaming climate risks/adaptation	Review of relevant and publicly available legislation and regulations	0	1 Water Code (primary) revised to integrate climate resilient Drinking Water Security and Safety Planning	1 Water Code (primary) formally ratified 2 Climate resilience focused water abstraction and water quality regulation (secondary) drafted and ratified	Utilities fully adopt new management systems to improve service delivery, including O&M planning.
	1.2 Number of water service providers within target zones using Drinking Water Safety and Security Planning	Water User Association and Water Management Committee reports,	0	6 target zone Water service providers using Water Security Plans that	15 target zone Water service providers using Water Security Plans that	Utilities fully adopt new management systems to improve service delivery, including O&M planning

⁶¹ Primary Legislation is created usually by an act (Act) of government and provides definition to an umbrella legal instrument. Secondary legislation (also referred to as delegated legislation or subordinate legislation) is law made by an executive authority under powers delegated from/by an enactment of primary legislation, which grants the executive agency power to implement and administer the requirements of that primary legislation.

		work plans and audits		can accommodate climate extremes (drought & flood)	can accommodate climate extremes (drought & flood)	
	1.3 Percentage of Water Management Committees with women leading discussions on the integration of climate-informed practices into water management	Review of the committee members registry, Water User Association and Water Management Committee Reports	0%	40%	100%	Sufficient women are willing to lead discussions in the committees and communities agree to women having an active role in committees
	1.4 Number of households contributing financially to the cost of climate resilient water services	Household survey; Annual reports of the local water providers	0	0 households contributing financially to the cost of climate resilient water services ⁶²	50,000 households contributing financially to the cost of climate resilient water services	Advisory information is properly targeted, reaches intended audiences and beneficiaries perceive improvement and benefits and have sufficient resources (financial) to priorities and pay for climate resilient water service delivery
2. Climate Informed Water Resources and Watershed Management including forecasting and	2.1 Number of IWRM Management Committees established with a climate resilience	Review of the acts establishing the IWRM committees	No formal IWRM watershed management committees exist.	15 catchment specific committees established that prioritize water	32 catchment-specific committees established that prioritize water	The communities participate with understand and respect IWRM processes.

⁶² Up to year 7 of project implementation, the Government of Comoros will support climate resilient water services; from year 8 onwards, there will be a reduced subsidized tariff, therefore, households will be able to contribute directly to the subsidized tariff

early warnings of climate risks	mandate in target watersheds			sector climate resilience	sector climate resilience	
	2.2 Level ⁶³ of integration of climate information products and services (CIPS) for EWS in watershed management by IRWM management committees	Surveys using scorecard to evaluate the level of integration at MT and EOP, WMC reports	Level=1	Level=2	Level=4	IRWM management committee members are aware of the benefits to using CIPS and accept to integrate them
	2.3 Number of meteorological workers with the capacity to analyse climate information and model flood forecasts	ANACM annual reports.	10 workers with relevant capacities	20 workers with relevant capacities	30 workers with relevant capacities	Enough workers in the ANACM with the appropriate skills level to undertake the trainings
	2.4 Number of female and male farmers receiving advisories for water management	Household and agricultural associations surveys	0	5,000 target female and male farmers receiving advisories for water management 2,550 Female 2,450 Male	10,000 target female and male farmers receiving advisories for water management 5,100 Female 4,900 Male	Agricultural policies and incentives consistent with the watershed action plans
3. Climate Resilient Water Supply Infrastructure	3.1 Number and value of physical assets made more resilient to climate variability and change, considering human benefits,	Field inspection reports; infrastructure assets registry	0	10 covered storage units, 5 treatment systems, 8 new and protected water	30 covered storage units, 14 treatment systems and 22 new and protected	Sufficient rainfall, groundwater and surface water can be collected / mobilized to help achieve water security.

⁶³ Level 1: No integration of climate information products; Level 2: low integration- able to identify the types and locations of climate hazard risks within the catchments; Level 3: medium integration- able to use drought forecasting to increase resilience of watershed management; Level 4: high integration: able to use of flood forecasts to increase resilience of watershed management

	reported and where applicable			sources for a value of US\$ 3,911,551.57	water sources for a value of US\$ 13,989,495.88	
	3.2 Number of households in Grand Comore receiving water throughout in dry season.	Household surveys, water service provider annual reports	0	15,000 total households (of which 7,650 are female headed households) with access to water in dry season	30,000 total households (of which 15,300 are female headed households) with access to water in dry season	Sufficient rainfall, groundwater and surface water can be mobilized to help achieve water security.
	3.3 Number of Households in Anjouan and Moheli receiving drinking water throughout storm events (disaggregated for female headed households).	Household surveys, water service provider annual reports	0	5,000 total households (of which 2,550 are female headed households) with access to water during storm events	20,000 total households (of which 10,200 are female headed households) with access to water during storm events	Sufficient rainfall, groundwater and surface water can be collected/ mobilised to help achieve water security
Activities	Description	Inputs			Description	
1.1. Prepare recommendations and legal guidance on the integration of climate change adaptation into the national (federal) and regional (state) water sector agencies governance	A recent national consultation process has developed a new draft legislative framework for water management in Comoros. The Code will guide the water sector for the next 10 years or more. However, the Water Code does not refer to climate change and does not recognize climate risks as a threat to safe water supply. The project will draft the	1.1.1 Legal consultancy for existing legislation review on Climate Resilience inclusion, drafting of Primary (Water Code) and Secondary legislation, regulations; and 3 rounds of consultation meetings with national agencies. 1.1.2 Consultancy providing recommendations on the climate resilient water planning,			Inclusion of climate risk reduction considerations across latest revisions of Water Code and the related secondary legislation, providing clear recommendations and guidance as well as operations of the relevant authorities.	

frameworks, regulations and operations.	guidance/recommendation on how to enshrine the requirement for climate risk reduction in the legal framework, as well as how to include it in institutional plans, work programmes, budgets and departmental structures. (Note: Any failure of the Code to be enacted will not prevent the other mainstreaming approaches being realized and legislative support will still be provided for later integration or its integration in the secondary legislation)	budgeting and operations with special focus on the project targeted areas (with special focus on gender mainstreaming) 1.1.3 Technical consultancy services on systematic approaches to water sector climate risk assessment and risk reduction (i.e. UNICEF Drinking Water Safety and Security Planning -DWSSP) and assistance on inclusion in work programmes and planning.	
1.2. Develop water sector climate change risk reduction awareness raising programme for national and state agencies and establish CCA knowledge and information exchange mechanisms	The project will support the government in the implementation of the new requirements/framework (1.1.) through knowledge capacity building and advocacy. This activity will mainstream climate risk reduction as a requirement of all national and state actors working in planning and regulation of water management, including water supply, water resources and watershed management, flood alleviation and drainage, and irrigation.	1.2.1 Design and delivery of water sector climate risk reduction approaches awareness raising campaign for government staff s (with focus on gender) - office based rolling programme of series of presentations, seminars and lectures. 1.2.2 Set-up of an open access (web based) central knowledge platform hub for accessing global/regional climate vulnerability assessments and adaptation awareness materials available to all authorities involved in the water sector and agriculture planning	Promotion of climate risk inclusions in water code to water sector senior staff, and broader risk reduction approaches using multimedia and briefings through rounds of consultations.
1.3. Develop and apply criteria for assessing socially sensitive water pricing mechanisms ensuring prices take into account the actual costs of production, storage and	This activity (funded by the Government of Comoros) will support the preparations for the transition into the new water pricing, taking fully into account the financial sustainability needs, social and economic vulnerabilities of the local population and the projected climate impact.	1.3.1 Consultancy services to conduct economic and social analysis/evaluation, on-the-ground survey, evaluation, analysis of water users and service providers across all 3 islands to determine acceptable tariff structures required to sustain climate resilient water supplies.	Data capture of actual and future production costs, metering reading of supplied water, and consultation with government stakeholders on tariff blocks, including subsidized survival tariff

processing required in view of the projected climate stresses			
1.4. Develop planning guidance on source protection and water quality standards in view of climate change, operating procedures during periods of drought/floods; and safety plans	This activity will help the water authorities to build their capacity and response mechanisms during periods of droughts and floods as well as to understand the possible impact of climate change on the water quality and measures that can be taken to ensure standards are met in all targeted zones.	1.4.1 Consultancy services to collect best practices on Drinking Water Safety and Security Planning (DWSSP), including source protection, drought and flood event O&M, and water quality monitoring requirements, tailored to Comoros national state sector planning and regulatory requirements - with consultations on each island. 1.4.2. Preparation of Water Security and Safety Plans under the Water Master Plan to respond to (i) intense precipitation (ii) increased or prolonged drought (iii) saline intrusion	Development of risk reduction procedures and protocols, standards through industry consultation and CCA expert inputs.
1.5. Design and conduct trainings on best practices and gender-sensitive techniques of climate change adaptation in the context of water management, health and nutrition among national, regional and local water stakeholders	This activity will provide a basis for the authorities (water sector and agriculture sector when applicable) to have an understanding the global best practice on the climate change adaptation techniques, taking gender sensitive approach to their possible application.	1.5.1 Consultancy services to develop and deliver climate change national and state (island) level training programme (4 x 3-day workshops to 20 agencies) on water adaptation best practices and gender-sensitivity in water management, health and nutrition	Tailored recommendations on national and international gender sensitive CCA techniques in water sector and consultations with national stakeholders.
1.6 Strengthen decentralized water resources management capacities to undertake climate risk reduction	Existing on-island municipal, communal, and rural water supply operators/ service providers have no experience or expertise in assessing climatic risks to water supply infrastructure and therefore cannot design, construct,	1.6.1. Training provision on DWSSP (with a sole focus on climate risk reduction) in the planning, and management of design, O&M and planning activities of water service providers in each island.	Develop and deliver formal training programmes for service providers including water demand management.

assessments and develop and deliver awareness campaigns and training programmes to Water Management Committees and users	operate or maintain infrastructure to protect it from climatic risks. This activity will use training to introduce the UNICEF developed Drinking Water Safety and Security Planning (DWSSP) to local water providers, and support them with their risk assessments, risk mitigation/reduction improvement schedules, risk reduction operation & maintenance procedures, and formalization of service provider drought and cyclone resilience planning, including the integration of these with emergency flood and long term drought forecasts.	1.6.2. Train management committee trainers (training of trainers model - 3 workshops in each island) in good practices in water conservation (leakage and theft reduction) and preservation of water resources (pollution minimization) 1.6.3. Train DGEF / DGEME and their decentralized island (DR) agencies on the planning and regulation of climate risk reduction approaches to water management at the communal level and implement the tariff standards for water to deliver climate resilience. 1.6.4. Organize 15 participatory workshops (one per zone, up to 20 persons per workshop) to train all members of water management committees to use UNICEF Drinking Water Safety and Security Planning climate risk reduction design and operation procedures	Combination of theory and in-field based capacity building/training on each island. Develop formal climate change adaptation response training & awareness raising for communities being supplied. Introduction of DWSSP concepts and its integration into institutional planning. Workshop training programme using best practice materials to promote climate risk reduction with service providers and communities
2.1. Establish climate resilience focused IWRM Committees and Watershed Climate Risk Reduction Action Plans in the project intervention areas	The small size, steepness and proximity to the sea means the watersheds and aquifers of Comoros contain limited water resources which drain out over short periods yet also reach flash flood peaks very quickly. Given the finite and fragile nature of these water resources, the ability to help buffer these resources to climatic extremes of drought and salinization and flood and erosion/landslides is critical to maximizing the availability of raw water for exploitation. This activity will strengthen the watershed management governance of the basins to identify climate risks and	2.1.1 Establish and formalize the IWRM management committees in each watershed, prioritizing climate resilience and adaptation approaches, using multi-stakeholder engagement partnership consultations 2.1.2. Carry out a mapping survey of areas vulnerable to climate risks, particularly those exposed to saltwater intrusion to allow analysis of risk scenarios 2.1.3. Raise IWRM committees understanding of climate risks and adaptation options that ensure access to adequate water quality	IWRM committees established from water shed communities and stakeholders, including best practice partnership operating rules and governance structures. Characterization using GIS topography and slope analysis, well inventories, and ground truthing of erosion and deforestation. Awareness raising on watershed climate risks and available adaptation options to reduce risks of water supply derogation.

	adaptation options, to develop formal risk reduction action plans and implementation programmes to augment dry season flows (through recharge and water retention) and attenuate reduce flood flow peaks and reduce erosion and turbidity.	2.1.4. Develop Climate Risk Reduction action plans for river basins, taking into account climate change impacts on rainfall patterns, floods, rising temperatures and increasing salinity 2.1.5. Provide expert consultant(s) for the development of Water Code Decrees to integrate IWRM best practices to address the risks of climate change and facilitate knowledge exchange and formal collaboration between IWRM Committees and DG / REF	Facilitator support to committees on developing realistic and prioritized watershed adaptation action plans. IWRM committees in water resource climate change risk reduction to be legally recognized.
2.2. Implement the water protection and risk mitigating measures on the ground/operation alize the risk reduction plans	The project will support operationalization of these plans and fund priority watershed ecosystem based adaptation measures to improve water quantity and quality.	2.2.1. Implement soil conservation measures at community level in watersheds of 15 target areas to reduce increase rainfall intensity erosion 2.2.2 Upgrade community-based recharge areas in 11 watersheds to protect drought flows through reforestation and to other appropriate techniques	Reforestation, re-vegetation, buffer no-go areas and improved farming practices of hills and river banks to promote anti-erosion and slope stabilization interventions in priority sites within each catchment. Water retention and run-off capture interventions in priority areas
2.3 Establish water source protection zones and raise public awareness on climate risk reduction benefits of watershed management	The ability to regulate watershed activities in such a small country as Comoros with more than 40 mini-watersheds is recognized as challenging. There are no currently legislative instruments to use to support watershed adaptation or conservation approaches, and little public understanding of the impacts of land use on aggravating or mitigating climate risks.	2.3.1. Legal advisory services to be provided to IWRM committees for the implementation of climate resilience establishing watershed-specific zoning regulatory frameworks for the protection of recharge areas 2.3.2. 8 training to Train IWRM committee trainers in mainstreaming climate change adaptation, integrated water resources management strategy, planning and	Legal support in designing bye-laws and other regulatory approaches to establishing protection zones, including formalizing compliance responsibilities. Strengthening capacity of IWRM committees to develop adaptation intervention programmes with formal planning and budgeting, linked to fee income streams e.g. eco-tourism, water service provider set-aside.

	This activity will support both formal regulatory and informal self-regulatory approaches to introducing and sustaining watershed adaptation measures by upskilling IWRM committees in management approaches, developing appropriate watershed/land use protection zone legislation, but also developing and delivering public awareness campaigns and workshops on adaptation practices. The current knowledge base of the vulnerability of the nation's water resources is inadequate. Without the ability to monitor and analyze groundwater and surface water availability during droughts and degradation during floods, it is not possible to inform appropriate water supply infrastructure resilience upgrades.	budgeting, business models, management fees and performance monitoring. 2.3.3. Organize 5 public awareness workshops on increasing watershed resilience to climate change and ensuring the preservation of water resources good water management practices, including soil and water conservation, and reforestation	Public awareness campaigns for catchment residents to demonstrate self-benefits of risk reduction land use practices to improving watershed drinking water quality
2.4 Establish water resource monitoring network and upgrade the existing monitoring infrastructure to enable the collection of the required climate/weather data	This activity will focus on characterization of the nation's groundwater and surface water resources with a particular focus on how they respond to climatic extremes. This will require the development of a national water resources monitoring network, including instrumentation and technical training on monitoring, as well as building capacity to analyze available water balances and understand climatic impacts on water resources and identification of adaptation options. The ability to forecast both drought impacts on water	2.4.1. Prepare technical design of optimized monitoring network and detailed specification and selection of equipment. 2.4.2. Acquire and install 6 agro-meteorological stations in the targeted areas of the three islands 2.4.3. Acquire and install 20 automatic rain gauges covering the three islands 2.4.4. Acquire and install radar to improve protection of water resources and storage units during periods of heavy rainfall	Detailed design of monitoring network including access and protection considerations. Installation of rain gauge and weather stations across all islands, including one rainfall radar system.

	resources and water supply as well as flood events on watersheds and water supplies is critical to climate risk reduction and will be developed using the monitoring data collected.	2.4.5. Acquire and install 10 flow measurement equipment on two islands 2.4.6. Acquire and install 43 piezometric probes for the monitoring of saline intrusion in Grand Comore and Anjouan and Moheli 2.4.7. Strengthen the structure (climate proof) of the national hydrometeo services infrastructure (offices, data storage, power back-up, telecomms links) to withstand with climate shocks and disasters	Design and installation of stream water level stations; and routine flow gauging to develop discharge curves. Drilling of multi-level piezometers and use salinity probes to monitor water level and salinity with depth.
2.5. Build the capacities of the meteorological services to analyze and produce drought and flood forecasts for targeted users, including for flood early warning system	The meteorological services currently do not possess the required technical and analytical capacity to assess the data and prepare forecasts and reports/notifications for early warnings for the targeted project areas. The proposed activity aims to overcome this barrier and equip the staff with the necessary skills to prepare the basis for the climate information product dissemination.	2.5.1. Design and provide training to the staff of the meteorological services to develop and apply dynamic and statistical tools for seasonal forecasting and develop a simple rainfall based flood early warning alert system (using real time rainfall gauges only) 2.5.2. Develop standardized operational procedures (SOPs) for the collection, processing and analysis of water cycle data as well as drought and flood alert thresholds 2.5.3. Design and provide training for 20 field workers on the installation and maintenance of measurement equipment 2.5.4. Expert training design and delivery for 10 staffs of meteorological services on (i) drought indexing, (ii) preparation of annual, quarterly and monthly rainfall prediction, (iii) water resource projections	Train staff in climate forecasting in cooperation with the Indian Ocean WMO centre in Reunion. Train staff in monitoring network operation, maintenance, data capture, checking, processing and analysis to develop rainfall predictions and water balances into drought and flood forecasts using regional expertise at WMO RSMC Reunion where possible. Establishing a communication line with the respective authorities (DGSC, DGME) responsible for the communication of analysis outcome to the local population

		based on water balance analysis; (iv) assessments/reporting to notify the relevant authorities on the droughts and floods (EW basis)	Selected members of the committees will serve as the future trainers to ensure that this training is not one-off and continuity is ensured.
2.6 Build the capacity of the key government, local authorities and committees to interpret the climate information and raise awareness of the local population to act upon the forecasts and EWS.	The development of forecasts above does not always result in their use. The ability of organisations (notably water service providers) but also the public and watershed users to respond in a timely fashion is also critical to reduce exposure to climatic risks. Key government departments and local authorities will be trained in interpreting water-related climate and weather forecasts to support their decision-making processes. This activity focuses also on strengthening the national capacity to disseminate forecasts and flood warnings, and to ensure they are as accessible and useable as possible across a range of target users e.g. water service providers and rural farmers.	2.6.1 IT connectivity services to connect Storm Early Warning Forecasts to existing Volcanic Observatory Seismic Warning System (radio and SMS texts) and conduct 8 trainings for the key government and local officials, committee trainers (that will be appointed within the committees) to interpret this and other climate information provided by the meteo services 2.6.2. Conducting awareness raising campaigns (through radio, TV and printed media) to sensitize communities to the use of forecasting and alert information on the impacts of climate risks on water resources 2.6.3. Design new university web-based modules (1 for undergraduate and 1 for graduate students to be rolled out within the relevant existing specializations/degrees such as engineering etc.) with focus on the climate change reduction 2.6.4. Acquire hydrogeological control equipment to supporting existing university research to enable them to have capacity to conduct water quality and specific groundwater analysis, documenting the effects of climate change on the water cycle, in particular the intrusion of saltwater	Media releases and products on seasonal forecasts, drought & flood risks and their alleviation. Develop outreach trainers for community training on forecast understanding and response options. Programme of community education & empowerment on forecasts & response planning. Utilize existing available courses for climate risk reduction education as the basis. Increase national university capacity on groundwater investigation, monitoring, analysis on climate impacts and risk reduction strategies and their effectiveness. Equipment includes: 6 No. manual water level meters and 6 No. meters, 10 No. automated water level and salinity dataloggers, 1 No. Electrical Resistivity (VES) survey equipment, 1 No. Electrical Conductivity (EM 34) equipment, 3 No. pipe datalogger flowmeters, 3 No. stream flowmeters, laboratory equipment for major and minor ions analyses, 2 No. GPS high

			precision survey equipment, 3 No. all weather all terrain laptops
3.1. Undertake climate risk assessments of existing groundwater abstraction wells to develop risk reduction pumping strategies, and construction of additional boreholes in zones at risk of drought water scarcity in Grande Comore	Groundwater is more resilient water resource than surface water flows and rainwater harvesting due to the storage held within the ground, however groundwater is still vulnerable to over-exploitation especially during dry periods. This activity focuses on maximizing the resilience of existing boreholes to drought periods through rigorous technical assessment of salinization, and where additional groundwater resources are identified to exploit these for the distribution to rural communities with only access to rainwater. The surface water supply infrastructure is vulnerable to increased frequency and strength of flood damage, erosion & turbidity, and water shortages due to low flows and/or salinization.	3.1.1. Site selection confirmation for new production borehole siting based upon review of groundwater piezometer installations. 3.1.2 Step test salinity pumping tests of 6 existing production boreholes in Zones 1 to 5 to determine optimum pumping rate 3.1.3 Design of 5 new production boreholes in Zones 1, 2, 3 and 4. 3.1.4 Drilling, testing and commissioning of new production boreholes	Climate risk reduction measures for groundwater resources exploitation in Grande Comore. Includes pumping tests of existing boreholes to optimize pumping regime to minimize pumped water salinity. Construction and commissioning of new boreholes will improve resilience of existing groundwater exploitation and provide dry season supply to RWH dominated communities.
3.2. Build infrastructure to increase resilience of water supply facilities to extended duration low flow periods, greater intensity flood flow damage and flood flow higher turbidity and bacteria loadings (Grande Comore,	This activity focuses on increasing the resilience of the stream water supply intake infrastructure to flood damage, reducing exposure of water treatment plants to debris and sediment laden water, and increasing access to stream low flows. This activity also will expand access to groundwater supplies on Grand Comore to provide drought resilience to rural communities reliant on rainwater, and reduce dry season risks of pumping	3.2.1 Grand Comore Island: Construction of 17 reservoir tanks, 5 treatment plants, 6 distribution network extensions and 4 irrigation impluviums 3.2.2. Anjouan Island Flood protection upgrades to 9 surface water intakes, 7 treatment plants, 11 storage tanks and 7 network extensions, and livestock and irrigation watering 110 basins and 95 troughs 3.2.3 Moheli Island	Improvements to water supply networks on Grand Comore to reduce vulnerability to drought and flood risks Improvements to water supply networks on Anjouan to reduce vulnerability of surface water schemes to drought and flood risks. Improvements to water supply networks on Moheli to reduce

Anjouan island and Moheli island)	saline groundwater by using storage tanks to reduce peak pumping.	Flood protection upgrades to 2 surface water intakes, 2 treatment plants, 3 storage tanks and 2 network extensions, and livestock and irrigation watering 28 basins and 12 troughs	vulnerability of surface water schemes to drought and flood risks.
3.3. Installation of flowmeters to support climate resilient tariff adjustments, and leakage reduction programmes to improve the water pricing and management system taking into account the additional costs associated with climatic hazards	This activity supports the setting of climate resilience tariffs by monitoring water provision. It also contributes to drought resilience through reducing system leakage and water loss.	3.3.1 Procurement and installation of 300 bulk flow meters, leakage detection equipment and flow monitoring data loggers 3.3.2 Training to the water providers on the leakage detection and leakage reduction programming, operations and maintenance of meters 3.3.3. Consultancy to analyze the consumption patterns and draft guidance for the water providers with respect to tariffs/pricing	Installation and use of flow meters to demonstrate improvements to drought and flood resilient water supply and enable robust setting tariffs to sustain climate risk reduction. Technical assistance to island water supply providers on water losses (leakage, theft, meter calibration) and on leakage reduction asset management. Technical assistance to national and island regulators on consumption patterns and necessary tariff rates.

H.2. Arrangements for Monitoring, Reporting and Evaluation

GCF funding will be used to ensure that monitoring and evaluation systems are put in place to track progress over the 6 years of project implementation towards the planned project outcomes and fund level impacts. This will be achieved through the means of verification outlined in Table H.1.2 above, where progress on each indicator from the baseline to the mid-point and end-point targets for those indicators will be tracked. The project will use the Randomized Control Trial approaches to monitor communities' adaptive capacity and exposure to climate risks, tracking their involvement in both water resources management, watershed management, water access, improvement of water supply delivery services and health condition integrating gender dimensions as outlined in the gender action plan. The questionnaire will include gathering baseline data on sources of water, capacity on integrating climate information on water related decision making, household economic activities, production yields (fishing, farming, aquaculture), monetary income and harvesting of natural resources. The questionnaire will also include analysis of additional variables contributing to vulnerability to climate change and adaptive capacity, including understanding of climate change impacts on water access, gender roles. The monitoring will use routine, concurrent and participatory processes. The project will carry out external mid-term and terminal evaluation (using International and national independent consultants) and the evaluations will be undertaken through a counterfactual analysis (or 'with or without' scenario) that allows a comparison between what happened and what would have happened in the absence of project interventions. For this analysis to take place, a comparison group who are/were not directly exposed to any of the project's intervention should be created. This will greatly aid in attributing changes in project outcomes to interventions undertaken by the project. For example, a detailed survey instrument, the Household Survey: Adaptive Capacity and Climate Vulnerability, will be used in the target landscapes.

The project is designed to engage the communities in continuous monitoring of the project activities as indicated in the ecosystems restoration efforts. The project monitoring will engage beneficiaries and mechanisms will be put in place to collect data through participation of local communities. To capture the impacts of the proposed interventions, the project includes the use of Randomized Control Trial approaches as part of its annual surveys (Household Survey: Adaptive Capacity and Climate Vulnerability) to assess the (i) Impact of improving the water supply delivery service in dry periods and (ii) impact of restoration on risk exposure.

The project results as outlined in the project results framework will be monitored and reported annually and evaluated periodically during project implementation to ensure the project effectively achieves these results.

Project-level monitoring and evaluation will be undertaken in compliance with UNDP requirements as outlined in the [UNDP POPP](#) and [UNDP Evaluation Policy](#). While these UNDP requirements are not outlined in this project document, the UNDP Country Office will work with the relevant project stakeholders to ensure UNDP M&E requirements are met in a timely fashion and to high quality standards. Additional mandatory GCF-specific M&E requirements will be undertaken in accordance with relevant GCF policies.

In addition to these mandatory UNDP and GCF M&E requirements, other M&E activities deemed necessary to support project-level adaptive management will be agreed during the Project Inception Workshop and will be detailed in the Inception Workshop Report. This will include the exact role of project target groups and other stakeholders in project M&E activities including national/regional institutes assigned to undertake project monitoring.

A detailed and budgeted monitoring and evaluation plan will be included in the UNDP project document. The development and the implementation of the M&E plan will be guided by the UNDP guidelines for project and programs M&E that can be accessed from the following link: (<http://web.undp.org/evaluation/handbook/documents/english/pme-handbook.pdf>). The guidelines provide information on the possible types and methods for evaluation, however that does not prescribe a concrete

method this project use. Also the evaluation of the project will be undertaken through the counterfactual analysis by using the Randomized Control Trial approaches.

M&E oversight and monitoring responsibilities:

National Project Manager: The NPM is responsible for day-to-day project management and regular monitoring of project results and risks, including social and environmental risks. The NPM will ensure that all project staff maintain a high level of transparency, responsibility and accountability in M&E and reporting of project results. The Project Manager will inform the Project Board, the UNDP Country Office and the UNDP-GEF Regional Technical Advisor of any delays or difficulties as they arise during implementation so that appropriate support and corrective measures can be adopted.

The NPM will develop annual work plans to support the efficient implementation of the project. The NPM will ensure that the standard UNDP and GCF M&E requirements are fulfilled to the highest quality. This includes, but is not limited to, ensuring the results framework indicators are monitored annually in time for evidence-based reporting in the Annual Project Report, and that the monitoring of risks and the various plans/strategies developed to support project implementation (e.g. Environmental and social management plan, gender action plan etc..) occur on a regular basis.

Project Board: The Project Board will take corrective action as needed to ensure the project achieves the desired results. The Project Board will hold project reviews to assess the performance of the project and appraise the Annual Work Plan for the following year. In the project's final year, the Project Board will hold an end-of-project review to capture lessons learned and discuss opportunities for scaling up and to highlight project results and lessons learned with relevant audiences. This final review meeting will also discuss the findings outlined in the project terminal evaluation report and the management response.

Project Implementing Partner: The Implementing Partner is responsible for providing all required information and data necessary for timely, comprehensive and evidence-based project reporting, including results and financial data, as necessary and appropriate. The Implementing Partner will strive to ensure project-level M&E is undertaken by national institutes and is aligned with national systems so that the data used by and generated by the project supports national systems.

UNDP Country Office: The UNDP Country Office in Moroni will support the NPM as needed, including through annual supervision missions. The annual supervision missions will take place per the schedule outlined in the annual work plan. Supervision mission reports will be circulated to the project team and Project Board within one month of the mission. The UNDP Country Office will initiate and organize key M&E activities including the Annual Project Report, the independent mid-term review and the independent terminal evaluation. The UNDP Country Office will also ensure that the standard UNDP and GCF M&E requirements are fulfilled to the highest quality.

The UNDP Country Office is responsible for complying with all UNDP project-level M&E requirements as outlined in the [UNDP POPP](#). This includes ensuring the UNDP Quality Assurance Assessment during implementation is undertaken annually; the regular updating of the ATLAS risk log; and, the updating of the UNDP gender marker on an annual basis based on gender mainstreaming progress reported in the Annual Project Report and the UNDP ROAR. Any quality concerns flagged during these M&E activities (e.g. Annual Project Report quality assessment ratings) must be addressed by the UNDP Country Office and the Project Manager.

The UNDP Country Office will support GCF staff (or their designate) during any missions undertaken in the country and support any ad-hoc checks or ex post evaluations that may be required by the GCF.

The UNDP Country Office will retain all project records for this project for up to seven years after project financial closure to support any ex-post reviews and evaluations undertaken by the UNDP Independent Evaluation Office (IEO) and/or the GCF.

UNDP-Global Environmental Finance Unit (UNDP-GEF): Additional M&E and implementation oversight, quality assurance and troubleshooting support will be provided by the UNDP-GEF Regional Technical Advisor and the UNDP-GEF Directorate as outlined in the management arrangement section above.

Audit arrangements:

The project will be audited annually per UNDP Financial Regulations and Rules and applicable audit policies on NIM implemented projects.⁶⁴ Additional audits may be undertaken at the request of the GCF.

Additional monitoring and reporting requirements:

Inception Workshop and Report: A project inception workshop will be held within four months after the project document has been signed by all relevant parties to, amongst others:

- a. Re-orient project stakeholders to the project strategy and discuss any changes in the overall context that influence project strategy and implementation;
- b. Discuss the roles and responsibilities of the project team, including reporting and communication lines and conflict resolution mechanisms;
- c. Review the results framework and finalize the indicators, means of verification and monitoring plan;
- d. Discuss reporting, monitoring and evaluation roles and responsibilities and finalize the M&E budget; identify national/regional institutes to be involved in project-level M&E;
- e. Identify how project M&E can support national monitoring of SDG indicators as relevant;
- f. Update and review responsibilities for monitoring the various project plans and strategies, including the risk log; Environmental and Social Management Plan and other safeguard requirements; the gender action plan; and other relevant strategies;
- g. Review financial reporting procedures and mandatory requirements, and agree on the arrangements for the annual audit; and
- h. Plan and schedule Project Board meetings and finalize the first-year annual work plan.

The NPM will prepare the inception workshop report no later than one month after the inception workshop. The inception workshop report will be cleared by the UNDP Country Office and the UNDP-GEF Regional Technical Adviser and will be approved by the Project Board. Inception report will be generated in line with the GCF template.

Annual Project Report: The NPM, the UNDP Country Office, and the UNDP-GEF Regional Technical Adviser will provide objective input to the annual project report covering the calendar year for each year of project implementation. The NPM will ensure that the indicators included in the project results framework are monitored annually in advance so that progress can be included in the report. Any environmental and social risks and related management plans will be monitored regularly, and progress will be included in the report.

The Annual Project Report will be shared with the Project Board. The UNDP Country Office will coordinate the input of other stakeholders to the report as appropriate. The quality rating of the previous year's report will be used to inform the preparation of the subsequent report.

Lessons learned and knowledge generation: Results from the project will be disseminated within and beyond the project intervention area through existing information sharing networks and forums. The project will identify and participate, as relevant and appropriate, in scientific, policy-based and/or any other networks, which may be of benefit to the project. The project will identify, analyze and share lessons learned that might be beneficial to the design and implementation of similar projects and disseminate these lessons widely. There will be continuous information exchange between this project and other projects of similar focus in the same country, region and globally.

⁶⁴ See guidance here: <https://info.undp.org/global/popp/frm/pages/financial-management-and-execution-modalities.aspx>

Independent Mid-term Review (MTR): An independent mid-term review process will begin after the third Annual Project Report has been submitted to the GCF.

The MTR findings and responses outlined in the management response will be incorporated as recommendations for enhanced implementation during the final half of the project's duration. The terms of reference, the review process and the MTR report will follow the standard templates and guidance prepared by the UNDP IEO for GEF-financed projects available on the [UNDP Evaluation Resource Center \(ERC\)](#). As noted in this guidance, the evaluation will be 'independent, impartial and rigorous'. The consultants that will be hired to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. Other stakeholders will be involved and consulted during the terminal evaluation process. Additional quality assurance support is available from the UNDP-GEF Directorate. The final MTR report will be available in English and will be cleared by the UNDP Country Office and the UNDP-GEF Regional Technical Adviser and approved by the Project Board.

Terminal Evaluation (TE): An independent terminal evaluation (TE) will take place upon completion of all major project outputs and activities. The terminal evaluation process will begin at least three months before operational closure of the project allowing the evaluation mission to proceed while the project team is still in place, yet ensuring the project is close enough to completion for the evaluation team to reach conclusions on key aspects such as project sustainability.

The NPM will remain on contract until the TE report and management response have been finalized. The terms of reference, the evaluation process and the final TE report will follow the standard templates and guidance prepared by the UNDP IEO for GEF-financed projects available on the [UNDP Evaluation Resource Center](#). As noted in this guidance, the evaluation will be 'independent, impartial and rigorous'. The consultants that will be hired to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. Additional quality assurance support is available from the UNDP-GEF Directorate. The final TE report will be cleared by the UNDP Country Office and the UNDP-GEF Regional Technical Adviser and will be approved by the Project Board. The TE report will be publicly available in English on the UNDP ERC.

The UNDP Country Office will include the planned project terminal evaluation in the UNDP Country Office evaluation plan and will upload the final terminal evaluation report in English and the corresponding management response to the UNDP Evaluation Resource Centre (ERC).

Final Report: The project's final Annual Project Report along with the terminal evaluation (TE) report and corresponding management response will serve as the final project report package. The final project report package shall be discussed with the Project Board during an end-of-project review meeting to discuss lesson learned and opportunities for scaling up.

I. Supporting Documents for Funding Proposal

- ☒ NDA No-objection Letter
- ☒ Feasibility Study
- ☒ Integrated Financial Model that provides sensitivity analysis of critical elements (xls format, if applicable)
- ☒ Confirmation letter or letter of commitment for co-financing commitment (If applicable)
- ☒ Project/Programme Confirmation/Term Sheet (including cost/budget breakdown, disbursement schedule, etc.) – see *the Accreditation Master Agreement, Annex I*
- ☒ Environmental and Social Impact Assessment (ESIA) or Environmental and Social Management Plan
(If applicable)
- ☒ Appraisal Report or Due Diligence Report with recommendations (If applicable)
- ☐ Evaluation Report of the baseline project (If applicable)
- ☒ Map indicating the location of the project/programme
- ☒ Timetable of project/programme implementation
- ☒ Project/Programme Confirmation
- ☒ Economic Analysis
- ☒ Procurement Plan
- ☒ Operations and Maintenance Plan
- ☒ Stakeholder consultation and engagement plan
- ☒ Gender Assessment and Action Plan
- ☒ HACT Assessment
- ☒ Project Design Process Meetings Documents
- ☒ Community Engagement Documents

* Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.

Vice-présidence

En charge du Ministère de l'Energie, de l'Agriculture, de la Pêche, de l'Environnement, de l'Aménagement du Territoire et de l'Urbanisme

Le Secrétaire Général,

Réf. N018-868 /VP-MEAPEATU/SG



نيابة الرئيس

وزارة الاقتصاد، والتخطيط، والطاقة، والصناعة،
والحرف، والسياحة، والاستثمار، والقطاع الخاص،
والشؤون العقارية

إدارة العامة للشؤون الطاقة والمناجم والمياه

المدير العام

Moroni le 01 Août 2018

To: The Green Climate Fund ("GCF")

Subject: Funding proposal for the GCF by UNDP Comoros regarding project « Ensuring climate resilient water supplies in the Comoros Islands »

Dear Madam, Sir,

We refer to the project« Ensuring climate resilient water supplies in the Comoros Islands »in the Union of Comoros as included in the funding proposal submitted by UNDP Comoros to us on October 2017

The undersigned is the focal point of the Union of Comoros.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we here by communicate our no-objection to the project« Ensuring climate resilient water supplies in the Comoros Islands »as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of the Union of Comoros has no-objection to the project« Ensuring climate resilient water supplies in the Comoros Islands »as included in the funding proposal;
- (b) The project« Ensuring climate resilient water supplies in the Comoros Islands »as included in the funding proposal is in conformity with the Union of Comoros' national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the project« Ensuring climate resilient water supplies in the Comoros Islands »as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project« Ensuring climate resilient water supplies in the Comoros Islands »as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the project "Ensuring climate resilient water supplies in the Comoros Islands »

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,


Youssouf Hamadi
GCF Focal Point



Environmental and social report(s) disclosure

Basic project/programme information	
Project/programme title	Ensuring climate resilient water supplies in the Comoros Islands
Accredited entity	United Nations Development Programme (UNDP)
Environmental and social safeguards (ESS) category	Category B
Specific project location or target country/ies of the programme, as the case may be	Union of Comoros (all three islands: Grande Comore, Anjouan and Moheli)
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	2018-09-13
Language(s) of disclosure	English, French, and Shikomori
Link to disclosure	English http://www.km.undp.org/content/dam/comoros/docs/Library/FP-UNDP-230518-5740-Annex VI b-ESMF-EN.pdf French http://www.km.undp.org/content/dam/comoros/docs/Library/FP-UNDP-230518-5740-Annex VI b-ESMF-FR.pdf Shikomori http://www.km.undp.org/content/comoros/fr/home/library/CGES-GCF.html An ESIA consistent with the requirements for a category B project is contained in the Environmental and Social Management Framework
Other link(s)	http://www.km.undp.org/content/comoros/fr/home/library/CGES-GCF.html
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	2018-09-13
Language(s) of disclosure	English, French, and Shikomori
Link to disclosure	English http://www.km.undp.org/content/dam/comoros/docs/Library/FP-UNDP-230518-5740-Annex VI b-ESMF-EN.pdf French http://www.km.undp.org/content/dam/comoros/docs/Library/FP-UNDP-230518-5740-Annex VI b-ESMF-FR.pdf

	Shikomori http://www.km.undp.org/content/comoros/fr/home/library/CGES-GCF.html An ESMP consistent with the requirements for a category B project is contained in the Environmental and Social Management Framework
Other link(s)	http://www.km.undp.org/content/comoros/fr/home/library/CGES-GCF.html https://www.gouvernement.km/actualites/cadre-de-gestion-environnementale-.html http://comorosparks.com/index.php?id=30 http://www.uccia-comores.com/assurer-un-approvisionnement-en-eau-resilient-au-climat-aux-comores/ https://www.facebook.com/Gouvkm/ https://www.facebook.com/UNDPComoros/
Resettlement Action Plan (RAP) (if applicable)	
Date of disclosure on accredited entity's website	
Any other relevant ESS reports and/or disclosures (if applicable)	
Description of report/disclosure	Social and Environmental Screening Template
Language(s) of disclosure	English, French, and Shikomori English http://www.km.undp.org/content/dam/comoros/docs/Library/FP-UNDP-230518-5740-Annex VI a-SESP-EN.pdf French http://www.km.undp.org/content/dam/comoros/docs/Library/FP-UNDP-230518-5740-Annex VI a-SESP-FR.pdf Shikomori http://www.km.undp.org/content/dam/comoros/docs/Library/FP-UNDP-230518-5740-Annex VI a-SESP-KM.pdf
Link to disclosure	http://www.km.undp.org/content/comoros/fr/home/library/MEES-GCF.html
Other link(s)	https://www.gouvernement.km/actualites/mod%C3%A8le-d%E2%80%99examen-pr%C3%A9alable-social-et-environnemental-sur-le-fonds-vert-pour-le-climat-1118.html https://www.facebook.com/UNDPComoros/
Disclosure in locations convenient to affected peoples	
Date	[]
Place	[]

Note: This form was prepared by the accredited entity stated above.